

Laws of Indices & Surds

Powers follow four laws; surds follow two. Almost every question here is a “no calculator” question — the working is the answer.



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THE TOPIC ON ONE MAP

Laws of Indices & Surds

01 — Indices

$$81^{\frac{1}{4}}$$

“in the form ax^n ”

$$a^{\frac{1}{3}} = -4$$

02 — Surds: simplify and expand

$$a\sqrt{b}$$

“show that”

03 — Rationalise the denominator

$$\frac{2}{\sqrt{5}-1}$$

$$a + b\sqrt{5}$$

04 — Equations with surd coefficients

$$x\sqrt{3} - 3 = x + \sqrt{3}$$

exact area

05 — Powers of x in the power

$$y = 3^x$$

$$p = 2^x$$

hence solve

01 Indices: Laws, Negative and Fractional Powers

Same base: add, subtract, multiply the powers. Negative means flip. The bottom of a fraction is the root.

WHEN YOU SEE → YOU DO

“write down the value of $81^{\frac{1}{4}}$ ”

→ root first, then power

“in the form ax^n ”, “ kx^n where k, n are simplified”

→ one number $\times$ one power of x

“given $y = \frac{1}{16}x^4$, express $y^{-\frac{3}{2}}$...”

→ number and x each get the power

“given $a^{\frac{1}{3}} = -4$, find a ”

→ undo: cube both sides

“use your answer to part (a)”

→ build the harder power from the easy one

$$a^m \times a^n = a^{m+n} \quad a^m \div a^n = a^{m-n} \quad (a^m)^n = a^{mn}$$

$$a^0 = 1 \quad a^{-n} = \frac{1}{a^n} \quad a^{\frac{m}{n}} = (\sqrt[n]{a})^m$$

WORKED MODEL • JUNE 2025

$p = \frac{1}{16}x^4$, $q = \frac{40}{x^3}$. Find pq^2 in the form kx^n .

1 $q^2 = 1600x^{-6}$

square every part

2 $\frac{1}{16} \times 1600 = 100$, $x^{4-6} = x^{-2}$

numbers, then powers

ANSWER $100x^{-2}$

IDEA 01

Practise: Indices

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Root first, then power

P1 • TOPIC 1 3 marks

Find the exact value of (a) $27^{\frac{2}{3}}$ (b) $16^{-\frac{3}{4}}$ (c) $\left(\frac{9}{4}\right)^{-\frac{1}{2}}$

Q2 DR ESLAM Form ax^n

P1 • TOPIC 1 3 marks

Write $\frac{6x^3 - 2\sqrt{x}}{4x^2}$ in the form $ax^p + bx^q$.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 1 3 marks

1 Write down the value of

(i) $81^{\frac{1}{4}}$

(ii) $8^{\frac{2}{3}}$

(iii) 5^{-2}

(3 marks)

Q4**PRACTICE** Practice • Medium**P1 • TOPIC 1 4 marks**

- 4 (a) Given that $y = \frac{1}{16}x^4$, express each of the following in the form ax^n , where a and n are constants.

$y^{\frac{1}{2}}$

(1 mark)

(b) y^{-1}

(1 mark)

(c) $y^{-\frac{3}{2}}$

(2 marks)

Q5**PRACTICE**

Practice • Very Hard

P1 • TOPIC 1 4 marks

- 4 (a) Given that $y = \frac{81}{16}x^{-12}$, express each of the following in the form ax^n , where a and n are constants.

$y^{\frac{3}{4}}$

(1 mark)

(b) $y^{-\frac{1}{2}}$

(1 mark)

(c) $\left(\frac{1}{y^2}\right)^{-3}$

(2 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA11 · October 2023 · Q2

P1 · TOPIC 1 4 marks

2. Given that

$$a = \frac{1}{64}x^2 \quad b = \frac{16}{\sqrt{x}}$$

express each of the following in the form kx^n where k and n are simplified constants.

(a) $a^{\frac{1}{2}}$ (1)

(b) $\frac{16}{b^3}$ (1)

(c) $\left(\frac{ab}{2}\right)^{-\frac{4}{3}}$ (2)

(Total for Question 2 is 4 marks)

Q7 **PAST PAPER** Pearson IAL WMA11 · June 2025 · Q1**P1 · TOPIC 1** **5 marks**

1. Given that

$$p = \frac{1}{16}x^4 \qquad q = \frac{40}{x^3}$$

express each of the following in the form kx^n where k and n are fully simplified constants.

(a) $p^{\frac{1}{2}}$ (1)

(b) $(pq)^{-1}$ (2)

(c) pq^2 (2)

(Total for Question 1 is 5 marks)

02 Surds: Simplify and Expand

Find the biggest square factor. Surds multiply like letters, and $\sqrt{a} \times \sqrt{a} = a$.

WHEN YOU SEE → YOU DO

“in the form $a\sqrt{b}$ ”

$$\rightarrow \sqrt{48} = \sqrt{16}\sqrt{3} = 4\sqrt{3}$$

“show that $2\sqrt{3} + 4\sqrt{12} - 3\sqrt{75} = -5\sqrt{3}$ ”

→ simplify every surd to the same $\sqrt{}$, then collect

“by expanding and simplifying, show that”

→ expand like brackets; $\sqrt{a}\sqrt{a} = a$

“ $\sqrt{a} + \sqrt{b} = \sqrt{a+b}$ is not true — give an example”

→ one numerical counter-example

WORKED MODEL • EXPAND

$$(\sqrt{12} - 3)(2 - \sqrt{75})$$

$$\textcircled{1} (2\sqrt{3} - 3)(2 - 5\sqrt{3})$$

simplify first

$$\textcircled{2} 4\sqrt{3} - 30 - 6 + 15\sqrt{3}$$

expand: $\sqrt{3}\sqrt{3} = 3$

$$\text{ANSWER } 19\sqrt{3} - 36$$

× **TRAP** $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$. $\sqrt{9+16} = 5$, but $\sqrt{9} + \sqrt{16} = 7$.

IDEA 02

Practise: Surds: Simplify and Expand

My questions first, then selected practice.

Q8 DR ESLAM Biggest square factor

P1 · TOPIC 1 3 marks

Simplify (a) $\sqrt{50} - \sqrt{18}$ (b) $(3 + \sqrt{2})(3 - \sqrt{2})$ **SELECTED PRACTICE****Q9** PRACTICE Practice · Easy

P1 · TOPIC 1 3 marks

3 Write the following in the form $a\sqrt{b}$

(i) $\sqrt{2} + \sqrt{8}$

(ii) $4\sqrt{3} - \sqrt{12} + 4\sqrt{48}$

(iii) $(2\sqrt{2})^3 + 3\sqrt{2}$

(3 marks)

Q10**PRACTICE**

Practice · Hard

P1 · TOPIC 1 5 marks

5 (a) Show that $2\sqrt{3} + 4\sqrt{12} - 3\sqrt{75} = -5\sqrt{3}$

(3 marks)

(b) By expanding and simplifying, show that

$$(\sqrt{3} - 5)(5 - \sqrt{3}) = 10\sqrt{3} - 28$$

(2 marks)

Q11**PRACTICE**

Practice · Very Hard

P1 · TOPIC 1 12 marks

8 (a) Expand $(a + b\sqrt{5})^2$.

(2 marks)

(b) A square has an area of $(49 + 12\sqrt{5}) \text{ m}^2$ and a side length of $(a + b\sqrt{5}) \text{ m}$.

Show that $ab = 6$, and explain why this proves that a and b must both be non-negative.

(2 marks)

(c) Show that $a^4 - 49a^2 + 180 = 0$.

(3 marks)

(d) By using the substitution $y = a^2$ or otherwise, solve the equation $a^4 - 49a^2 + 180 = 0$.
Hence determine the side length of the square.

(5 marks)

03 Rationalise the Denominator

No surd may stay on the bottom. Multiply top and bottom by the **conjugate** — the same bracket with the sign flipped.

WHEN YOU SEE → YOU DO

“rationalise the denominator”, $\frac{3}{\sqrt{2}}$

$$\rightarrow \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$\frac{2}{3 + \sqrt{5}}$$

$$\rightarrow \times \frac{3 - \sqrt{5}}{3 - \sqrt{5}} \text{ — bottom becomes } 9 - 5$$

“in the form $a + b\sqrt{5}$ where a, b are rational”

→ rationalise, then split the fraction

1

Conjugate

Flip the sign in the middle of the bottom.

2

Bottom

$$(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b \text{ — no surd left.}$$

3

Top

Expand, then split into $p + q\sqrt{b}$.

WORKED MODEL • JUNE 2022

$$\frac{4\sqrt{5} - 5}{7 - 3\sqrt{5}}$$

$$1 \times \frac{7 + 3\sqrt{5}}{7 + 3\sqrt{5}}$$

conjugate

$$2 \text{ bottom: } 49 - 45 = 4; \text{ top: } 28\sqrt{5} + 60 - 35 - 15\sqrt{5}$$

expand

ANSWER $\frac{25}{4} + \frac{13}{4}\sqrt{5}$

IDEA 03

Practise: Rationalise the Denominator

My questions first, then selected practice, then real past papers.

Q12 DR ESLAM Conjugate

P1 · TOPIC 1 3 marks

Rationalise (a) $\frac{6}{\sqrt{3}}$ (b) $\frac{4}{\sqrt{5}-1}$

SELECTED PRACTICE

Q13 PRACTICE Practice · Easy

P1 · TOPIC 1 2 marks

4 Rationalise the denominator of $\frac{3}{\sqrt{2}}$.

(2 marks)

Q14 **PRACTICE** Practice · Easy

P1 • TOPIC 1 3 marks

9 Simplify

$$\frac{3 - 2\sqrt{3}}{4 - \sqrt{3}}$$

(3 marks)

Q15 **PRACTICE** Practice · Very Hard

P1 • TOPIC 1 5 marks

6 (a) $\sqrt{a} - \sqrt{b} = \sqrt{a-b}$ is not true in general. Give an example of an a and a b for which it is true.

(1 mark)

(b) Show that $\frac{2 - \sqrt{3}}{1 + \sqrt{3}} = a + b\sqrt{3}$, where a and b are rational numbers.

(4 marks)

PAST PAPERS

Q16 PAST PAPER Pearson IAL WMA11 • June 2022 • Q3

P1 • TOPIC 1 5 marks

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

- (i) Show that $\frac{\sqrt{180} - \sqrt{80}}{\sqrt{5}}$ is an integer and find its value.

(2)

- (ii) Simplify

$$\frac{4\sqrt{5} - 5}{7 - 3\sqrt{5}}$$

giving your answer in the form $a + b\sqrt{5}$ where a and b are rational numbers.

(3)

(Total 5 marks)

Q17 PAST PAPER Pearson IAL WMA11 · June 2023 · Q6

P1 · TOPIC 1 7 marks

6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Expand and simplify

$$\left(r - \frac{1}{r}\right)^2$$

(2)

(b) Express $\frac{1}{3 + 2\sqrt{2}}$ in the form $p + q\sqrt{2}$ where p and q are integers.

(2)

(c) Use the results of parts (a) and (b), or otherwise, to show that

$$\sqrt{3 + 2\sqrt{2}} - \frac{1}{\sqrt{3 + 2\sqrt{2}}} = 2$$

(3)

(Total for Question 6 is 7 marks)

Q18**PAST PAPER**

Pearson IAL WMA11 • October 2023 • Q3

P1 • TOPIC 1 6 marks

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Write $\frac{8 - \sqrt{15}}{2\sqrt{3} + \sqrt{5}}$ in the form $a\sqrt{3} + b\sqrt{5}$ where a and b are integers to be found.

(3)

(b) Hence, or otherwise, solve

$$(x + 5\sqrt{3})\sqrt{5} = 40 - 2x\sqrt{3}$$

giving your answer in simplest form.

(3)

(Total for Question 3 is 6 marks)

Q19**PAST PAPER**

Pearson IAL WMA11 · October 2024 · Q2

P1 · TOPIC 1 7 marks

2. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(i) Simplify fully

$$\frac{3y^3(2x^4)^3}{4x^2y^4} \quad (3)$$

(ii) Find the exact value of a such that

$$\frac{16}{\sqrt{3} + 1} = a\sqrt{27} + 4$$

Write your answer in the form $p\sqrt{3} + q$ where p and q are fully simplified rational constants.

(4)

(Total for Question 2 is 7 marks)

04 Equations with Surd Coefficients

Treat $\sqrt{3}$ like a number: collect the x terms, factorise x out, divide — then rationalise.

WHEN YOU SEE → YOU DO

“solve $x\sqrt{3} - 3 = x + \sqrt{3}$, answer in the form $p + q\sqrt{3}$ ”

→ collect, factorise, divide, rationalise

area / length with surds (“exact value”)

→ set up the formula, divide, rationalise

“(a + b $\sqrt{5}$)² = 49 + 12 $\sqrt{5}$ ”

→ compare rational parts and surd parts

WORKED MODEL • JUNE 2024

$$x\sqrt{3} - 3 = x + \sqrt{3}$$

$$1 \quad x\sqrt{3} - x = 3 + \sqrt{3}$$

collect x

$$2 \quad x(\sqrt{3} - 1) = 3 + \sqrt{3} \Rightarrow x = \frac{3 + \sqrt{3}}{\sqrt{3} - 1}$$

factorise, divide

$$3 \quad x \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} : \frac{6 + 4\sqrt{3}}{2}$$

rationalise

ANSWER $x = 3 + 2\sqrt{3}$

IDEA 04

Practise: Equations with Surds

My questions first, then selected practice, then real past papers.

Q20 DR ESLAM Factorise x out

P1 · TOPIC 1 4 marks

Solve $x\sqrt{2} + 3 = 2x + \sqrt{2}$, giving x in the form $a + b\sqrt{2}$.

$x =$ _____

SELECTED PRACTICE

Q21 PRACTICE Practice · Medium

P1 · TOPIC 1 4 marks

7 Solve the equation $6 - x\sqrt{7} = \frac{2x}{\sqrt{7}}$, giving your answer in the form $\frac{a\sqrt{7}}{b}$ where a and b are integers.

(4 marks)

Q22**PRACTICE**

Practice · Hard

P1 · TOPIC 1 4 marks

- 7 Solve the equation $66 + x\sqrt{75} = \frac{4x}{\sqrt{3}}$, giving your answer in the form $a\sqrt{b}$ where a and b are integers.

(4 marks)**Q23****PRACTICE**

Practice · Very Hard

P1 · TOPIC 1 5 marks

- 7 Solve the equation $\sqrt{20} + \frac{\sqrt{5}}{2x} = \frac{1}{x\sqrt{45}}$

(5 marks)

PAST PAPERS

Q24 PAST PAPER Pearson IAL WMA11 · January 2022 · Q3

P1 · TOPIC 1 7 marks

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(i)

$$f(x) = (x + \sqrt{2})^2 + (3x - 5\sqrt{8})^2$$

Express $f(x)$ in the form $ax^2 + bx\sqrt{2} + c$ where a , b and c are integers to be found. (3)

(ii) Solve the equation

$$\sqrt{3}(4y - 3\sqrt{3}) = 5y + \sqrt{3}$$

giving your answer in the form $p + q\sqrt{3}$ where p and q are simplified fractions to be found. (4)

(Total 7 marks)

Q25 PAST PAPER Pearson IAL WMA11 • June 2024 • Q2

P1 • TOPIC 1 6 marks

2. (i) Given that $m = 2^n$, express each of the following in simplest form in terms of m .

(a) 2^{n+3} (1)

(b) 16^{3n} (2)

- (ii) In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Solve the equation

$$x\sqrt{3} - 3 = x + \sqrt{3}$$

giving your answer in the form $p + q\sqrt{3}$ where p and q are integers. (3)

(Total for Question 2 is 6 marks)

05 Powers of x in the Power: Substitute

9^x , 3^{x+2} , 3^{x-1} all hide one letter, $p = 3^x$. Rewrite everything in p and it becomes a quadratic.

WHEN YOU SEE → YOU DO

“given $y = 3^x$, express 3^{3x} ... in terms of y ”

$$\rightarrow 3^{3x} = (3^x)^3 = y^3$$

“by substituting $p = 2^x$, show that”

$$\rightarrow 4^x = p^2, \quad 2^{x+3} = 8p, \quad 2^{x-1} = \frac{p}{2}$$

“hence solve”

$$\rightarrow \text{solve for } p, \text{ then } 2^x = p \rightarrow x$$

p negative

$$\rightarrow \text{reject: } 3^x > 0 \text{ always}$$

WORKED MODEL • JANUARY 2024

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4, \quad p = 2^x$$

$$1 \quad 2p^2 - 8p = \frac{17}{2}p - 4$$

rewrite in p

$$2 \quad 4p^2 - 33p + 8 = 0 \Rightarrow p = 8 \text{ or } \frac{1}{4}$$

quadratic

ANSWER $2^x = 8 \Rightarrow x = 3, \quad 2^x = \frac{1}{4} \Rightarrow x = -2$

IDEA 05

Practise: Substitute $p = 3^x$

My questions first, then real past papers.

Q26 DR ESLAM Substitute

P1 · TOPIC 1 4 marks

By substituting $p = 3^x$, solve $9^x - 10 \times 3^x + 9 = 0$. $x =$ _____

PAST PAPERS

Q27 **PAST PAPER** Pearson IAL WMA11 • January 2020 • Q2**P1 • TOPIC 1** **5 marks**

2. Given $y = 3^x$, express each of the following in terms of y . Write each expression in its simplest form.

(a) 3^{3x} (1)

(b) $\frac{1}{3^{x-2}}$ (2)

(c) $\frac{81}{9^{2-3x}}$ (2)

(Total 5 marks)

Q28 **PAST PAPER** Pearson IAL WMA11 · January 2023 · Q5**P1 · TOPIC 1** **6 marks**

5. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) By substituting $p = 3^x$, show that the equation

$$3 \times 9^x + 3^{x+2} = 1 + 3^{x-1}$$

can be rewritten in the form

$$9p^2 + 26p - 3 = 0 \quad (3)$$

(b) Hence solve

$$3 \times 9^x + 3^{x+2} = 1 + 3^{x-1} \quad (3)$$

(Total for Question 5 is 6 marks)

Q29**PAST PAPER**

Pearson IAL WMA11 · January 2024 · Q4

P1 · TOPIC 1 6 marks

4. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) By substituting $p = 2^x$, show that the equation

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4$$

can be written in the form

$$4p^2 - 33p + 8 = 0 \quad (3)$$

(b) Hence solve

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4 \quad (3)$$

(Total for Question 4 is 6 marks)

Q30**PAST PAPER**

Pearson IAL WMA11 · January 2025 · Q4

P1 · TOPIC 1 9 marks

4. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) Given that

$$y = a^x \quad \text{where } a \text{ is a positive constant}$$

express, in simplest form, in terms of y and a

(a) a^{3x+1} (1)

(b) $\frac{5}{(3a^{1-x})^{-2}}$ (3)

- (ii) (a) Use the substitution $p = 9^t$ to show that the equation

$$3(3^{4t+2} + 1) = 82 \times 9^t$$

can be rewritten as

$$27p^2 - 82p + 3 = 0$$
 (2)

- (b) Hence solve

$$3(3^{4t+2} + 1) = 82 \times 9^t$$
 (3)

(Total for Question 4 is 9 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------------|---|
| 01 | Indices | Add · subtract · multiply powers. Negative = flip, fraction = root. |
| 02 | Surds | Biggest square factor. $\sqrt{a}\sqrt{a} = a$. $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$. |
| 03 | Rationalise | × conjugate. Bottom: $a^2 - b$. |
| 04 | Surd equations | Collect, factorise x , divide, rationalise. |
| 05 | $p = 3^x$ | Rewrite in $p \rightarrow$ quadratic $\rightarrow$ back to x . Reject $p \leq 0$. |

BOTTOM LINE Same base $\rightarrow$ laws · biggest square factor $\rightarrow$ simplify · surd on the bottom $\rightarrow$ conjugate · power in the power $\rightarrow$ substitute.

Answers

Laws of Indices & Surds • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • INDICES

- Q1** Dr Eslam • Root first, then power (a) 9 (b) $\frac{1}{8}$ (c) $\frac{2}{3}$
- Q2** Dr Eslam • Form ax^n $\frac{3}{2}x - \frac{1}{2}x^{-\frac{3}{2}}$
- Q3** Practice • Easy (i) 3 (ii) 4 (iii) $\frac{1}{25}$
- Q4** Practice • Medium (a) $\frac{1}{4}x^2$ (b) $16x^{-4}$ (c) $64x^{-6}$
- Q5** Practice • Very Hard (a) $\frac{27}{8}x^{-9}$ (b) $\frac{4}{9}x^6$ (c) $\frac{64}{729}x^{18}$
- Q6** Pearson IAL WMA11 • October 2023 • Q2 (a) $\frac{1}{8}x$ (b) $\frac{1}{256}x^{\frac{3}{2}}$ (c) $16x^{-2}$
- Q7** Pearson IAL WMA11 • June 2025 • Q1 (a) $\frac{1}{4}x^2$ (b) $\frac{2}{5}x^{-1}$ (c) $100x^{-2}$

IDEA 02 • SURDS: SIMPLIFY AND EXPAND

- Q8** Dr Eslam • Biggest square factor (a) $2\sqrt{2}$ (b) 7
- Q9** Practice • Easy (i) $3\sqrt{2}$ (ii) $18\sqrt{3}$ (iii) $19\sqrt{2}$
- Q10** Practice • Hard (a) $2\sqrt{3} + 8\sqrt{3} - 15\sqrt{3}$ (b) $5\sqrt{3} - 3 - 25 + 5\sqrt{3}$
- Q11** Practice • Very Hard (a) $a^2 + 5b^2 + 2ab\sqrt{5}$ (d) $a^2 = 4$ or 45; side $2 + 3\sqrt{5}$ m

IDEA 03 • RATIONALISE THE DENOMINATOR

- Q12** Dr Eslam • Conjugate (a) $2\sqrt{3}$ (b) $1 + \sqrt{5}$
- Q13** Practice • Easy $\frac{3\sqrt{2}}{2}$
- Q14** Practice • Easy $\frac{6-5\sqrt{3}}{13}$
- Q15** Practice • Very Hard (a) e.g. $b = 0$ (b) $-\frac{5}{2} + \frac{3}{2}\sqrt{3}$
- Q16** Pearson IAL WMA11 • June 2022 • Q3 (i) $6 - 4 = 2$ (ii) $\frac{25}{4} + \frac{13}{4}\sqrt{5}$
- Q17** Pearson IAL WMA11 • June 2023 • Q6 (a) $r^2 - 2 + r^{-2}$ (b) $3 - 2\sqrt{2}$ (c) $\sqrt{3 + 2\sqrt{2}} = 1 + \sqrt{2}$
- Q18** Pearson IAL WMA11 • October 2023 • Q3 (a) $3\sqrt{3} - 2\sqrt{5}$ (b) $x = 15\sqrt{3} - 10\sqrt{5}$
- Q19** Pearson IAL WMA11 • October 2024 • Q2 (i) $\frac{6x^{10}}{y}$ (ii) $a = \frac{8}{3} - \frac{4}{3}\sqrt{3}$

IDEA 04 • EQUATIONS WITH SURDS

- Q20** Dr Eslam • Factorise x out $x(\sqrt{2} - 2) = \sqrt{2} - 3 \Rightarrow x = 2 + \frac{\sqrt{2}}{2}$
- Q21** Practice • Medium $x = \frac{2\sqrt{7}}{3}$
- Q22** Practice • Hard $x = -6\sqrt{3}$
- Q23** Practice • Very Hard $x = -\frac{13}{60}$
- Q24** Pearson IAL WMA11 • January 2022 • Q3 (i) $10x^2 - 58x\sqrt{2} + 202$ (ii) $y = \frac{57}{23} + \frac{41}{23}\sqrt{3}$
- Q25** Pearson IAL WMA11 • June 2024 • Q2 (i) $8m, m^{12}$ (ii) $x = 3 + 2\sqrt{3}$

IDEA 05 • SUBSTITUTE $p = 3^x$

- Q26** Dr Eslam • Substitute $p = 1$ or 9 : $x = 0$ or $x = 2$
- Q27** Pearson IAL WMA11 • January 2020 • Q2 (a) y^3 (b) $\frac{9}{y}$ (c) y^6
- Q28** Pearson IAL WMA11 • January 2023 • Q5 (b) $p = \frac{1}{9}$ ($p = -3$ rejected): $x = -2$
- Q29** Pearson IAL WMA11 • January 2024 • Q4 (b) $x = 3$ or $x = -2$
- Q30** Pearson IAL WMA11 • January 2025 • Q4 (i) $ay^3; \frac{45a^2}{y^2}$ (ii) $p = 3$ or $\frac{1}{27}$: $t = \frac{1}{2}$ or $-\frac{3}{2}$

Quadratics

Solve it, complete the square, read the discriminant. Every quadratic question is one of these three — and the graph ties them together.



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THE TOPIC ON ONE MAP

Quadratics

01 — Solve: calculator first

find the roots surd form word problem

02 — Complete the square

$a(x + b)^2 + c$ turning point line of symmetry

03 — The discriminant

two distinct equal roots no real roots

04 — Hidden quadratics

$x^4 - 13x^2 + 36$ 2^{2x} $\sqrt{x}$

05 — Graphs and models

sketch maximum height

01 Solve: Calculator First, Then Show It

Get the roots on the calculator first. Clean roots → factorise. Messy roots → formula or completing the square.

WHEN YOU SEE → YOU DO

“solve”, “find the roots”

→ calculator roots, then the working

“using algebra” / “calculator solutions not acceptable”

→ show factorising or the formula

“exact” / “surd form”

→ keep the $\sqrt{}$

a word problem (perimeter, area)

→ form two equations → substitute → quadratic

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (\text{brackets around a negative } b)$$

WORKED MODEL · JUNE 2023

Rectangle: perimeter 350, area 7350, $x > y$.

1 $x + y = 175, xy = 7350$

two equations

2 $x(175 - x) = 7350 \Rightarrow x^2 - 175x + 7350 = 0$

substitute

3 $(x - 105)(x - 70) = 0$

factorise

ANSWER $x = 105, y = 70$

IDEA 01

Practise: Solve

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Calculator, then show it

P1 · TOPIC 2 3 marks

Solve (a) $x^2 - 6x + 5 = 0$ (b) $2x^2 + 3x - 4 = 0$ (exact)

SELECTED PRACTICE

Q2 **PRACTICE** Practice · Easy

P1 · TOPIC 2 3 marks

1 Expand and simplify

(i) $(x + 4)(2x - 3)$

(ii) $(3x - 4)(3x + 4)$

(iii) $(2x + 1)^2$

(3 marks)

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 2 3 marks**4** Solve

(i) $x^2 + 8x - 9 = 0$

(ii) $3x^2 - 13x + 4 = 0$

(iii) $4x^2 - 6x - 5 = 0$

(3 marks)

Q4**PRACTICE**

Practice · Very Hard

P1 · TOPIC 2 5 marks**12 (a)** Factorise $x^2 + 6x + 9$ **(1 mark)****(b)** Factorise $x^2 + 6xy + 9y^2$ **(2 marks)****(c)** Find a relationship between x and y such that $x^2 + 6xy + 9y^2 = 0$ **(2 marks)**

PAST PAPERS

Q5

PAST PAPER

Pearson IAL WMA11 · June 2023 · Q2

P1 · TOPIC 2 6 marks

2. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A rectangular sports pitch has length x metres and width y metres, where $x > y$

Given that the perimeter of the pitch is 350 m,

(a) write down an equation linking x and y (1)

Given also that the area of the pitch is 7350 m^2

(b) write down a second equation linking x and y (1)

(c) hence find the value of x and the value of y (4)

(Total for Question 2 is 6 marks)

02 Completing the Square and the Turning Point

The same two-step structure from O-Level: list a , b , c , $\frac{b}{2a}$, then $a\left(x + \frac{b}{2a}\right)^2 - a\left(\frac{b}{2a}\right)^2 + c$.

WHEN YOU SEE → YOU DO

“in the form $a(x + b)^2 + c$ ”, “ $a + b(x + c)^2$ ”

→ complete the square

“minimum/maximum point”, “turning point”

→ flip the inside, keep the outside

“minimum value of $f(x)$ ”

→ the number outside the bracket

“line of symmetry”

→ x = the flipped inside number

“use inequalities to define R ”

→ one inequality per boundary: $x \geq 0$,
 $y \leq \text{curve}$, $y \geq \text{line}$

WORKED MODEL · OCTOBER 2023

$$2x^2 - 12x + 14$$

1 $a = 2$, $b = -12$, $c = 14$, $\frac{b}{2a} = -3$

list

2 $2(x - 3)^2 - 2(9) + 14$

structure

ANSWER $2(x - 3)^2 - 4$, $P(3, -4)$

× **TRAP** A negative a flips the curve: $-6x^2 + \dots$ has a **maximum**.

IDEA 02

Practise: Completing the Square

My questions first, then selected practice, then real past papers.

Q6 DR ESLAM Turning point

P1 · TOPIC 2 3 marks

Write $x^2 - 6x + 5$ in the form $(x + p)^2 + q$ and state the turning point of $y = x^2 - 6x + 5$.

SELECTED PRACTICE**Q7** PRACTICE Practice · Easy

P1 · TOPIC 2 3 marks

3 Complete the square for

(i) $x^2 + 8x - 4$

(ii) $2x^2 + 12x - 5$

(iii) $5x^2 - 3x + 2$

(3 marks)

Q8**PRACTICE**

Practice · Hard

P1 · TOPIC 2**6 marks**

- 1 (a) Write the quadratic function $y = 4x^2 + 8x - 5$ in the form $y = a(x + b)^2 + c$ where a , b and c are integers to be found.

(2 marks)

- (b) Write down the minimum point on the graph of $y = 4x^2 + 8x - 5$.

(1 mark)

- (c) Sketch the graph of $y = 4x^2 + 8x - 5$, clearly labelling the minimum point and any point where the graph intersects the coordinate axes.

(3 marks)

Q9**PRACTICE**

Practice · Very Hard

P1 · TOPIC 2 4 marks

7 (a) Show that the equation $ax^2 + bx + c = 0$ can be written in the form

$$a\left(x + \frac{b}{2a}\right)^2 - \frac{(b^2 - 4ac)}{4a} = 0$$

(2 marks)

(b) Hence show that $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

(2 marks)

PAST PAPERS

Q10 PAST PAPER Pearson IAL WMA11 · January 2022 · Q2

P1 · TOPIC 2 6 marks

2. $f(x) = 11 - 4x - 2x^2$

- (a) Express
- $f(x)$
- in the form

$$a + b(x + c)^2$$

where a , b and c are integers to be found.

(3)

- (b) Sketch the graph of the curve
- C
- with equation
- $y = f(x)$
- , showing clearly the coordinates of the point where the curve crosses the
- y
- axis.

(2)

- (c) Write down the equation of the line of symmetry of
- C
- .

(1)

(Total 6 marks)

Q11 **PAST PAPER** Pearson IAL WMA11 · June 2023 · Q3**P1 · TOPIC 2** **6 marks**

3. (a) Express $3x^2 + 12x + 13$ in the form

$$a(x + b)^2 + c$$

where a , b and c are integers to be found.

(3)

- (b) Hence sketch the curve with equation $y = 3x^2 + 12x + 13$

On your sketch show clearly

- the coordinates of the y intercept
- the coordinates of the turning point of the curve

(3)**(Total for Question 3 is 6 marks)**

Q12**PAST PAPER**

Pearson IAL WMA11 · October 2023 · Q11

P1 · TOPIC 2 10 marks

11.

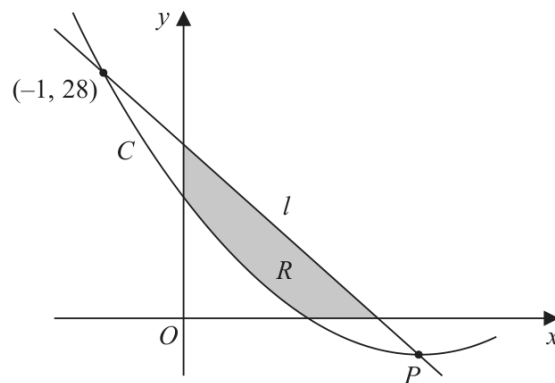
**Figure 5**

Figure 5 shows part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^2 - 12x + 14$$

- (a) Write $2x^2 - 12x + 14$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

Given that C has a minimum at the point P

- (b) state the coordinates of P

(1)

The line l intersects C at $(-1, 28)$ and at P as shown in Figure 5.

- (c) Find the equation of l giving your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

The finite region R , shown shaded in Figure 5, is bounded by the x -axis, l , the y -axis, and C .

- (d) Use inequalities to define the region R .

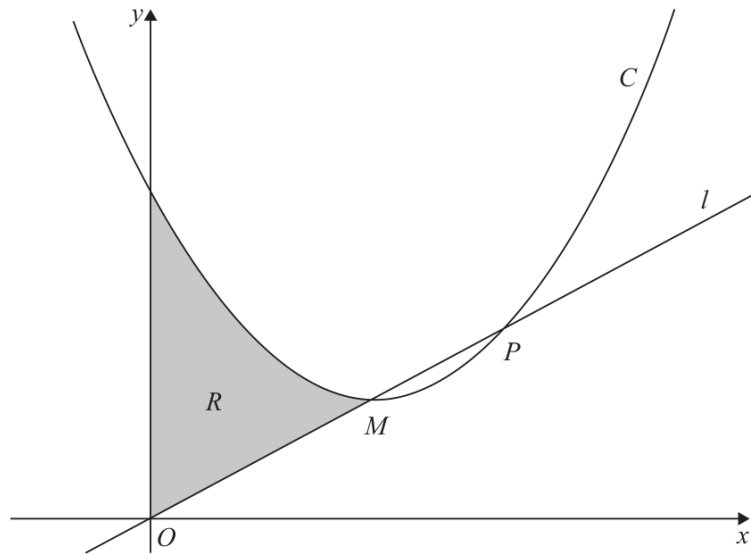
(3)**(Total for Question 11 is 10 marks)**

Q13**PAST PAPER**

Pearson IAL WMA11 · October 2025 · Q5

P1 · TOPIC 2 10 marks

5.

**Figure 1**

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

- (a) Express $2x^2 - 16x + 50$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

Figure 1 shows a sketch of the curve C with equation $y = 2x^2 - 16x + 50$

Given that in Figure 1

- M is the minimum point on C
- line l passes through the origin and intersects C at the points M and P

- (b) find, using algebra and showing your working, the x coordinate of P .

(5)

The region R is shown shaded in Figure 1.

- (c) Use inequalities to fully define R .

(2)

(Total for Question 5 is 10 marks)

Q14 PAST PAPER Pearson IAL WMA11 · June 2026 · Q5

P1 · TOPIC 2 7 marks

5. The curve C has equation

$$y = -2x^2 + 12x + 7$$

- (a) Write $-2x^2 + 12x + 7$ in the form $a(x + b)^2 + c$, where a , b and c are constants. (3)

The point P is the maximum point on C .

- (b) Find the coordinates of P . (2)

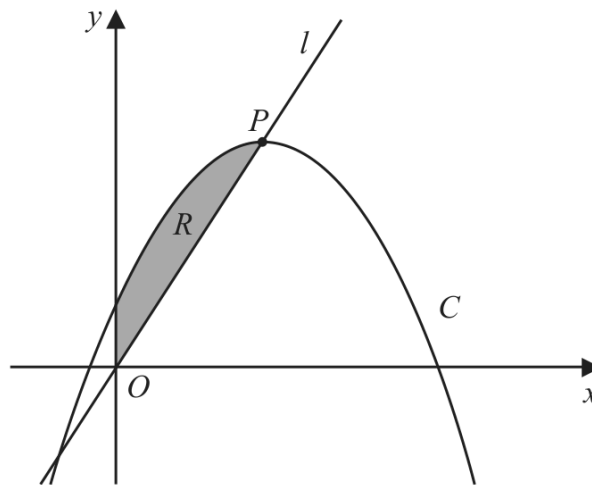


Figure 1

Figure 1 shows a sketch of the curve C and a line l .

The line l passes through the point P and the origin O .

The region R , shown shaded in Figure 1, is bounded by C , l and the y -axis.

- (c) Use inequalities to define R . (2)

Question 5 continued

(Total for Question 5 is 7 marks)

03 The Discriminant $b^2 - 4ac$

It counts the roots without solving. Most exam questions give the number of roots and ask for k .

WHEN YOU SEE → YOU DO

“two distinct real roots” → $b^2 - 4ac > 0$

“equal roots”, “one root”, “touches”, “meet at a single point” → $b^2 - 4ac = 0$

“no real roots”, “does not cross” → $b^2 - 4ac < 0$

“real roots” → $b^2 - 4ac \geq 0$

two curves meet → curve = curve → one quadratic → its discriminant

1

Rearrange to $ax^2 + bx + c = 0$

Read a , b , c — they may contain k .

2

Write the inequality

$b^2 - 4ac >, =, \text{ or } < 0$.

3

Solve the quadratic inequality in k

Critical values, sketch, choose inside or outside.

WORKED MODEL · JANUARY 2023

$kx^2 + 6kx + 5 = 0$ has no real roots.

1 $36k^2 - 20k < 0$

$b^2 - 4ac < 0$

2 $4k(9k - 5) < 0$, critical values $0, \frac{5}{9}$

factorise

ANSWER $0 < k < \frac{5}{9}$

IDEA 03

Practise: The Discriminant

My questions first, then selected practice, then real past papers.

Q15 DR ESLAM Equal roots

P1 · TOPIC 2 2 marks

$x^2 + kx + 4 = 0$ has equal roots. Find k .

$k =$ _____

Q16 DR ESLAM No real roots

P1 · TOPIC 2 3 marks

Find the values of k for which $x^2 + 4x + k = 0$ has no real roots.

SELECTED PRACTICE

Q17**PRACTICE**

Practice · Easy

P1 · TOPIC 2

3 marks

5 Write down the value of the discriminant of

(i) $x^2 - 3x + 4$

(ii) $4x + 3 - 2x^2$

(iii) $5 - 8x + 2x^2$

(3 marks)**Q18****PRACTICE**

Practice · Medium

P1 · TOPIC 2

2 marks

7 The equation $y = x^2 + px + q$ has no real roots. Show that $p^2 < 4q$.

(2 marks)

Q19**PRACTICE**

Practice · Very Hard

P1 · TOPIC 2**5 marks**

8 (a) The function $f(x)$ is defined by $f(x) = (k-1)x^2 - (k-2)x - 2k$, $x \in \mathbb{R}$.

The function $g(x)$ is defined by $g(x) = (k-1)x^2 - 3kx + k + 1$, $x \in \mathbb{R}$.

k is a non-zero constant and $k \neq 1$.

The graphs of $y = f(x)$ and $y = g(x)$ intersect once.

Find the x -coordinate of the intersection, giving your answer in terms of k .

(3 marks)

(b) In the case when $k = 3$, find the coordinates of the point of intersection of $y = f(x)$ and $y = g(x)$.

(2 marks)

PAST PAPERS

Q20 **PAST PAPER** Pearson IAL WMA11 · January 2023 · Q4**P1 · TOPIC 2** **4 marks**

4. Given that the equation

$$kx^2 + 6kx + 5 = 0 \quad \text{where } k \text{ is a non zero constant}$$

has no real roots, find the range of possible values for k .**(4)****(Total for Question 4 is 4 marks)**

Q21**PAST PAPER**

Pearson IAL WMA11 · October 2023 · Q6

P1 · TOPIC 2 6 marks

6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The equation

$$4(p - 2x) = \frac{12 + 15p}{x + p} \quad x \neq -p$$

where p is a constant, has two distinct real roots.

(a) Show that

$$3p^2 - 10p - 8 > 0 \quad (3)$$

(b) Hence, using algebra, find the range of possible values of p (3)

(Total for Question 6 is 6 marks)

Q22 **PAST PAPER** Pearson IAL WMA11 · June 2024 · Q4**P1 · TOPIC 2** **6 marks**

4. The curve C_1 has equation

$$y = x^2 + kx - 9$$

and the curve C_2 has equation

$$y = -3x^2 - 5x + k$$

where k is a constant.

Given that C_1 and C_2 meet at a single point P

(a) show that

$$k^2 + 26k + 169 = 0 \quad (3)$$

(b) Hence find the coordinates of P (3)

(Total for Question 4 is 6 marks)

Q23**PAST PAPER**

Pearson IAL WMA11 · June 2025 · Q10

P1 · TOPIC 2 7 marks**10.**

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

$$(k-1)x^6 + 4x^3 + (k-4) = 0 \quad \text{where } k \text{ is a constant}$$

- (a) Find the exact solutions to the given equation for $k = 4.5$

(3)

- (b) Find the set of possible values of k for which the given equation has no real roots.

(4)**(Total for Question 10 is 7 marks)**

04 Hidden Quadratics: Substitute

Look for a power that is double another: x^4 and x^2 , $x^{\frac{2}{3}}$ and $x^{\frac{1}{3}}$, x and $\sqrt{x}$, 4^x and 2^x .

WHEN YOU SEE → YOU DO

$$x^4 - 13x^2 + 36$$

$$\longrightarrow u = x^2$$

$$x - 6x^{\frac{1}{2}} + 4, 5\sqrt{x} + 3 = 2x$$

$$\longrightarrow u = \sqrt{x}$$

$$2^{2x} + 64 = 20(2^x)$$

$$\longrightarrow u = 2^x$$

$$\sqrt{x} \text{ or } 2^x \text{ negative}$$

$$\longrightarrow \text{reject it}$$

WORKED MODEL · OCTOBER 2019

$$x - 6x^{\frac{1}{2}} + 4 = 0$$

$$1 \quad u^2 - 6u + 4 = 0, \quad u = x^{\frac{1}{2}}$$

substitute

$$2 \quad u = 3 \pm \sqrt{5}$$

solve

$$\text{ANSWER} \quad x = (3 \pm \sqrt{5})^2 = 14 \pm 6\sqrt{5}$$

IDEA 04

Practise: Hidden Quadratics

My questions first, then selected practice, then real past papers.

Q24**DR ESLAM**Substitute $u = x^2$

P1 · TOPIC 2

3 marks

Solve $x^4 - 5x^2 + 4 = 0$.

$x =$ _____

SELECTED PRACTICE**Q25****PRACTICE**

Practice · Medium

P1 · TOPIC 2

3 marks

10 Solve the equation $x^4 - 13x^2 + 36 = 0$.

(3 marks)

Q26**PRACTICE**

Practice · Hard

P1 · TOPIC 2 9 marks

9 (a) Solve the equation $5\sqrt{x} + 3 = 2x$.

(3 marks)

(b) Solve $x^{\frac{2}{3}} + 2x^{\frac{1}{3}} = 8$.

(3 marks)

(c) Solve the equation $2^{2x} + 64 = 20(2^x)$.

(3 marks)

Q27**PRACTICE**

Practice · Very Hard

P1 · TOPIC 2 6 marks

9 (a) Solve the equation $8\sqrt{x} = 48 - x$.

(3 marks)

(b) Solve the equation $2^{4x} + 64 = 20(2^{2x})$.

(3 marks)

PAST PAPERS

Q28 **PAST PAPER** Pearson IAL WMA11 · October 2019 · Q8**P1 · TOPIC 2** **5 marks**

8. Solve, using algebra, the equation

$$x - 6x^{\frac{1}{2}} + 4 = 0$$

Fully simplify your answers, writing them in the form $a + b\sqrt{c}$, where a , b and c are integers to be found.

(5)**(Total 5 marks)**

Q29**PAST PAPER**

Pearson IAL WMA11 · January 2020 · Q5

P1 · TOPIC 2 7 marks

5. (a) Find, using algebra, all solutions of

$$20x^3 - 50x^2 - 30x = 0 \quad (3)$$

- (b) Hence find all real solutions of

$$20(y+3)^{\frac{3}{2}} - 50(y+3) - 30(y+3)^{\frac{1}{2}} = 0 \quad (4)$$

(Total 7 marks)

Q30**PAST PAPER**

Pearson IAL WMA11 · October 2025 · Q4

P1 · TOPIC 2 7 marks

4. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

- (i) Using the laws of indices, solve

$$3^{2p-5} = \frac{1}{9}\sqrt{27}$$

(3)

- (ii) Find the real roots of the equation

$$x^2 + 1 = \frac{36}{x^2 - 4}$$

(4)**(Total for Question 4 is 7 marks)**

05 Graphs and Models

A sketch needs three things: shape, the axis crossings, the turning point. Models ask you to read those same points in context.

WHEN YOU SEE → YOU DO

“sketch ... showing where it crosses the axes”

→ shape (a), y -intercept, roots, turning point

“ $y = f(x + 2)$ ”, “ $y = f(x) + 6$ ”

→ move 2 left · move 6 up

“height of the cliff”, “when does it hit the sea”

→ $t = 0$ · the positive root

“maximum height”, “is the bridge wide enough?”

→ turning point · distance between the roots

“find the points of intersection”

→ curve = line → solve

WORKED MODEL · STONE FROM A CLIFF

$$h = 24 + 2t - 0.5t^2$$

1 $t = 0$: cliff 24 m

start

2 $-\frac{1}{2}(t - 2)^2 + 26$

complete the square

3 $0 = 24 + 2t - 0.5t^2 \Rightarrow t = 2 + 2\sqrt{13}$

positive root

ANSWER max 26 m; hits the sea at 9.21 s

IDEA 05

Practise: Graphs and Models

My questions first, then selected practice, then real past papers.

Q31 DR ESLAM Sketch

P1 · TOPIC 2 3 marks

Sketch $y = x^2 - 6x + 5$, showing the axis crossings and the turning point.

SELECTED PRACTICE

Q32

PRACTICE

Practice · Easy

P1 · TOPIC 2 6 marks

6 (a) (i) Write down the y -axis intercept on the graph of $y = 2x^2 + 5x - 3$.

(ii) Find the roots of $y = 2x^2 + 5x - 3$.

(3 marks)

(b) Sketch the graph of $y = 2x^2 + 5x - 3$, labelling all points where the graph crosses the coordinate axes.

(3 marks)**Q33**

PRACTICE

Practice · Hard

P1 · TOPIC 2 3 marks

6 Sketch the graph of $y = 12x^2 - 5x - 72$, labelling any points where the graph crosses the coordinate axes. (You do not need to label the turning point.)

(3 marks)

Q34**PRACTICE**

Practice · Very Hard

P1 · TOPIC 2 7 marks

- 11 (a)** A stone is thrown vertically upwards from the top of a cliff. The path of the stone is modelled by the quadratic function $h(t) = 52 + 3t - 0.5t^2$, $t \geq 0$, where h is the height, in meters, of the stone above the sea and t is the time in seconds since the stone was thrown.

Write down the height of the cliff from which the stone was thrown.

(1 mark)

- (b)** Find the maximum height the stone reaches above the sea.

(2 marks)

- (c)** How long does it take for the stone to hit the sea?

(2 marks)

- (d)** How long does the stone stay above it's starting height for?

(2 marks)

PAST PAPERS

Q35 PAST PAPER Pearson IAL WMA11 · January 2023 · Q8

P1 · TOPIC 2 8 marks

8.

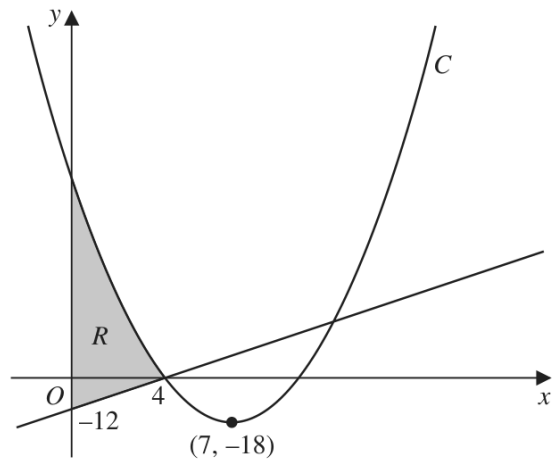


Figure 2

Figure 2 shows a sketch of the straight line l and the curve C .

Given that l cuts the y -axis at -12 and cuts the x -axis at 4 , as shown in Figure 2,

- (a) find an equation for l , writing your answer in the form $y = mx + c$, where m and c are constants to be found.

(2)

Given that C

- has equation $y = f(x)$ where $f(x)$ is a quadratic expression
- has a minimum point at $(7, -18)$
- cuts the x -axis at 4 and at k , where k is a constant

- (b) deduce the value of k ,

(1)

- (c) find $f(x)$.

(3)

The region R is shown shaded in Figure 2.

- (d) Use inequalities to define R .

(2)

(Total for Question 8 is 8 marks)

Q36 **PAST PAPER** Pearson IAL WMA11 · January 2024 · Q9**P1 · TOPIC 2 10 marks**

9. The curve C_1 has equation $y = f(x)$.

Given that

- $f(x)$ is a quadratic expression
 - C_1 has a maximum turning point at $(2, 20)$
 - C_1 passes through the origin
- (a) sketch a graph of C_1 showing the coordinates of any points where C_1 cuts the coordinate axes,

(2)

(b) find an expression for $f(x)$.

(3)

The curve C_2 has equation $y = x(x^2 - 4)$

Curve C_1 and C_2 meet at the origin, and at the points P and Q

Given that the x coordinate of the point P is negative,

(c) using algebra and showing all stages of your working, find the coordinates of P

(5)

(Total for Question 9 is 10 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 Solve** Calculator first. Clean → factorise, messy → formula.
- 02 Complete the square** Two-step structure. Turning point: flip inside, keep outside.
- 03 Discriminant** > 0 two · $= 0$ one · < 0 none. Then a quadratic inequality in k .
- 04 Hidden quadratics** $u = x^2, \sqrt{x}, 2^x$... reject impossible u .
- 05 Graphs & models** Shape, crossings, turning point. Regions: one inequality per edge.

BOTTOM LINE Roots → solve · turning point → complete the square · how many roots → discriminant.

Answers

Quadratics · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · SOLVE

- Q1** Dr Eslam · Calculator, then show it (a) $x = 1, 5$ (b) $x = \frac{-3 \pm \sqrt{41}}{4}$
- Q2** Practice · Easy (i) $2x^2 + 5x - 12$ (ii) $9x^2 - 16$ (iii) $4x^2 + 4x + 1$
- Q3** Practice · Easy (i) $1, -9$ (ii) $\frac{1}{3}, 4$ (iii) $\frac{3 \pm \sqrt{29}}{4}$
- Q4** Practice · Very Hard (a) $(x + 3)^2$ (b) $(x + 3y)^2$ (c) $x = -3y$
- Q5** Pearson IAL WMA11 · June 2023 · Q2 (a) $2x + 2y = 350$ (b) $xy = 7350$ (c) $x = 105, y = 70$

IDEA 02 · COMPLETING THE SQUARE

- Q6** Dr Eslam · Turning point $(x - 3)^2 - 4; (3, -4)$
- Q7** Practice · Easy (i) $(x+4)^2 - 20$ (ii) $2(x+3)^2 - 23$ (iii) $5\left(x - \frac{3}{10}\right)^2 + \frac{31}{20}$
- Q8** Practice · Hard (a) $4(x+1)^2 - 9$ (b) $(-1, -9)$ (c) roots $-\frac{5}{2}, \frac{1}{2}$; y-int -5
- Q9** Practice · Very Hard divide by a , complete the square, root
- Q10** Pearson IAL WMA11 · January 2022 · Q2 (a) $13 - 2(x + 1)^2$ (b) max $(-1, 13)$, y-int 11 (c) $x = -1$
- Q11** Pearson IAL WMA11 · June 2023 · Q3 (a) $3(x + 2)^2 + 1$ (b) y-int 13 , min $(-2, 1)$
- Q12** Pearson IAL WMA11 · October 2023 · Q11 (a) $2(x - 3)^2 - 4$ (b) $(3, -4)$ (c) $y = -8x + 20$ (d) $x \geq 0, y \geq 0, y \leq 20 - 8x, y \geq 2x^2 - 12x + 14$
- Q13** Pearson IAL WMA11 · October 2025 · Q5 (a) $2(x - 4)^2 + 18$ (b) $x = \frac{25}{4}$ (c) $x \geq 0, y \geq \frac{9}{2}x, y \leq 2x^2 - 16x + 50$
- Q14** Pearson IAL WMA11 · June 2026 · Q5 (a) $25 - 2(x - 3)^2$ (b) $P(3, 25)$ (c) $x \geq 0, y \geq \frac{25}{3}x, y \leq -2x^2 + 12x + 7$

IDEA 03 · THE DISCRIMINANT

- Q15** Dr Eslam · Equal roots $k^2 - 16 = 0 \Rightarrow k = \pm 4$
- Q16** Dr Eslam · No real roots $16 - 4k < 0 \Rightarrow k > 4$
- Q17** Practice · Easy (i) -7 (ii) 40 (iii) 24
- Q18** Practice · Medium $p^2 - 4q < 0$
- Q19** Practice · Very Hard (a) $x = \frac{3k+1}{2(k+1)}$ (b) $\left(\frac{5}{4}, -\frac{33}{8}\right)$
- Q20** Pearson IAL WMA11 · January 2023 · Q4 $0 < k < \frac{5}{9}$
- Q21** Pearson IAL WMA11 · October 2023 · Q6 (b) $p < -\frac{2}{3}$ or $p > 4$
- Q22** Pearson IAL WMA11 · June 2024 · Q4 (b) $k = -13$: $P(1, -21)$
- Q23** Pearson IAL WMA11 · June 2025 · Q10 (a) $x = -1, -\frac{1}{\sqrt[3]{7}}$ (b) $k < 0$ or $k > 5$

IDEA 04 · HIDDEN QUADRATICS

- Q24** Dr Eslam · Substitute $u = x^2$ $u = 1, 4$: $x = \pm 1, \pm 2$
- Q25** Practice · Medium $x = \pm 2, \pm 3$
- Q26** Practice · Hard (a) $x = 9$ (b) $x = 8, -64$ (c) $x = 2, 4$
- Q27** Practice · Very Hard (a) $x = 16$ (b) $x = 1, 2$
- Q28** Pearson IAL WMA11 · October 2019 · Q8 $x = 14 \pm 6\sqrt{5}$
- Q29** Pearson IAL WMA11 · January 2020 · Q5 (a) $0, 3, -\frac{1}{2}$ (b) $y = 6$ or $y = -3$
- Q30** Pearson IAL WMA11 · October 2025 · Q4 (i) $p = \frac{9}{4}$ (ii) $x = \pm 2\sqrt{2}$

IDEA 05 · GRAPHS AND MODELS

- Q31** Dr Eslam · Sketch cup; (0, 5), (1, 0), (5, 0); minimum (3, -4)
- Q32** Practice · Easy (a) -3; roots $\frac{1}{2}$, -3 (b) cup through (-3, 0), $(\frac{1}{2}, 0)$, (0, -3)
roots $-\frac{9}{4}$, $\frac{8}{3}$; y-int -72
- Q33** Practice · Hard (a) 52 m (b) 56.5 m (c) $3 + \sqrt{113} \approx 13.6$ s (d) 6 s
- Q34** Practice · Very Hard (a) $y = 3x - 12$ (b) $k = 10$ (c) $2x^2 - 28x + 80$ (d) $x \geq 0, y \geq 0, y \leq 2x^2 - 28x + 80$
- Q35** Pearson IAL WMA11 · January 2023 · Q8 (b) $f(x) = 20x - 5x^2$ (c) $P(-8, -480)$
- Q36** Pearson IAL WMA11 · January 2024 · Q9

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Simultaneous Equations

Two linear equations: eliminate. One linear and one quadratic: make one letter the subject and substitute. Then ERCS and solve.

THE TOPIC ON ONE MAP

Simultaneous Equations

01 Linear: elimination

match a coefficient

a constant k

02 Linear and quadratic: substitute

x^2 , xy

show $x^2 - 2x - 35 = 0$

03 One solution: the discriminant

tangent

two / no solutions

04 Models and harder forms

$H^3 = pt^2 + q$

exact distance PQ

01 Linear: Elimination

Calculator first — you know the answer. Then show it: match one coefficient, add or subtract, back-substitute.

WHEN YOU SEE → YOU DO

“use elimination”

→ multiply to match a coefficient

signs of the matched terms are different

→ ADD

signs are the same

→ SUBTRACT

a constant k in both equations

→ solve as normal — the answer contains k

WORKED MODEL • ELIMINATION

$$7x + 4y = 17, \quad 3x - 2y = 11$$

1 $7x + 4y = 17$ and $6x - 4y = 22$

second $\times 2$

2 $13x = 39 \Rightarrow x = 3$

add

ANSWER $x = 3, y = -1$

IDEA 01

Practise: Linear: Elimination

My questions first, then selected practice.

Q1 **DR ESLAM** Add or subtract

P1 · TOPIC 3 3 marks

Solve $x + 2y = 7$, $3x - y = 7$.

SELECTED PRACTICE

Q2 **PRACTICE** Practice · Easy

P1 · TOPIC 3 2 marks

1 Solve the simultaneous equations

$$x + y = 8$$

$$x - y = 4$$

(2 marks)

Q3**PRACTICE**

Practice · Medium

P1 · TOPIC 3 4 marks

- 1 Use elimination to solve the simultaneous equations

$$7x + 4y = 17$$

$$3x - 2y = 11$$

(4 marks)

Q4**PRACTICE**

Practice · Medium

P1 · TOPIC 3**5 marks****6 (a)**

$$4x - ky = 13$$

$$3x + 2ky = 7$$

are simultaneous equations, where k is a constant.

Show that $x = 3$.

(3 marks)

(b) Find an expression for y in terms of the constant k .

(1 mark)

(c) Given that $y = 3$, find the value of k .

(1 mark)

02 Linear and Quadratic: Substitute

Rearrange the linear one, substitute into the quadratic, ERCS, solve, then back into the **linear** one for the pairs.

WHEN YOU SEE → YOU DO

one equation has x^2, y^2 or xy

→ substitution — never elimination

“by eliminating y , show that $x^2 - 2x - 35 = 0$ ”

→ make y the subject of the linear, substitute

“hence solve”

→ solve the quadratic, then find each y

“in the form $a \pm b\sqrt{3}$ ”

→ quadratic formula, exact

“state which x goes with which y ”

→ write the answers as pairs

1

Rearrange the linear

Choose the letter with coefficient 1 to avoid fractions.

2

Substitute and ERCS

Expand, rearrange to $= 0$, collect.

3

Solve and pair

Two x values → two y values from the **linear** equation.

WORKED MODEL • JANUARY 2026

$$y - x = 4, \quad 2x^2 - xy = 8$$

1 $2x^2 - x(x + 4) = 8 \Rightarrow x^2 - 4x - 8 = 0$

substitute $y = x + 4$

2 $x = 2 \pm 2\sqrt{3}$

formula, exact

ANSWER $(2 + 2\sqrt{3}, 6 + 2\sqrt{3}), (2 - 2\sqrt{3}, 6 - 2\sqrt{3})$

IDEA 02

Practise: Linear and Quadratic: Substitute

My questions first, then selected practice, then real past papers.

Q5 DR ESLAM Substitute

P1 · TOPIC 3 4 marks

Solve $y = x + 1$, $x^2 + y^2 = 13$.

SELECTED PRACTICE**Q6** PRACTICE Practice · Easy

P1 · TOPIC 3 4 marks

6 Solve the simultaneous equations

$$y = 2x - 1$$

$$x^2 + y^2 - 2 = 0$$

(4 marks)

Q7**PRACTICE**

Practice · Medium

P1 · TOPIC 3**7 marks**

5 (a) By eliminating y from the equations

$$x^2 - 8y = -40$$

$$3x + 2y = 4$$

show that $x^2 + 12x + 24 = 0$.

(2 marks)

(b) Hence solve the simultaneous equations

$$x^2 - 8y = -40$$

$$3x + 2y = 4$$

giving x and y in the form $a \pm b\sqrt{3}$, where a and b are integers.

(5 marks)

Q8**PRACTICE**

Practice · Hard

P1 · TOPIC 3

7 marks

5 (a) By eliminating y from the equations

$$15x^2 - 4y^2 = 12$$

$$4x + 2y = 1$$

show that $x^2 - 8x + 13 = 0$.

(2 marks)

(b) Hence solve the simultaneous equations

$$15x^2 - 4y^2 = 12$$

$$4x + 2y = 1$$

giving x and y in the form $a \pm b\sqrt{3}$, where a and b are rational numbers.

(5 marks)

PAST PAPERS

Q9

PAST PAPER

Pearson IAL WMA11 · January 2026 · Q2

P1 · TOPIC 3

6 marks

2.

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

(a) Given

$$y - x = 4$$

and

$$2x^2 - xy = 8$$

show that

$$x^2 - 4x - 8 = 0$$

(2)

(b) Hence solve the simultaneous equations

$$y - x = 4$$

$$2x^2 - xy = 8$$

writing the answers in the form $p + q\sqrt{3}$, where p and q are integers to be found.

(4)

(Total for Question 2 is 6 marks)

03 One Solution: the Discriminant

“The line touches the curve”, “exactly one solution” — the quadratic you get after substituting has $b^2 - 4ac = 0$.

WHEN YOU SEE → YOU DO

“exactly one solution”, “the line is a tangent”

$$\longrightarrow b^2 - 4ac = 0$$

“two solutions / no solutions”

$$\longrightarrow > 0 \text{ / } < 0$$

“find the solution for that k ”

$$\longrightarrow \text{repeated root } x = -\frac{b}{2a}$$

WORKED MODEL • ONE SOLUTION

$$x^2 - 2y = 10, \quad 2x - y = k$$

$$1 \quad x^2 - 2(2x - k) = 10 \Rightarrow x^2 - 4x + 2(k - 5) = 0$$

substitute

$$2 \quad 16 - 8(k - 5) = 0 \Rightarrow k = 7$$

$$b^2 - 4ac = 0$$

$$\text{ANSWER} \quad k = 7, \quad x = 2, \quad y = -3$$

IDEA 03

Practise: One Solution: the Discriminant

My questions first, then selected practice.

Q10 DR ESLAM Tangent line

P1 · TOPIC 3 3 marks

Find k such that the line $y = 2x + k$ touches the curve $y = x^2 + 4$.

$k =$ _____

SELECTED PRACTICE

Q11**PRACTICE**

Practice · Medium

P1 · TOPIC 3 8 marks

7 (a)

$$x^2 - 2y = 10$$

$$2x - y = k$$

are simultaneous equations, where k is a constant.

By eliminating y from the equations show that $x^2 - 4x + 2(k - 5) = 0$.

(3 marks)

- (b) By considering the discriminant of $x^2 - 4x + 2(k - 5) = 0$ find the value of k for which the simultaneous equations have only one solution.

(2 marks)

- (c) Find the solution to the simultaneous equations for the value of k that you found in part (b).

(3 marks)

Q12**PRACTICE**

Practice · Hard

P1 · TOPIC 3 8 marks**7 (a)**

$$x^2 - y + 3x = k$$

$$20x - 4y = -5$$

are simultaneous equations, where k is a constant.

Given that the simultaneous equations have exactly one solution, find the value of the constant k .

(5 marks)

- (b)** Find the solution to the simultaneous equations for the value of k that you found in part (a).

(3 marks)

Q13**PRACTICE**

Practice · Very Hard

P1 · TOPIC 3**9 marks****7 (a)**

$$x^2 + 2y^2 = 25$$

$$x - y = k$$

are simultaneous equations, where k is a constant.

Find the respective sets of values for k for which the simultaneous equations have one, two, and no solutions.

(6 marks)

- (b)** Given that the simultaneous equations have exactly one solution, find all possible pairs (x, y) that might correspond to that solution. Give all your values for x and y in the form $a\sqrt{6}$, where a is a rational number.

(3 marks)

04 Models and Harder Forms

A model $H^3 = pt^2 + q$ with two data points is two linear equations in p and q . Quartics $2x^4 - 7x^2 - 15 = 0$ are hidden quadratics.

WHEN YOU SEE → YOU DO

“given $H^3 = pt^2 + q$... when $t = 3, H = 2$ ” → substitute both points → solve for p, q

“ $2x^4 - 15x^2 - 50 = 0$ ” → let $u = x^2$

“the exact distance PQ ” → $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

a curve with xy → make y the subject: $y = \frac{\dots}{x}$

WORKED MODEL • JANUARY 2021

$H^3 = pt^2 + q$; $t = 3, H = 2$ and $t = 5, H = 2.4$.

1 $8 = 9p + q, 13.824 = 25p + q$

two data points

2 $p = 0.364, q = 4.72$

subtract

3 $125 = 0.364 T^2 + 4.72$

height 5 m

ANSWER $T = 18.2$ years

IDEA 04

Practise: Models and Harder Forms

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE**Q14**

PRACTICE

Practice • Medium

P1 • TOPIC 3 6 marks

- 8 You are asked to advise a client on which parcel delivery service to use to deliver parcels of differing sizes. Linear Deliveries Inc. charges a flat rate of £2.25 per parcel, plus 40p times the mass of the parcel in kilograms. Square Deal Delivery Solutions charges a flat rate of £4 per parcel, plus 1p times the *square* of the parcel's mass in kilograms. Under what circumstances would you advise your client to use each of the two delivery services? Be sure to show clear mathematical justifications for your answer.

(6 marks)

Q15**PRACTICE**

Practice · Hard

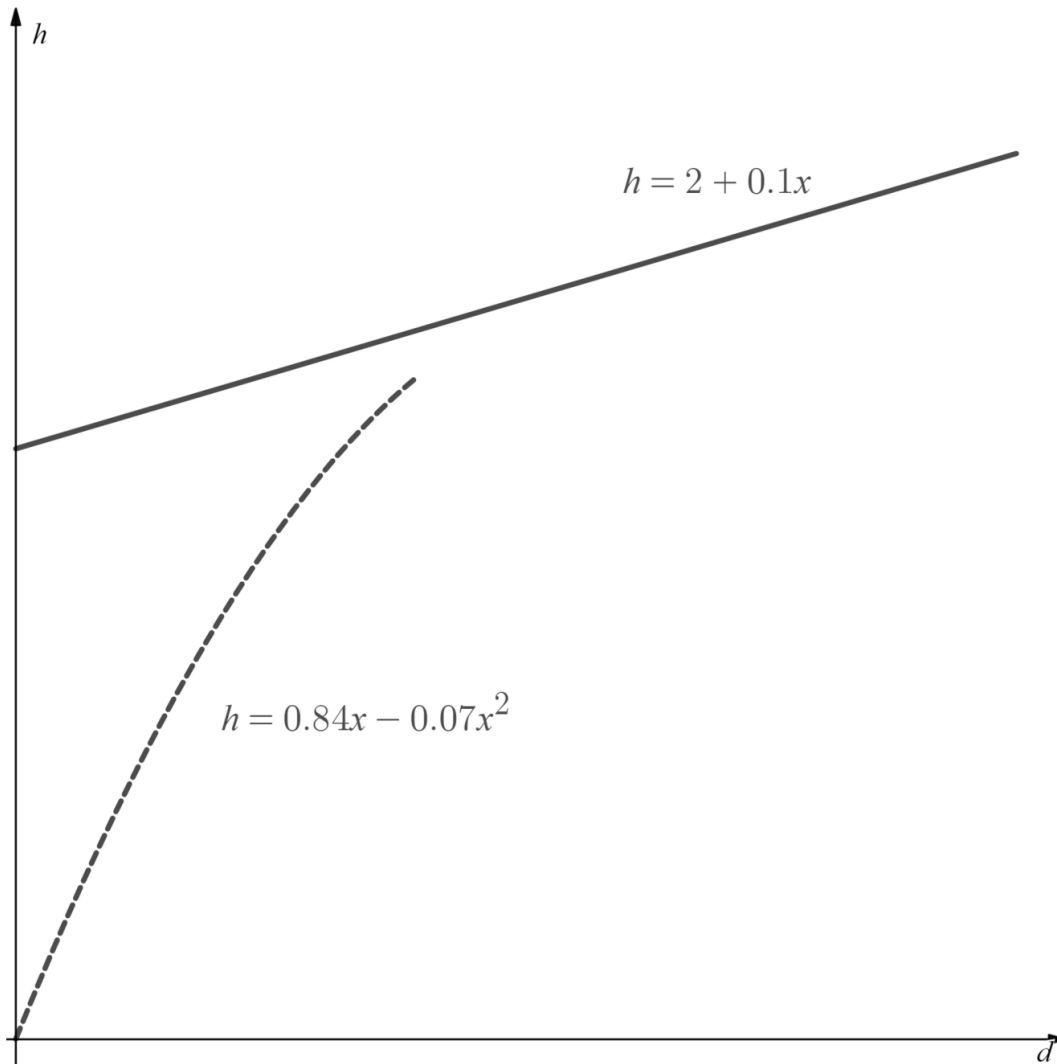
P1 · TOPIC 3 5 marks

- 8 A firework is launched inside a large shed with a sloping roof. In relation to the horizontal distance from the point it was launched, the height of the firework, h m, can be modelled by the quadratic equation

$$h = 0.84x - 0.07x^2$$

The sloping roof of the shed can be modelled with the equation

$$h = 2 + 0.1x$$



Determine whether, according to the model, the firework will hit the roof of the shed before escaping out the open end of the shed on the right of the diagram.

(5 marks)

PAST PAPERS

Q16 PAST PAPER Pearson IAL WMA11 · January 2021 · Q2

P1 · TOPIC 3 6 marks

2. A tree was planted.

Exactly 3 years after it was planted, the height of the tree was 2 m.

Exactly 5 years after it was planted, the height of the tree was 2.4 m.

Given that the height, H metres, of the tree, t years after it was planted, can be modelled by the equation

$$H^3 = pt^2 + q$$

where p and q are constants,

- (a) find, to 3 significant figures where necessary, the value of p and the value of q .

(4)

Exactly T years after the tree was planted, its height was 5 m.

- (b) Find the value of T according to the model, giving your answer to one decimal place.

(2)

(Total 6 marks)

Q17 PAST PAPER Pearson IAL WMA11 · June 2022 · Q6

P1 · TOPIC 3 7 marks

6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Given that

$$2xy - 3x^2 = 50$$

and

$$y - x^3 + 6x = 0$$

show that

$$2x^4 - 15x^2 - 50 = 0 \quad (2)$$

(b) Hence solve the simultaneous equations

$$2xy - 3x^2 = 50$$

$$y - x^3 + 6x = 0$$

Give your answers in fully simplified surd form.

(5)

(Total 7 marks)

Q18**PAST PAPER**

Pearson IAL WMA11 · October 2023 · Q8

P1 · TOPIC 3 7 marks

8. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The curve C_1 has equation

$$xy = \frac{15}{2} - 5x \quad x \neq 0$$

The curve C_2 has equation

$$y = x^3 - \frac{7}{2}x - 5$$

(a) Show that C_1 and C_2 meet when

$$2x^4 - 7x^2 - 15 = 0 \quad (2)$$

Given that C_1 and C_2 meet at points P and Q

(b) find, using algebra, the exact distance PQ (5)

(Total for Question 8 is 7 marks)

Q19**PAST PAPER**

Pearson IAL WMA11 · January 2025 · Q3

P1 · TOPIC 3 5 marks

3. The population of a town was monitored.

Exactly 5 years after monitoring began, the population was 58 000

Exactly 10 years after monitoring began, the population was 65 000

Given that the population of the town, P thousand, t years after monitoring began can be modelled by the equation

$$P^2 = a + bt^3$$

where a and b are constants,

- (a) find the value of a and the value of b .

(3)

According to the model, exactly T years after monitoring began, the population was 85 000

Making your method clear,

- (b) find the value of T , giving your answer to one decimal place.

(Solutions relying entirely on calculator technology are not acceptable.)

(2)**(Total for Question 3 is 5 marks)**

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | | |
|-----------|-------------------------|---|--|
| 01 | Linear | | Match a coefficient $\rightarrow$ different signs add, same signs subtract. |
| 02 | Linear quadratic | + | Subject from the linear $\rightarrow$ substitute $\rightarrow$ ERCS $\rightarrow$ pairs. |
| 03 | One solution | | Discriminant of the substituted quadratic = 0. |
| 04 | Models | | Two data points $\rightarrow$ two equations in p, q . $x^4 \rightarrow u = x^2$. |

BOTTOM LINE Calculator first $\rightarrow$ eliminate (linear) or substitute (quadratic) $\rightarrow$ ERCS $\rightarrow$ pair every x with its y .

Answers

Simultaneous Equations • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • LINEAR: ELIMINATION

- Q1** Dr Eslam • Add or subtract $x = 3, y = 2$
- Q2** Practice • Easy $x = 6, y = 2$
- Q3** Practice • Medium $x = 3, y = -1$
- Q4** Practice • Medium (b) $y = -\frac{1}{k}$ (c) $k = -\frac{1}{3}$

IDEA 02 • LINEAR AND QUADRATIC: SUBSTITUTE

- Q5** Dr Eslam • Substitute $(2, 3)$ and $(-3, -2)$
- Q6** Practice • Easy $(1, 1)$ and $(-\frac{1}{5}, -\frac{7}{5})$
- Q7** Practice • Medium (b) $x = -6 \pm 2\sqrt{3}, y = 11 \mp 3\sqrt{3}$
- Q8** Practice • Hard (b) $x = 4 \mp \sqrt{3}, y = -\frac{15}{2} \pm 2\sqrt{3}$
- Q9** Pearson IAL WMA11 • January 2026 • Q2 (b) $(2 + 2\sqrt{3}, 6 + 2\sqrt{3}), (2 - 2\sqrt{3}, 6 - 2\sqrt{3})$

IDEA 03 • ONE SOLUTION: THE DISCRIMINANT

- Q10** Dr Eslam • Tangent line $x^2 - 2x + 4 - k = 0: 4 - 4(4 - k) = 0 \Rightarrow k = 3$
- Q11** Practice • Medium (b) $k = 7$ (c) $x = 2, y = -3$
- Q12** Practice • Hard (a) $k = -\frac{9}{4}$ (b) $x = 1, y = \frac{25}{4}$
- Q13** Practice • Very Hard one: $k = \pm \frac{5\sqrt{6}}{2}$; two: $|k| < \frac{5\sqrt{6}}{2}$; none: $|k| > \frac{5\sqrt{6}}{2}$

IDEA 04 • MODELS AND HARDER FORMS

Q14 Practice • Medium

equal cost at $m = 5$ and $m = 35$ kg: Square Deal is cheaper for $5 < m < 35$, Linear otherwise

Q15 Practice • Hard

$0.07x^2 - 0.74x + 2 = 0$ has no real roots ($b^2 - 4ac < 0$): it escapes without hitting the roof
(a) $p = 0.364$, $q = 4.72$ (b) $T = 18.2$

Q16 Pearson IAL WMA11 • January 2021 • Q2

(b) $x = \sqrt{10}$, $y = 4\sqrt{10}$; $x = -\sqrt{10}$, $y = -4\sqrt{10}$

Q17 Pearson IAL WMA11 • June 2022 • Q6

(b) $PQ = \sqrt{65}$

Q18 Pearson IAL WMA11 • October 2023 • Q8

(a) $a = 3241$, $b = \frac{123}{125}$ (b) $T = 15.9$

Q19 Pearson IAL WMA11 • January 2025 • Q3

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Inequalities

Linear: flip the sign when you divide by a negative. Quadratic: critical values, sketch, choose. Regions: one inequality per edge.

THE TOPIC ON ONE MAP

Inequalities

01 — Linear inequalities

÷ negative: flip set notation

02 — Quadratic inequalities

roots → sketch → read “satisfied by both”

03 — Inequalities for k

no real roots real roots

04 — Regions on a graph

define R above / below

01 Linear Inequalities

Exactly like an equation — with one rule: multiplying or dividing by a **negative** flips the sign.

WHEN YOU SEE → YOU DO

“solve $3x + 4 \leq 5(x - 1)$ ”

→ expand, collect, divide

dividing by a negative

→ flip: $>$ becomes $<$

“in set notation”

→ $\{x : x \geq 3\} \cdot \{x : -1 < x \leq 4\}$

$\frac{4}{z} < 8$ — unknown on the bottom

→ multiply by z^2 (always positive), never by z

WORKED MODEL • JANUARY 2026

$$\frac{4}{z} < 8$$

1 $4z < 8z^2 \Rightarrow 8z^2 - 4z > 0$

× z^2 , sign safe

2 $4z(2z - 1) > 0$, critical values $0, \frac{1}{2}$

factorise

ANSWER $z < 0$ or $z > \frac{1}{2}$

IDEA 01

Practise: Linear Inequalities

My questions first, then selected practice.

Q1 **DR ESLAM** Flip the sign

P1 · TOPIC 4 2 marks

Solve (a) $3x - 2 > 7$ (b) $5 - 2x \leq 11$

SELECTED PRACTICE**Q2** **PRACTICE** Practice · Easy

P1 · TOPIC 4 3 marks

1 Solve the inequalities:

(i) $2x \geq 8$

(ii) $3 + 2x < 11$

(iii) $5 + x > 4x - 1$

(3 marks)

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 4**4 marks**

2 Solve the inequalities:

(i) $2x - 9 \geq 5(x - 3)$

(ii) $3(5 - x) < 2(9 - 2x)$

(4 marks)**Q4****PRACTICE**

Practice · Medium

P1 · TOPIC 4**3 marks**

1 Solve the inequality $3x + 4 \leq 5(x - 1)$.

(3 marks)

02 Quadratic Inequalities

Three steps: everything on one side, critical values (calculator), sketch the cup and read off.

WHEN YOU SEE → YOU DO

“> 0” with a cup shape

→ outside the roots: $x < \alpha$ or $x > \beta$

“< 0” with a cup shape

→ between the roots: $\alpha < x < \beta$

“using algebra”, “not calculator technology”

→ show the factorising or formula

two inequalities “satisfied by both”

→ number line, take the overlap

1

One side = 0

Rearrange so the x^2 term is positive.

2

Critical values

Solve the equation (calculator first).

3

Sketch and choose

Above the axis (> 0) = outside · below (< 0) = inside.

WORKED MODEL · JUNE 2023

$$4x^2 - 3x + 7 \geq 4x + 9$$

1 $4x^2 - 7x - 2 \geq 0$

one side

2 $(4x + 1)(x - 2) \geq 0$, $x = -\frac{1}{4}, 2$

critical values

ANSWER $x \leq -\frac{1}{4}$ or $x \geq 2$

IDEA 02

Practise: Quadratic Inequalities

My questions first, then selected practice, then real past papers.

Q5 DR ESLAM Inside the roots

P1 · TOPIC 4 3 marks

Solve $x^2 - x - 12 \leq 0$.

Q6 DR ESLAM Outside the roots

P1 · TOPIC 4 3 marks

Solve $x^2 - 4x + 3 > 0$.

SELECTED PRACTICE

Q7**PRACTICE**

Practice · Easy

P1 · TOPIC 4 6 marks

3 (a) Write down the solutions to $(x - 3)(x - 8) = 0$.

(2 marks)

(b) Sketch the graph of $y = (x - 3)(x - 8)$, clearly showing the coordinates of the points where the graph intercepts the x -axis.

(2 marks)

(c) Hence, or otherwise, solve the inequality $(x - 3)(x - 8) < 0$.

(2 marks)

Q8**PRACTICE**

Practice · Medium

P1 · TOPIC 4**5 marks**

5 Find the values of x that satisfy the inequalities

$$x^2 + 3x > 4$$

$$4x + 1 > 4$$

(5 marks)**Q9****PRACTICE**

Practice · Very Hard

P1 · TOPIC 4**4 marks**

1 Solve the simultaneous inequalities

$$t^2 - 2t - 15 < 0 \text{ and}$$

$$t^2 + 14 \leq 9t.$$

(4 marks)

PAST PAPERS

Q10 PAST PAPER Pearson IAL WMA11 • June 2023 • Q1

P1 • TOPIC 4 4 marks

1. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Solve the inequality

$$4x^2 - 3x + 7 \geq 4x + 9$$

(4)

(Total for Question 1 is 4 marks)

03 Inequalities for k : the Discriminant

A condition on the number of roots becomes a quadratic inequality in k .

WHEN YOU SEE → YOU DO

“does not cross or touch”, “no real roots”

$$\longrightarrow b^2 - 4ac < 0$$

“two distinct real roots”

$$\longrightarrow b^2 - 4ac > 0$$

“real roots”

$$\longrightarrow b^2 - 4ac \geq 0$$

a line and a curve

$\longrightarrow$ line = curve $\rightarrow$ one quadratic $\rightarrow$ discriminant

WORKED MODEL • JANUARY 2020

$y = k(2x - 1)$ does not cross or touch $y = x^2 + 2x + 11$.

1 $x^2 + (2 - 2k)x + 11 + k = 0$

curve = line

2 $(2 - 2k)^2 - 4(11 + k) < 0 \Rightarrow k^2 - 3k - 10 < 0$

discriminant < 0

ANSWER $-2 < k < 5$

IDEA 03

Practise: Inequalities for k

My questions first, then selected practice, then real past papers.

Q11 DR ESLAM No real roots

P1 · TOPIC 4 3 marks

$x^2 + 2x + k = 0$ has no real roots. Find the range of k .

SELECTED PRACTICE**Q12** PRACTICE Practice · Easy

P1 · TOPIC 4 3 marks

7 (a) Write down, in terms of k , the discriminant of $x^2 + 8x + 4k$.

(1 mark)

(b) Hence find the values of for k which the equation $x^2 + 8x + 4k = 0$ has two real and distinct solutions.

(2 marks)

Q13**PRACTICE**

Practice · Medium

P1 · TOPIC 4 4 marks

- 3 The equation $kx^2 + 2kx + 4 = 0$, where k is a constant, has two distinct real roots.

Find the possible value(s) of k .

(4 marks)**Q14****PRACTICE**

Practice · Hard

P1 · TOPIC 4 4 marks

- 3 The equation $(kx)^2 + (k - 2)x + 1 = 0$, where k is a constant, has two distinct real roots. Find the possible values of k .

(4 marks)

PAST PAPERS

Q15 PAST PAPER Pearson IAL WMA11 · January 2020 · Q8

P1 · TOPIC 4 6 marks

8. The straight line l has equation $y = k(2x - 1)$, where k is a constant.

The curve C has equation $y = x^2 + 2x + 11$

Find the set of values of k for which l does not cross or touch C .

(6)

(Total 6 marks)

04 Regions on a Graph

Every edge of the region is one inequality. Solid line $\leq/\geq$, dashed line $</>$.

WHEN YOU SEE → YOU DO

“use inequalities to define R ”

→ one inequality per boundary

region below a line/curve

→ $y \leq \dots$

region above

→ $y \geq \dots$

a vertical edge $x = a$

→ $x \geq a$ or $x \leq a$

“represent $x + 5 \leq y \leq x(x - 3)$ graphically”

→ shade where the curve is above the line

WORKED MODEL • JUNE 2023

R_1 : $0 \leq y \leq 2$, $y \leq 10 - 2x$, $y \leq kx$; area $\frac{27}{4}$; $x = a$ through $y = 2$ on $y = kx$.

1 trapezium: $\frac{1}{2} \cdot 2 \cdot (5 + (4 - a)) = \frac{27}{4}$

area

2 $a = \frac{9}{4}$, $k = \frac{2}{a} = \frac{8}{9}$

solve

ANSWER R_2 : $x \geq \frac{9}{4}$, $y \geq \frac{8}{9}x$, $y \leq 10 - 2x$

IDEA 04

Practise: Regions on a Graph

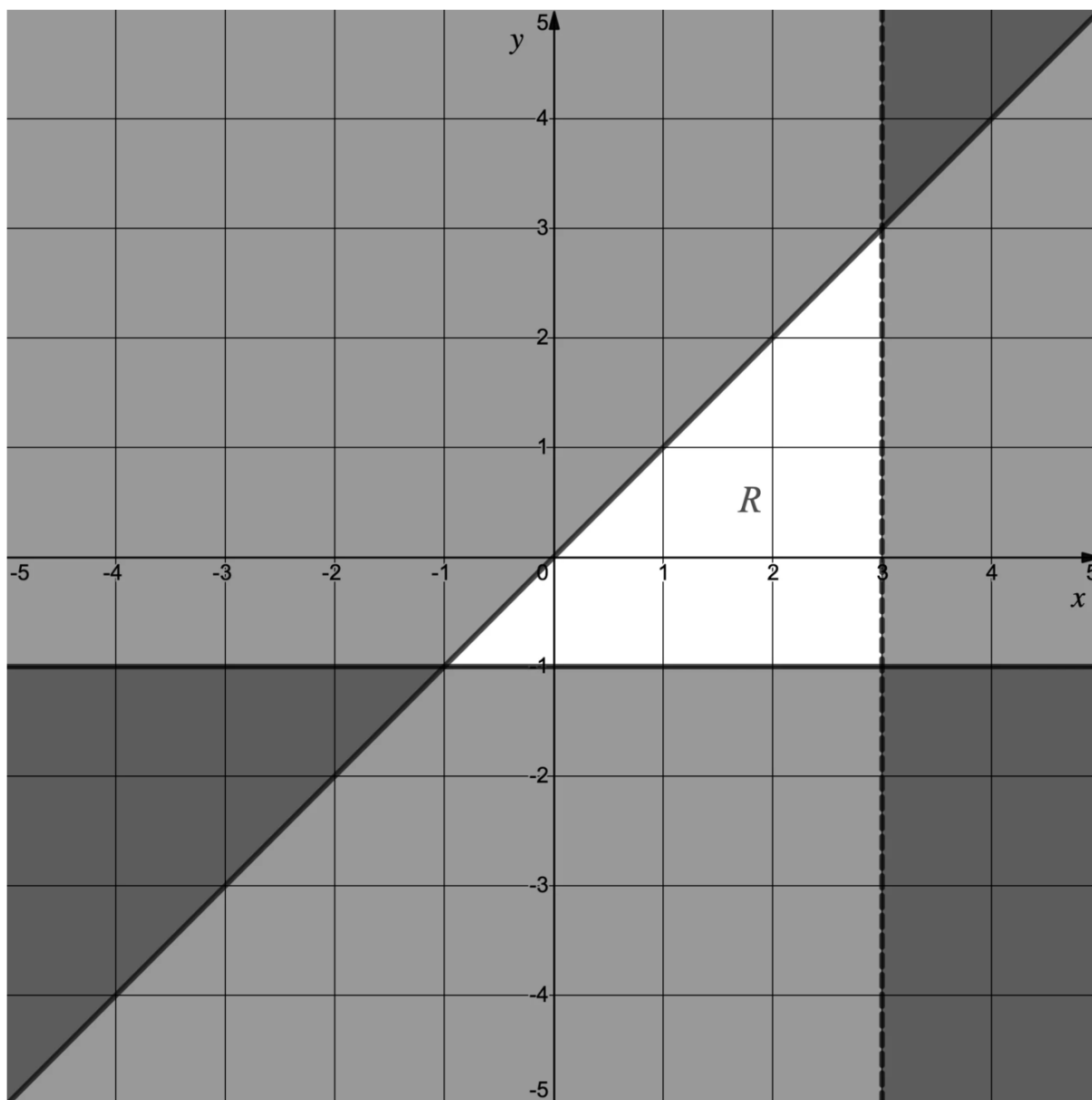
My questions first, then selected practice, then real past papers.

SELECTED PRACTICE

Q16 PRACTICE Practice · Easy

P1 · TOPIC 4 3 marks

- 8 Write down the three inequalities that define the region R shown in the diagram below.



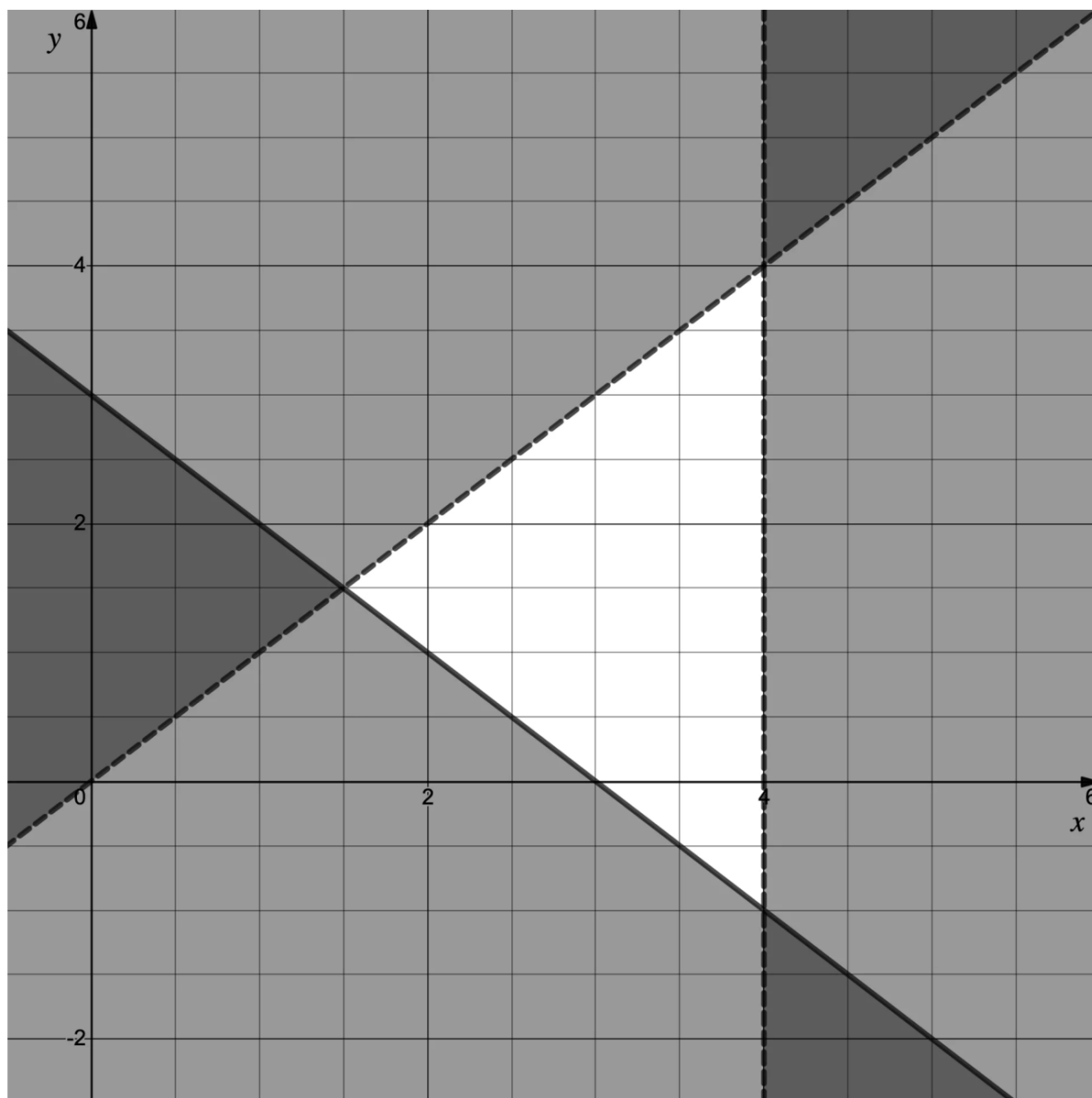
(3 marks)

Q17**PRACTICE**

Practice • Medium

P1 • TOPIC 4 4 marks

8 Write down the inequalities that define the region R shown in the diagram below.



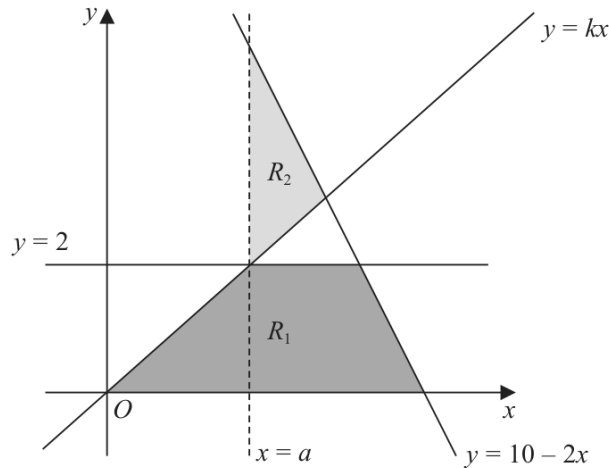
(4 marks)

PAST PAPERS

Q18 PAST PAPER Pearson IAL WMA11 · June 2023 · Q7

P1 · TOPIC 4 6 marks

7.

**Figure 2**

The region R_1 , shown shaded in Figure 2, is defined by the inequalities

$$0 \leq y \leq 2 \quad y \leq 10 - 2x \quad y \leq kx$$

where k is a constant.

The line $x = a$, where a is a constant, passes through the intersection of the lines $y = 2$ and $y = kx$

Given that the area of R_1 is $\frac{27}{4}$ square units,

(a) find

- (i) the value of a
- (ii) the value of k

(4)

(b) Define the region R_2 , also shown shaded in Figure 2, using inequalities.

(2)

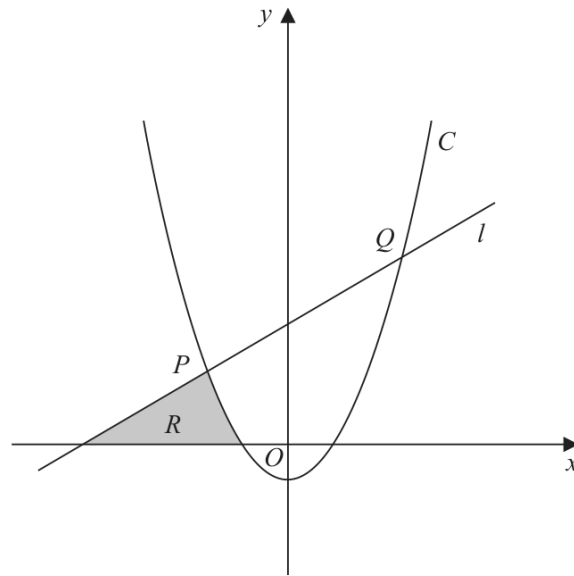
(Total for Question 7 is 6 marks)

Q19**PAST PAPER**

Pearson IAL WMA11 • June 2024 • Q6

P1 • TOPIC 4 7 marks

6.

**Figure 3**

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 3 shows

- the line l with equation $y - 5x = 75$
- the curve C with equation $y = 2x^2 + x - 21$

The line l intersects the curve C at the points P and Q , as shown in Figure 3.

- (a) Find, using algebra, the coordinates of P and the coordinates of Q .

(4)

The region R , shown shaded in Figure 3, is bounded by C , l and the x -axis.

- (b) Use inequalities to define the region R .

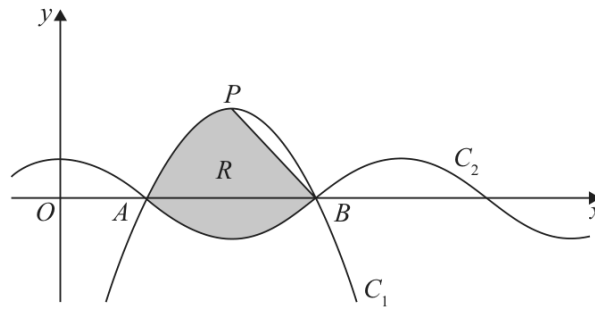
(3)

(Total for Question 6 is 7 marks)

Q20 PAST PAPER Pearson IAL WMA11 · January 2025 · Q9

P1 · TOPIC 4 13 marks

9.

**Figure 2**

- (a) Express $6x - \frac{27}{4} - x^2$ in the form $a + b(x + c)^2$ where a , b and c are constants to be found.

(3)

Figure 2 shows part of a sketch of curve C_1 with equation

$$y = 6x - \frac{27}{4} - x^2$$

Given that the point P is the maximum point on C_1

- (b) state the coordinates of P

(2)

Figure 2 also shows part of a sketch of curve C_2 with equation

$$y = \cos(kx)$$

where k is a constant and x is measured in radians.

Given that C_1 and C_2 intersect on the x -axis at point A and at point B , as shown in Figure 2,

- (c) (i) state the x coordinate of B
 (ii) state the value of k
 (iii) state the period of C_2

(3)

The line segment L joins P and B .

The region R , shown shaded in Figure 2, is bounded by L , C_1 and C_2

- (d) Use inequalities to define R .

(5)

(Total for Question 9 is 13 marks)

Q21 PAST PAPER Pearson IAL WMA11 • January 2026 • Q3

P1 • TOPIC 4 5 marks

3. (i) Find the range of values of
- z
- for which

$$\frac{4}{z} < 8$$

giving the answer in simplest form.

(2)

- (ii)

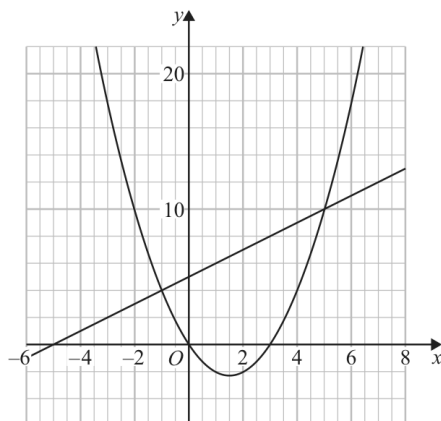


Figure 1

Figure 1 shows a plot of

- the line with equation $y = x + 5$
- the curve with equation $y = x(x - 3)$

On the opposite page there is a copy of Figure 1 labelled Diagram 1.

On Diagram 1 represent the following inequality graphically.

$$x + 5 \leq y \leq x(x - 3) \quad \text{for } x \geq -2$$

(3)

Question 3 continued

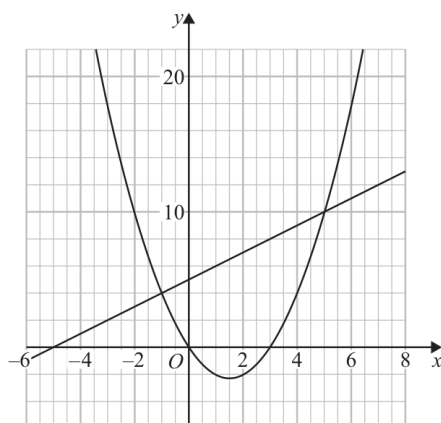


Diagram 1

(Total for Question 3 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------------|---|
| 01 | Linear | Like an equation; negative divide flips. Unknown on the bottom: $\times$ square. |
| 02 | Quadratic | One side $\rightarrow$ critical values $\rightarrow$ sketch: > 0 outside, < 0 inside. |
| 03 | For k | Number of roots $\rightarrow$ discriminant $\rightarrow$ quadratic inequality in k . |
| 04 | Regions | One inequality per edge. Solid $\leq$, dashed $<$. |

BOTTOM LINE Critical values $\rightarrow$ sketch $\rightarrow$ read the region. Never divide by an unknown whose sign you don't know.

Answers

Inequalities • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • LINEAR INEQUALITIES

- | | | |
|-----------|--------------------------|---|
| Q1 | Dr Eslam • Flip the sign | (a) $x > 3$ (b) $x \geq -3$ |
| Q2 | Practice • Easy | (i) $x \geq 4$ (ii) $x < 4$ (iii) $x < 2$ |
| Q3 | Practice • Easy | (i) $x \leq 2$ (ii) $x > -3$ |
| Q4 | Practice • Medium | $x \geq \frac{9}{2}$ |

IDEA 02 • QUADRATIC INEQUALITIES

- | | | |
|------------|---------------------------------------|-------------------------------------|
| Q5 | Dr Eslam • Inside the roots | $-3 \leq x \leq 4$ |
| Q6 | Dr Eslam • Outside the roots | $x < 1$ or $x > 3$ |
| Q7 | Practice • Easy | (a) 3, 8 (c) $3 < x < 8$ |
| Q8 | Practice • Medium | $x > 1$ |
| Q9 | Practice • Very Hard | $2 \leq t < 5$ |
| Q10 | Pearson IAL WMA11 • June 2023
• Q1 | $x \leq -\frac{1}{4}$ or $x \geq 2$ |

IDEA 03 • INEQUALITIES FOR K

Q11 Dr Eslam · No real roots $4 - 4k < 0 \Rightarrow k > 1$

Q12 Practice · Easy (a) $64 - 16k$ (b) $k < 4$

Q13 Practice · Medium $k < 0$ or $k > 4$

Q14 Practice · Hard $-2 < k < \frac{2}{3}, k \neq 0$

Q15 Pearson IAL WMA11 · January 2020 · Q8 $-2 < k < 5$

IDEA 04 · REGIONS ON A GRAPH

Q16 Practice · Easy $y \geq -1, y \leq x, x < 3$

Q17 Practice · Medium $y < x, y \geq 3 - x, x < 4$

Q18 Pearson IAL WMA11 · June 2023 · Q7 (a) $a = \frac{9}{4}, k = \frac{8}{9}$ (b) $x \geq \frac{9}{4}, y \geq \frac{8}{9}x, y \leq 10 - 2x$

Q19 Pearson IAL WMA11 · June 2024 · Q6 (a) $P(-6, 45), Q(8, 115)$ (b) $y \geq 0, y \leq 5x + 75, y \leq 2x^2 + x - 21$

Q20 Pearson IAL WMA11 · January 2025 · Q9 (a) $\frac{9}{4} - (x-3)^2$ (b) $P(3, \frac{9}{4})$ (c) $x = \frac{9}{2}, k = \frac{\pi}{3}$, period 6 (d) $y \geq \cos \frac{\pi x}{3}, y \leq 6x - \frac{27}{4} - x^2, y \leq \frac{27}{4} - \frac{3}{2}x$

Q21 Pearson IAL WMA11 · January 2026 · Q3 (i) $z < 0$ or $z > \frac{1}{2}$ (ii) shade between the line and the curve for $-2 \leq x \leq -1$ and $x \geq 5$

Polynomials

Expand, factorise, and read a cubic straight off its graph. Roots give brackets; one point gives the number in front.



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THE TOPIC ON ONE MAP

Polynomials

01 — Expand and factorise

three brackets

factorise completely

02 — Solving cubics

show all stages

hidden cubics

$f(-1) = 32$

03 — Cubic from a graph

cuts / touches

$f(x - 3) = 0$

01 Expand and Factorise

Expand: every term $\times$ every term. Factorise: HCF first, then the CC Rule on what is left.

WHEN YOU SEE $\rightarrow$ YOU DO

“expand and simplify” three brackets

$\rightarrow$ two first, then the third

“factorise completely” a cubic with no constant

$\rightarrow$ take out x , then factorise the quadratic

“(3x + y)(2x - 3y)(x - 2y) $\equiv$ ax³ + bx²y + ...”

$\rightarrow$ expand, compare coefficients

x⁵y - xy⁵

$\rightarrow$ HCF, then DOTS twice

WORKED MODEL · FACTORISE COMPLETELY

$$3x^3 - 51x^2 + 126x$$

1 $3x(x^2 - 17x + 42)$

HCF 3x

2 roots of the quadratic: 3, 14

calculator

ANSWER $3x(x - 3)(x - 14)$

IDEA 01

Practise: Expand and Factorise

My questions first, then selected practice.

Q1 DR ESLAM HCF, then CC Rule

P1 · TOPIC 5 2 marks

Factorise completely (a) $x^3 - 4x$ (b) $2x^3 + x^2 - 6x$

SELECTED PRACTICE**Q2** PRACTICE Practice · Easy

P1 · TOPIC 5 5 marks

2 Expand and simplify

(i) $(2x + 3)(x - 4)$

(ii) $2p(p + 3)(p - 2)$

(iii) $(y - 1)(y - 2)^2$

(5 marks)

Q3**PRACTICE**

Practice · Hard

P1 · TOPIC 5**2 marks**

5 Factorise completely $15x^3 + 19x^2 - 10x$.

(2 marks)**Q4****PRACTICE**

Practice · Very Hard

P1 · TOPIC 5**3 marks**

5 Factorise completely $x^5y - xy^5$.

(3 marks)

02 Solving Cubics and Hidden Cubics

Factorise to brackets, each bracket = 0. A power pattern like y , $y^{\frac{2}{3}}$, $y^{\frac{1}{3}}$ is the same cubic in disguise.

WHEN YOU SEE → YOU DO

“solve $f(x) = 0$ ” (show all stages)

→ factorise fully, each bracket = 0

“hence solve $3y + 26y^{\frac{2}{3}} - 9y^{\frac{1}{3}} = 0$ ”

→ $x = y^{\frac{1}{3}}$ — the same cubic

“ $3(9^{3z}) + 26(9^{2z}) - 9(9^z) = 0$ ”

→ $x = 9^z$; reject $x \leq 0$

“given $f(-1) = 32$, show $a = 3$ ”

→ substitute, solve for a

WORKED MODEL · OCTOBER 2022

$$f(x) = x^2(2x + 1) - 15x$$

1 $x(2x^2 + x - 15) = x(2x - 5)(x + 3) = 0$

factorise

2 $x = 0, \frac{5}{2}, -3$

roots

3 hence $y^{\frac{4}{3}}(2y^{\frac{2}{3}} + 1) - 15y^{\frac{2}{3}} = 0$: $x = y^{\frac{2}{3}} = \frac{5}{2}$

$y > 0$

ANSWER $y = \left(\frac{5}{2}\right)^{\frac{3}{2}} = \frac{5\sqrt{10}}{4}$

IDEA 02

Practise: Solving Cubics

My questions first, then real past papers.

Q5 **DR ESLAM** Solve a cubic

P1 · TOPIC 5 3 marks

Given that $x = 1$ is one root, solve $x^3 - 7x + 6 = 0$ by factorising.

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA11 · June 2021 · Q2

P1 · TOPIC 5 10 marks

2. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = ax^3 + (6a + 8)x^2 - a^2x$$

where a is a positive constant.Given $f(-1) = 32$ (a) (i) show that the only possible value for a is 3(ii) Using $a = 3$ solve the equation

$$f(x) = 0 \quad (5)$$

(b) Hence find all real solutions of

$$(i) \quad 3y + 26y^{\frac{2}{3}} - 9y^{\frac{1}{3}} = 0$$

$$(ii) \quad 3(9^{3z}) + 26(9^{2z}) - 9(9^z) = 0 \quad (5)$$

(Total 10 marks)

Q7**PAST PAPER**

Pearson IAL WMA11 • October 2022 • Q4

P1 • TOPIC 5**6 marks**

4.

**In this question you must show detailed reasoning.
Solutions relying on calculator technology are not acceptable.**

$$f(x) = x^2(2x + 1) - 15x$$

(a) Solve

$$f(x) = 0$$

(4)

(b) Hence solve

$$y^{\frac{4}{3}}\left(2y^{\frac{2}{3}} + 1\right) - 15y^{\frac{2}{3}} = 0 \quad y > 0$$

giving your answer in simplified surd form.

(2)**(Total for Question 4 is 6 marks)**

03 Cubic from a Graph

Crossing the x -axis at $a \rightarrow (x - a)$. **Touching** at $a \rightarrow (x - a)^2$. Then one more point finds the number in front, k .

WHEN YOU SEE $\rightarrow$ YOU DO

“cuts the x -axis at $-4, -2, 3$ ”

$\rightarrow y = k(x + 4)(x + 2)(x - 3)$

“touches the x -axis at 5 ”

$\rightarrow$ repeated root $(x - 5)^2$

“cuts the y -axis at 10 ” / “passes through P ”

$\rightarrow$ substitute to find k

“roots of $f(\frac{1}{3}x) = 0$ ”, “ $f(x - 3) = 0$ ”

$\rightarrow$ stretch $\times 3$ · shift right 3

curve meets a line

$\rightarrow$ curve = line; you already know one root $\rightarrow$ divide it out

WORKED MODEL · JANUARY 2026

Cuts at -2 , touches at 5 , y -intercept 10 .

1 $f(x) = k(x + 2)(x - 5)^2$

roots $\rightarrow$ brackets

2 $10 = k(2)(25) \Rightarrow k = \frac{1}{5}$

y -intercept

3 $\frac{1}{5}(x + 2)(x - 5)^2 = 10(x + 2) \Rightarrow (x - 5)^2 = 50$

meets $y = 10(x + 2)$

ANSWER $x = -2, \quad x = 5 \pm 5\sqrt{2}$

IDEA 03

Practise: Cubic from a Graph

My questions first, then real past papers.

Q8 DR ESLAM Roots to equation

P1 · TOPIC 5 3 marks

A cubic cuts the x-axis at -1 , 2 and 3 and passes through $(0, 6)$. Find its equation.

Q9 DR ESLAM Touching root

P1 · TOPIC 5 3 marks

A cubic touches the x-axis at $x = 2$, cuts it at $x = -1$ and passes through $(0, 8)$. Find its equation in factorised form.

PAST PAPERS

Q10 PAST PAPER Pearson IAL WMA11 · June 2021 · Q8

P1 · TOPIC 5 9 marks

8. The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

- (a) Write $3x^2 + 6x + 9$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

The point P is the minimum point of C_1

- (b) Deduce the coordinates of P .

(1)

A different curve C_2 has equation

$$y = Ax^3 + Bx^2 + Cx + D$$

where A , B , C and D are constants.

Given that C_2

- passes through P
- intersects the x -axis at -4 , -2 and 3

- (c) find, making your method clear, the values of A , B , C and D .

(5)

(Total 9 marks)

4.

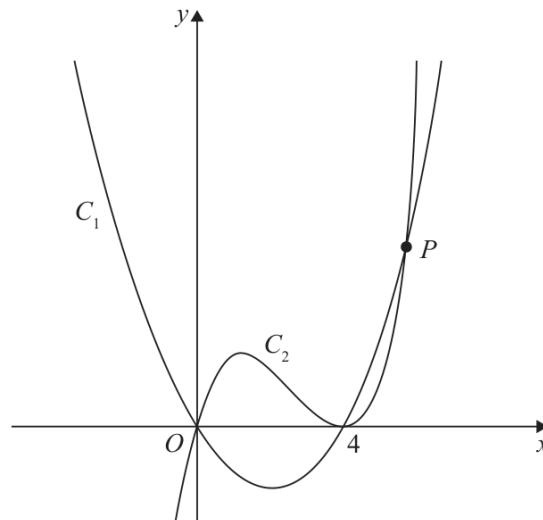


Figure 1

Figure 1 shows a sketch of part of the curves C_1 and C_2

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a quadratic function
- cuts the x -axis at the origin and at $x = 4$
- has a minimum turning point at $(2, -4.8)$

(a) find $f(x)$

(3)

Given that C_2

- has equation $y = g(x)$ where $g(x)$ is a cubic function
- cuts the x -axis at the origin and meets the x -axis at $x = 4$
- passes through the point $(6, 7.2)$

(b) find $g(x)$

(3)

The curves C_1 and C_2 meet in the first quadrant at the point P , shown in Figure 1.

(c) Use algebra to find the coordinates of P .

(4)

(Total for Question 4 is 10 marks)

Q12 PAST PAPER Pearson IAL WMA11 · January 2026 · Q6

P1 · TOPIC 5 9 marks

6.

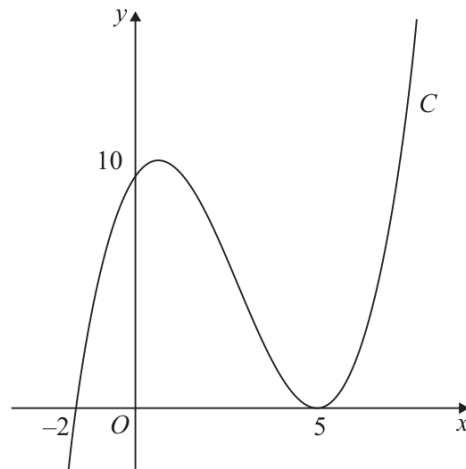
**Figure 3**

Figure 3 shows a sketch of the curve C with equation $y = f(x)$ where $f(x)$ is a cubic function in x .

The curve C

- cuts the x -axis at $(-2, 0)$ and cuts the y -axis at $(0, 10)$
- touches the x -axis at $(5, 0)$

as shown in Figure 3.

(a) Deduce the roots of the equation

(i) $f\left(\frac{1}{3}x\right) = 0$

(ii) $f(x - 3) = 0$

(2)

(b) Find an expression for $f(x)$. You should leave your answer in factorised form.

(3)

The curve C intersects the straight line $y = 10(x + 2)$ at exactly three points.

(c) Use algebra to find the exact x coordinates of the three points of intersection.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

(Total for Question 6 is 9 marks)

Q13 PAST PAPER Pearson IAL WMA11 · June 2026 · Q6

P1 · TOPIC 5 7 marks

6.

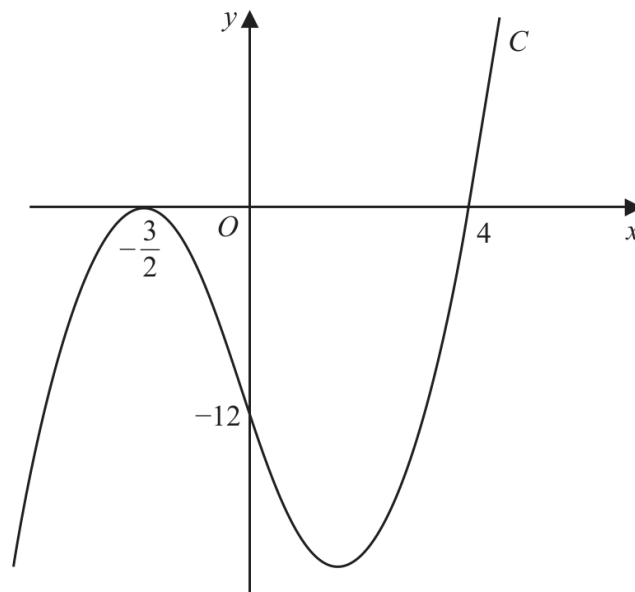
**Figure 2**

Figure 2 shows a sketch of the curve C with equation $y = f(x)$.

The curve C

- touches the x -axis at $-\frac{3}{2}$
- cuts the x -axis at 4
- cuts the y -axis at -12

Given that $f(x)$ is a cubic function,

(a) find $f(x)$. You may leave the answer in a factorised form.

(3)

The line with equation $y = 2x - 12$ cuts C at $(0, -12)$ and at two further points, P and Q .

(b) Use algebra to find the exact x coordinates of P and Q .

Give each answer in simplest form.

(Solutions relying entirely on calculator technology are not acceptable.)

(4)**Question 6 continued****(Total for Question 6 is 7 marks)**

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 Expand / fac-
torise** Every term $\times$ every term. HCF first, then CC Rule.
Compare coefficients.
- 02 Solve** Factorise fully, each bracket = 0. Hidden cubic:
 $x = y^{\frac{1}{3}}$, $x = 9^z$.
- 03 From a graph** Cut $\rightarrow (x - a)$, touch $\rightarrow (x - a)^2$, a point $\rightarrow k$.

BOTTOM LINE Roots $\rightarrow$ brackets $\rightarrow$ one point for k . Touching means squared.

Answers

Polynomials • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • EXPAND AND FACTORISE

- Q1** Dr Eslam • HCF, then CC Rule (a) $x(x+2)(x-2)$ (b) $x(2x-3)(x+2)$
- Q2** Practice • Easy (i) $2x^2 - 5x - 12$ (ii) $2p^3 + 2p^2 - 12p$ (iii) $y^3 - 5y^2 + 8y - 4$
- Q3** Practice • Hard $x(3x+5)(5x-2)$
- Q4** Practice • Very Hard $xy(x-y)(x+y)(x^2+y^2)$

IDEA 02 • SOLVING CUBICS

- Q5** Dr Eslam • Solve a cubic $(x-1)(x-2)(x+3): x = 1, 2, -3$
- Q6** Pearson IAL WMA11 • June 2021 • Q2 (a)(ii) $x = 0, \frac{1}{3}, -9$ (b)(i) $y = 0, \frac{1}{27}, -729$ (ii) $z = -\frac{1}{2}$
- Q7** Pearson IAL WMA11 • October 2022 • Q4 (a) $x = 0, \frac{5}{2}, -3$ (b) $y = \frac{5\sqrt{10}}{4}$

IDEA 03 • CUBIC FROM A GRAPH

- Q8** Dr Eslam · Roots to equation $y = (x + 1)(x - 2)(x - 3) = x^3 - 4x^2 + x + 6$
- Q9** Dr Eslam · Touching root $y = 2(x + 1)(x - 2)^2$
- Q10** Pearson IAL WMA11 · June 2021 · Q8 (a) $3(x + 1)^2 + 6$ (b) $P(-1, 6)$ (c) $A = -\frac{1}{2}$, $B = -\frac{3}{2}$, $C = 5$, $D = 12$
- Q11** Pearson IAL WMA11 · October 2024 · Q4 (a) $f(x) = \frac{6}{5}x^2 - \frac{24}{5}x$ (b) $g(x) = \frac{3}{10}x(x - 4)^2$ (c) $P(8, 38.4)$
- Q12** Pearson IAL WMA11 · January 2026 · Q6 (a)(i) $x = -6, 15$ (ii) $x = 1, 8$ (b) $f(x) = \frac{1}{5}(x + 2)(x - 5)^2$ (c) $x = -2, 5 \pm 5\sqrt{2}$
- Q13** Pearson IAL WMA11 · June 2026 · Q6 (a) $f(x) = \frac{1}{3}(2x + 3)^2(x - 4)$ (b) $x = \frac{1 \pm \sqrt{46}}{2}$

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Graphs of Functions

Know the shapes, mark the crossings, and count intersections with a discriminant. Sketch first — the picture tells you the algebra.

THE TOPIC ON ONE MAP

Graphs of Functions

01 — Cubic sketches

$$(2x + 5)(x - 3)^2 \quad f(x) \leq 0$$

02 — Reciprocal graphs

$$y = \frac{k}{x} \quad y = \frac{k}{x^2} \quad \text{asymptote}$$

03 — Intersections and roots

coordinates of P, Q number of roots

04 — Tangent / cross / miss

$$b^2 - 4ac \quad \text{at least one point}$$

01 Cubic Sketches from Factorised Form

Roots from the brackets, y-intercept from $x = 0$, shape from the sign of x^3 .
A squared bracket **touches**.

WHEN YOU SEE → YOU DO

“sketch $y = (2x + 5)(x - 3)^2$ ”

→ cuts at $-\frac{5}{2}$, touches at 3, y-int $f(0)$

“deduce the values of x for which $f(x) \leq 0$ ”

→ read below the axis from the sketch

“expand to $ax^3 + bx^2 + cx + d$ ” then
“gradient at P ”

→ differentiate the expansion

“translated two units in the positive x direction”

→ $g(x) = f(x - 2)$

WORKED MODEL • OCTOBER 2019

$$f(x) = (2x + 5)(x - 3)^2$$

1 cuts at $x = -\frac{5}{2}$, touches at $x = 3$

brackets

2 $f(x) = 2x^3 - 7x^2 - 12x + 45$, $P(0, 45)$

expand, $x = 0$

ANSWER $f(x) \leq 0 \iff x \leq -\frac{5}{2} \text{ or } x = 3$

IDEA 01

Practise: Cubic Sketches

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Squared bracket touches

P1 · TOPIC 6 3 marks

Sketch $y = (x + 1)(x - 2)^2$, showing where it meets the axes.

SELECTED PRACTICE

Q2 **PRACTICE** Practice · Easy

P1 · TOPIC 6 5 marks

6 (a) Express $2x^3 + 2x^2 - 12x$ in the form $ax(x + b)(x + c)$, where a , b and c are integers to be found.

(2 marks)

(b) Hence sketch the graph of $y = 2x^3 + 2x^2 - 12x$ labelling any points where the graph intersects the coordinate axes.

(3 marks)

Q3**PRACTICE**

Practice · Medium

P1 · TOPIC 6 3 marks

- 1 Sketch the graph of $y = (x + 3)^3$ labelling any points where the graph intersects the coordinate axes.

(3 marks)**Q4****PRACTICE**

Practice · Medium

P1 · TOPIC 6 3 marks

- 5 Sketch the graph of $y = 2x^2(x + 3)$ labelling any points where the graph intersects the coordinate axes.

(3 marks)

PAST PAPERS

Q5 PAST PAPER Pearson IAL WMA11 · October 2019 · Q10

P1 · TOPIC 6 10 marks

10.

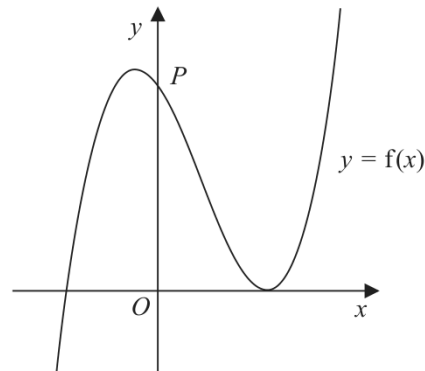


Figure 6

Figure 6 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (2x + 5)(x - 3)^2$$

- (a) Deduce the values of x for which $f(x) \leq 0$ (2)

The curve crosses the y -axis at the point P , as shown.

- (b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a , b , c and d are integers to be found.

(3)

- (c) Hence, or otherwise, find

- (i) the coordinates of P ,
 (ii) the gradient of the curve at P . (2)

The curve with equation $y = f(x)$ is translated two units in the positive x direction to a curve with equation $y = g(x)$.

- (d) (i) Find $g(x)$, giving your answer in a simplified factorised form.
 (ii) Hence state the y intercept of the curve with equation $y = g(x)$. (3)

(Total 10 marks)

Q6**PAST PAPER**

Pearson IAL WMA11 · January 2020 · Q10

P1 · TOPIC 6 8 marks

10. The curve C_1 has equation $y = f(x)$, where

$$f(x) = (4x - 3)(x - 5)^2$$

- (a) Sketch C_1 showing the coordinates of any point where the curve touches or crosses the coordinate axes.

(3)

(b) Hence or otherwise

- (i) find the values of x for which $f\left(\frac{1}{4}x\right) = 0$

- (ii) find the value of the constant p such that the curve with equation $y = f(x) + p$ passes through the origin.

(2)

A second curve C_2 has equation $y = g(x)$, where $g(x) = f(x + 1)$

- (c) (i) Find, in simplest form, $g(x)$. You may leave your answer in a factorised form.

- (ii) Hence, or otherwise, find the y intercept of curve C_2

(3)**(Total 8 marks)**

Q7**PAST PAPER**

Pearson IAL WMA11 · January 2021 · Q8

P1 · TOPIC 6 9 marks

8.

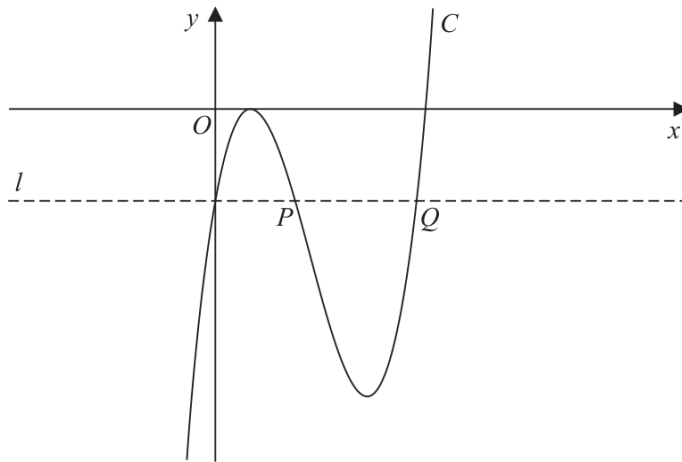
**Figure 4**

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x - 2)^2 (x - 4)$$

- (a) Deduce the values of x for which $f(x) > 0$

(1)

- (b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a , b , c and d are integers to be found.

(3)

The line l , also shown in Figure 4, passes through the y intercept of C and is parallel to the x -axis.

The line l cuts C again at points P and Q , also shown in Figure 4.

- (c) Using algebra and showing your working, find the length of line PQ . Write your answer in the form $k\sqrt{3}$, where k is a constant to be found.

(Solutions relying entirely on calculator technology are not acceptable.)

(5)**(Total 9 marks)**

Q8 **PAST PAPER** Pearson IAL WMA11 · October 2021 · Q6**P1 · TOPIC 6 10 marks**

6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = f(x)$ where

$$f(x) = 2(x + 1)(x - 3)^2$$

(a) Sketch a graph of C .

Show on your graph the coordinates of the points where C cuts or meets the coordinate axes.

(3)

(b) Write $f(x)$ in the form $ax^3 + bx^2 + cx + d$, where a, b, c and d are constants to be found.

(3)

(c) Hence, find the equation of the tangent to C at the point where $x = \frac{1}{3}$

(4)

(Total 10 marks)

Q9**PAST PAPER**

Pearson IAL WMA11 · June 2025 · Q7

P1 · TOPIC 6 10 marks

7.

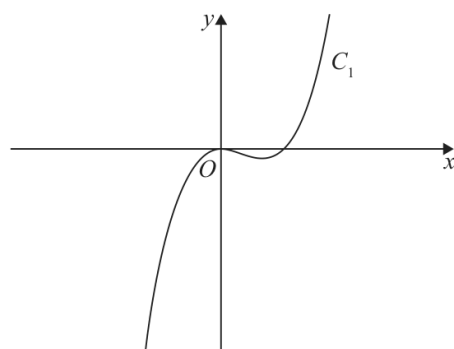
**Figure 3**

Figure 3 shows a sketch of part of the curve C_1

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a cubic function
- touches the x -axis at the origin and cuts the x -axis at $x = 4$
- passes through the point $(10, 120)$

(a) find $f(x)$

(3)

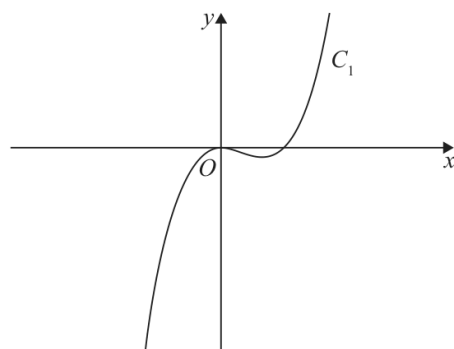
The curve C_2 has equation $y = 1.2x(8 - x)$

On the following page there is a copy of Figure 3 called Diagram 1.

(b) On Diagram 1 sketch a graph of the curve C_2

(2)

(c) Use algebra to find the coordinates of the points where C_1 and C_2 intersect.
Show each stage of your working.

(5)**Diagram 1**

(Total for Question 7 is 10 marks)

02 Reciprocal Graphs $y = \frac{k}{x}$, $y = \frac{k}{x^2}$

Two branches. Asymptotes: where the bottom is zero, and the number added on.

WHEN YOU SEE → YOU DO

$$y = \frac{4}{x - k}$$

→ vertical asymptote $x = k$; y-int $-\frac{4}{k}$

$$y = \frac{4}{x} + k$$

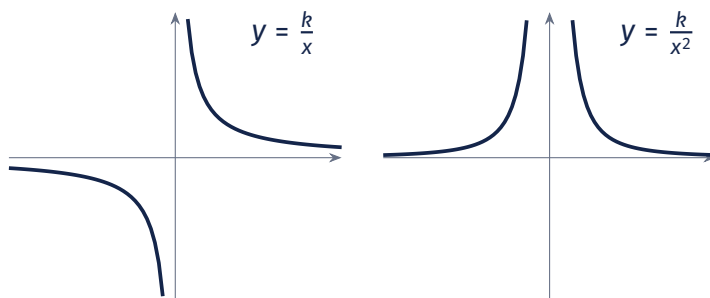
→ horizontal asymptote $y = k$; x-int $-\frac{4}{k}$

$$y = \frac{k}{x^2}$$

→ both branches above the axis

“state the equation of the asymptote”

→ write $x = \dots$ or $y = \dots$



IDEA 02

Practise: Reciprocal Graphs

My questions first, then selected practice, then real past papers.

Q10 DR ESLAM Asymptotes

P1 · TOPIC 6 2 marks

State the asymptotes of (a) $y = \frac{3}{x-2}$ (b) $y = \frac{1}{x} + 4$

SELECTED PRACTICE**Q11** PRACTICE Practice · Easy

P1 · TOPIC 6 3 marks

- 3 Sketch the graph of $y = \frac{1}{x}$, labelling any points where the graph intersects the coordinate axes and stating the equations of any asymptotes.

(3 marks)

Q12**PRACTICE**

Practice · Medium

P1 · TOPIC 6 2 marks

- 2 Sketch the graph of $y = \frac{1}{x^2}$ and write down the equations of any asymptotes.

(2 marks)**Q13****PRACTICE**

Practice · Medium

P1 · TOPIC 6 3 marks

- 4 y is inversely proportional to x . When $x = 3$, $y = 12$. Find the constant of proportionality and sketch the graph of y against x .

(3 marks)

PAST PAPERS

Q14 PAST PAPER Pearson IAL WMA11 · January 2021 · Q6

P1 · TOPIC 6 10 marks

6. (a) Sketch the curve with equation

$$y = -\frac{k}{x} \quad k > 0 \quad x \neq 0 \quad (2)$$

- (b) On a separate diagram, sketch the curve with equation

$$y = -\frac{k}{x} + k \quad k > 0 \quad x \neq 0$$

stating the coordinates of the point of intersection with the x -axis and, in terms of k , the equation of the horizontal asymptote.

(3)

- (c) Find the range of possible values of
- k
- for which the curve with equation

$$y = -\frac{k}{x} + k \quad k > 0 \quad x \neq 0$$

does not touch or intersect the line with equation $y = 3x + 4$

(5)

(Total 10 marks)

Q15**PAST PAPER**

Pearson IAL WMA11 · October 2022 · Q6

P1 · TOPIC 6 6 marks

6. (a) Given that k is a positive constant such that $0 < k < 4$ sketch, on **separate axes**, the graphs of

(i) $y = (2x - k)(x + 4)^2$

(ii) $y = \frac{k}{x^2}$

showing the coordinates of any points where the graphs cross or meet the coordinate axes, leaving coordinates in terms of k , where appropriate.

(5)

- (b) State, with a reason, the number of roots of the equation

$$(2x - k)(x + 4)^2 = \frac{k}{x^2}$$

(1)**(Total for Question 6 is 6 marks)**

Q16 PAST PAPER Pearson IAL WMA11 • October 2023 • Q4

P1 • TOPIC 6 7 marks

4.

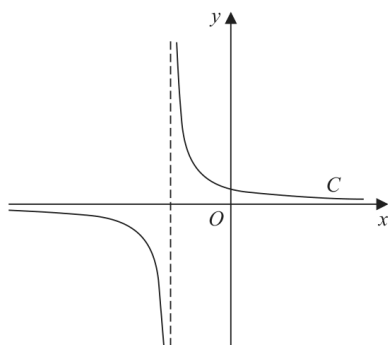
**Figure 1**

Figure 1 shows a sketch of part of the curve C with equation $y = \frac{1}{x+2}$

(a) State the equation of the asymptote of C that is parallel to the y -axis. (1)

(b) Factorise fully $x^3 + 4x^2 + 4x$ (2)

A copy of Figure 1, labelled Diagram 1, is shown on the next page.

(c) On Diagram 1, add a sketch of the curve with equation

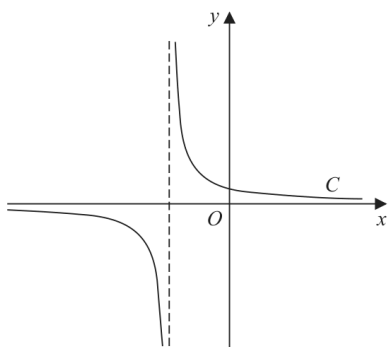
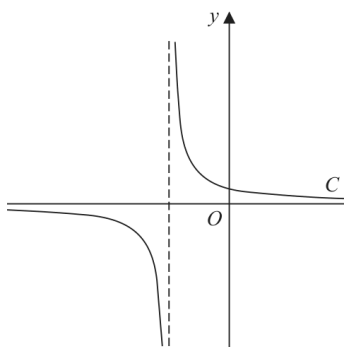
$$y = x^3 + 4x^2 + 4x$$

On your sketch, state clearly the coordinates of each point where this curve cuts or meets the coordinate axes. (3)

(d) Hence state the number of real solutions of the equation

$$(x+2)(x^3 + 4x^2 + 4x) = 1$$

giving a reason for your answer. (1)

**Diagram 1****copy of Diagram 1**

Only use the copy of Diagram 1 if you need to redraw your answer to part (c).

(Total for Question 4 is 7 marks)

03 Intersections and How Many Roots

Where two graphs meet: set them equal. How many solutions: count the crossings on a sketch.

WHEN YOU SEE → YOU DO

“find the coordinates of P and Q ”

→ curve = line → solve → pair

“state the number of roots, with a reason”

→ sketch both sides, count crossings

“use inequalities to define R ”

→ one inequality per edge

“the line passes through the y -intercept, parallel to the x -axis”

→ $y = f(0)$

WORKED MODEL • JANUARY 2022

$$x + y = 6 \text{ and } y = 6x - 2x^2 + 1$$

$$1 \quad 6 - x = 6x - 2x^2 + 1 \Rightarrow 2x^2 - 7x + 5 = 0$$

equal

$$2 \quad x = 1, \frac{5}{2}$$

solve

ANSWER $P(1, 5), Q\left(\frac{5}{2}, \frac{7}{2}\right)$

IDEA 03

Practise: Intersections

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE**Q17****PRACTICE**

Practice · Easy

P1 · TOPIC 6 3 marks

- 8 By sketching the graphs of $y = x^3$ and $y = \frac{1}{x}$ on the same diagram show that there are two real solutions to the equation $x^3 = \frac{1}{x}$.

(3 marks)

Q18**PRACTICE**

Practice · Medium

P1 · TOPIC 6 4 marks

6 (a) On the same diagram, sketch the graphs of $y = x(x+2)(x-1)$ and $y = \frac{1}{x}$.

(3 marks)

(b) Use your graph to determine the number of solutions to the equation

$$x(x+2)(x-1) = \frac{1}{x}.$$

(1 mark)

PAST PAPERS

Q19 PAST PAPER Pearson IAL WMA11 • October 2019 • Q3

P1 • TOPIC 6 10 marks

3. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

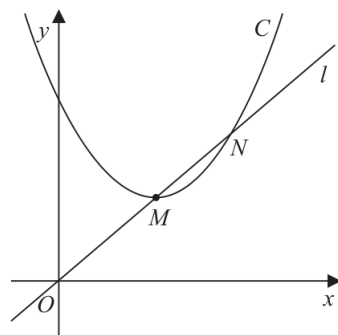


Figure 2

Figure 2 shows a sketch of the curve C with equation $y = x^2 - 5x + 13$

The point M is the minimum point of C .

The straight line l passes through the origin O and intersects C at the points M and N as shown.

Find, showing your working,

- (a) the coordinates of M , (3)
- (b) the coordinates of N . (5)

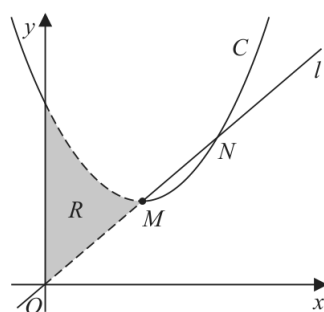


Figure 3

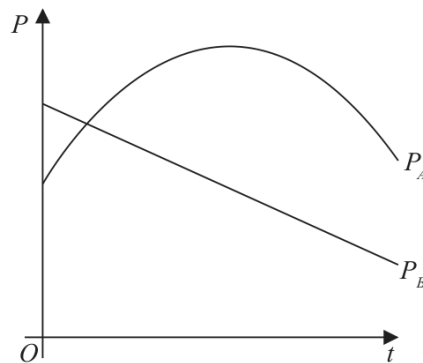
Figure 3 shows the curve C and the line l . The finite region R , shown shaded in Figure 3, is bounded by C , l and the y -axis.

(Total 10 marks)

Q20 PAST PAPER Pearson IAL WMA11 · June 2021 · Q5

P1 · TOPIC 6 8 marks

5.

**Figure 2**

The share value of two companies, company *A* and company *B*, has been monitored over a 15-year period.

The share value P_A of **company A**, in millions of pounds, is modelled by the equation

$$P_A = 53 - 0.4(t - 8)^2 \quad t \geq 0$$

where t is the number of years after monitoring began.

The share value P_B of **company B**, in millions of pounds, is modelled by the equation

$$P_B = -1.6t + 44.2 \quad t \geq 0$$

where t is the number of years after monitoring began.

Figure 2 shows a graph of both models.

Use the equations of one or both models to answer parts (a) to (d).

- Find the difference between the share value of **company A** and the share value of **company B** at the point monitoring began. (2)
- State the maximum share value of **company A** during the 15-year period. (1)
- Find, using algebra and showing your working, the times during this 15-year period when the share value of **company A** was greater than the share value of **company B**. (4)
- Explain why the model for **company A** should not be used to predict its share value when $t = 20$. (1)

(Total 8 marks)

4.

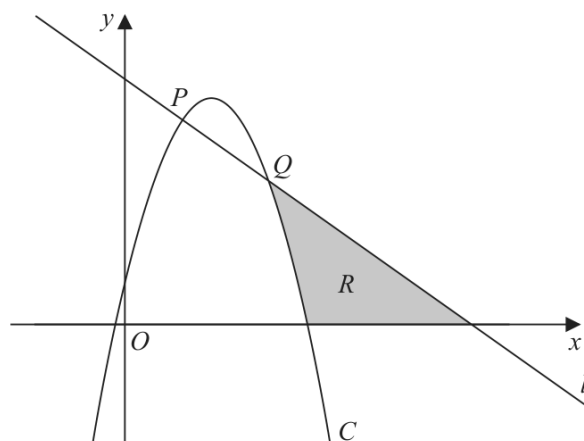


Figure 1

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 1 shows a line l with equation $x + y = 6$ and a curve C with equation $y = 6x - 2x^2 + 1$

The line l intersects the curve C at the points P and Q as shown in Figure 1.

- (a) Find, using algebra, the coordinates of P and the coordinates of Q . (4)

The region R , shown shaded in Figure 1, is bounded by C , l and the x -axis.

- (b) Use inequalities to define the region R . (3)

(Total 7 marks)

04 Tangent / Cross / Miss: the Discriminant

Line meets a reciprocal or quadratic: multiply out, rearrange to a quadratic, then use $b^2 - 4ac$.

WHEN YOU SEE → YOU DO

"the line is a tangent to C "	→ $b^2 - 4ac = 0$
"does not touch or intersect"	→ $b^2 - 4ac < 0$
"intersects at two distinct points"	→ $b^2 - 4ac > 0$
"at least one point of intersection"	→ $b^2 - 4ac \geq 0$
" k is a positive constant"	→ keep only $k > 0$ in the final range
"meets at 4 distinct points" (x^4)	→ $u = x^2$: two positive roots in u

WORKED MODEL • OCTOBER 2019

$y = \frac{4}{x} + k$ and the tangent $y = 10 - 2x$.

1 $4 + kx = 10x - 2x^2 \Rightarrow 2x^2 + (k - 10)x + 4 = 0$

× x , rearrange

2 $(k - 10)^2 - 32 = 0$

tangent

ANSWER $k = 10 \pm 4\sqrt{2}$

IDEA 04

Practise: Tangent, Cross or Miss

My questions first, then real past papers.

Q22 DR ESLAM Tangent

P1 · TOPIC 6 3 marks

Find k so that the line $y = 3x - k$ is a tangent to $y = x^2$.

$k =$ _____

PAST PAPERS

Q23 **PAST PAPER** Pearson IAL WMA11 • October 2019 • Q6**P1 • TOPIC 6** **8 marks**

6. The curve C has equation $y = \frac{4}{x} + k$, where k is a positive constant.
- (a) Sketch a graph of C , stating the equation of the horizontal asymptote and the coordinates of the point of intersection with the x -axis. (3)

The line with equation $y = 10 - 2x$ is a tangent to C .

- (b) Find the possible values for k . (5)

(Total 8 marks)

Q24 **PAST PAPER** Pearson IAL WMA11 · January 2022 · Q10**P1 · TOPIC 6** **9 marks**

10. The curve C has equation

$$y = \frac{1}{x^2} - 9$$

(a) Sketch the graph of C .

On your sketch

- show the coordinates of any points of intersection with the coordinate axes
- state clearly the equations of any asymptotes

(4)

The curve D has equation $y = kx^2$ where k is a constant.

Given that C meets D at 4 distinct points,

(b) find the range of possible values for k .

(5)

(Total 9 marks)

Q25 **PAST PAPER** Pearson IAL WMA11 · January 2024 · Q7**P1 · TOPIC 6** **9 marks**

7. (a) Sketch the graph of the curve C with equation

$$y = \frac{4}{x - k}$$

where k is a positive constant.

Show on your sketch

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

Given that the straight line with equation $y = 9 - x$ does not cross or touch C

- (b) find the range of values of k .

(5)

(Total for Question 7 is 9 marks)

Q26 **PAST PAPER** Pearson IAL WMA11 · June 2024 · Q8**P1 · TOPIC 6** **7 marks**

8. The curve C_1 has equation

$$y = x(4 - x^2)$$

- (a) Sketch the graph of C_1 showing the coordinates of any points of intersection with the coordinate axes.

(3)

The curve C_2 has equation $y = \frac{A}{x}$ where A is a constant.

- (b) Show that the x coordinates of the points of intersection of C_1 and C_2 satisfy the equation

$$x^4 - 4x^2 + A = 0$$

(1)

- (c) Hence find the range of possible values of A for which C_1 meets C_2 at 4 distinct points.

(3)**(Total for Question 8 is 7 marks)**

Q27**PAST PAPER**

Pearson IAL WMA11 · October 2024 · Q6

P1 · TOPIC 6 9 marks

6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Sketch the curve C with equation

$$y = \frac{1}{2-x} \quad x \neq 2$$

State on your sketch

- the equation of the vertical asymptote
- the coordinates of the intersection of C with the y -axis

(3)

The straight line l has equation $y = kx - 4$, where k is a constant.

Given that l cuts C at least once,

(b) (i) show that

$$k^2 - 5k + 4 \geq 0$$

(ii) find the range of possible values for k .

(6)

(Total for Question 6 is 9 marks)

Q28**PAST PAPER**

Pearson IAL WMA11 · January 2025 · Q7

P1 · TOPIC 6 8 marks

7. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = \frac{2}{x} - k$$

where k is a **positive** constant.

- (a) Sketch the graph of C .

Show on your sketch

- the coordinates of any points of intersection of C with the coordinate axes
- the equation of the horizontal asymptote to C

stating each in terms of k .

(3)

The line l has equation $y = -kx - 6$

Given that l intersects C at 2 distinct points,

- (b) find the range of possible values of k .

(5)

(Total for Question 7 is 8 marks)

Q29 **PAST PAPER** Pearson IAL WMA11 · October 2025 · Q6**P1 · TOPIC 6** **9 marks**

6. (a) Sketch the graph of the curve C with equation

$$y = \frac{4k}{x - 2k}$$

where k is a positive constant.

On your sketch show

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

The straight line l has equation

$$y = 6 - 2x$$

Given that there is at least one point of intersection between l and C ,

(b) find the range of possible values of k .

(5)

(Total for Question 6 is 9 marks)

Q30 **PAST PAPER** Pearson IAL WMA11 · June 2026 · Q10**P1 · TOPIC 6** **11 marks**

10. The curve C has equation

$$y = \frac{3k}{x - 2k} \quad x \neq 2k$$

where k is a positive constant.

(a) Sketch a graph of C .

Show on your sketch the coordinates of the point where C crosses a coordinate axis.

(3)

The line l has equation $x + y = 12$

Given that l is a tangent to C ,

(b) show that

$$k^2 - 15k + 36 = 0$$

(4)

The line l meets C at a point P .

Given that P lies below the x -axis,

(c) find the coordinates of P .

(Solutions relying entirely on calculator technology are not acceptable.)

(4)

Question 10 continued

(Total for Question 10 is 11 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------|---|
| 01 | Cubics | Roots from brackets, squared bracket touches, y-int at $x = 0$. |
| 02 | Reciprocals | Asymptotes: bottom = 0, and the added number. |
| 03 | Intersections | Set equal $\rightarrow$ solve $\rightarrow$ pair. Count crossings for "how many". |
| 04 | Discriminant | Tangent = 0 • miss < 0 • two points > 0. Keep $k > 0$. |

BOTTOM LINE Sketch $\rightarrow$ set equal $\rightarrow$ quadratic $\rightarrow$ discriminant decides tangent, cross or miss.

Answers

Graphs of Functions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • CUBIC SKETCHES

- Q1** Dr Eslam • Squared bracket touches cuts at $(-1, 0)$, touches at $(2, 0)$, y-int $(0, 4)$
- Q2** Practice • Easy (a) $2x(x + 3)(x - 2)$ (b) cuts at $-3, 0, 2$
- Q3** Practice • Medium touches/crosses at $(-3, 0)$ (point of inflection), y-int $(0, 27)$
- Q4** Practice • Medium touches at $(0, 0)$, cuts at $(-3, 0)$
- Q5** Pearson IAL WMA11 • October 2019 • Q10 (a) $x \leq -\frac{5}{2}$ or $x = 3$ (b) $2x^3 - 7x^2 - 12x + 45$ (c) $P(0, 45)$, gradient -12 (d) $g(x) = (2x+1)(x-5)^2$; y-int 25
- Q6** Pearson IAL WMA11 • January 2020 • Q10 (a) touches $(5, 0)$, cuts $(\frac{3}{4}, 0)$, $(0, -75)$ (b) $x = 3, 20$; $p = 75$ (c) $g(x) = (4x + 1)(x - 4)^2$; 16
- Q7** Pearson IAL WMA11 • January 2021 • Q8 (a) $x > 4$ (b) $9x^3 - 48x^2 + 52x - 16$ (c) $PQ = \frac{4}{3}\sqrt{3}$
- Q8** Pearson IAL WMA11 • October 2021 • Q6 (a) $(-1, 0)$, touches $(3, 0)$, $(0, 18)$ (b) $2x^3 - 10x^2 + 6x + 18$ (c) $y = \frac{512}{27}$ (gradient 0)
- Q9** Pearson IAL WMA11 • June 2025 • Q7 (a) $f(x) = \frac{1}{5}x^2(x - 4)$ (c) $(0, 0)$, $(6, 14.4)$, $(-8, -153.6)$

IDEA 02 • RECIPROCAL GRAPHS

- Q10** Dr Eslam • Asymptotes (a) $x = 2, y = 0$ (b) $x = 0, y = 4$
no intercepts; asymptotes $x = 0, y = 0$
- Q11** Practice • Easy asymptotes $x = 0, y = 0$; both branches above the x -axis
- Q12** Practice • Medium $k = 36: y = \frac{36}{x}$
- Q13** Practice • Medium (b) x -int $(1, 0)$, asymptote $y = k$ (c) $10 - 2\sqrt{21} < k < 10 + 2\sqrt{21}$
- Q14** Pearson IAL WMA11 • January 2021 • Q6 (b) 2 roots: the curves cross twice
- Q15** Pearson IAL WMA11 • October 2022 • Q6 (a) $x = -2$ (b) $x(x + 2)^2$ (d) 2 solutions
- Q16** Pearson IAL WMA11 • October 2023 • Q4

IDEA 03 • INTERSECTIONS

- Q17** Practice • Easy the curves cross twice (once in each of quadrants 1 and 3)
- Q18** Practice • Medium (b) 2 solutions
- Q19** Pearson IAL WMA11 • October 2019 • Q3 (a) $M\left(\frac{5}{2}, \frac{27}{4}\right)$ (b) $N\left(\frac{26}{5}, \frac{351}{25}\right)$
- Q20** Pearson IAL WMA11 • June 2021 • Q5 (a) 16.8 million (b) 53 million (c) $10 - \sqrt{58} < t < 10 + \sqrt{58}$, i.e. $2.38 < t < 15$
- Q21** Pearson IAL WMA11 • January 2022 • Q4 (a) $P(1, 5), Q\left(\frac{5}{2}, \frac{7}{2}\right)$

IDEA 04 • TANGENT, CROSS OR MISS

- Q22** Dr Eslam • Tangent $x^2 - 3x + k = 0: 9 - 4k = 0 \Rightarrow k = \frac{9}{4}$
- Q23** Pearson IAL WMA11 • October 2019 • Q6 (a) $y = k$, x-int $\left(-\frac{4}{k}, 0\right)$ (b) $k = 10 \pm 4\sqrt{2}$
- Q24** Pearson IAL WMA11 • January 2022 • Q10 (a) x-int $\pm\frac{1}{3}$, asymptotes $x = 0, y = -9$ (b) $-\frac{81}{4} < k < 0$
- Q25** Pearson IAL WMA11 • January 2024 • Q7 (a) y-int $-\frac{4}{k}$, asymptote $x = k$ (b) $5 < k < 13$
- Q26** Pearson IAL WMA11 • June 2024 • Q8 (a) $(-2, 0), (0, 0), (2, 0)$ (c) $0 < A < 4$
- Q27** Pearson IAL WMA11 • October 2024 • Q6 (a) $x = 2, (0, \frac{1}{2})$ (b)(ii) $k \leq 1$ or $k \geq 4$
- Q28** Pearson IAL WMA11 • January 2025 • Q7 (a) x-int $\frac{2}{k}$, asymptote $y = -k$ (b) $0 < k < 2$ or $k > 18$
- Q29** Pearson IAL WMA11 • October 2025 • Q6 (a) asymptote $x = 2k$, y-int -2 (b) $0 < k \leq \frac{1}{2}$ or $k \geq \frac{9}{2}$
- Q30** Pearson IAL WMA11 • June 2026 • Q10 (a) y-int $-\frac{3}{2}$ (c) $k = 12: P(18, -6)$

DR ESLAM AHMED
eliteigcse.com

Transformations of Functions

Inside the bracket moves x — the opposite way.
Outside moves y — the way you see. Move the key points, then draw.

THE TOPIC ON ONE MAP

Transformations of Functions

01 — Map a point

$f(x + 3)$ $f(3x)$ find a

02 — Sketch the transformed curve

asymptote touches the axis y -intercept

03 — Transformed curves meeting

$f(2x) = f(x) + 3$ $f(x) < 6$

$f(x) + a$	translate $\begin{pmatrix} 0 \\ a \end{pmatrix}$	$y + a$
$f(x + a)$	translate $\begin{pmatrix} -a \\ 0 \end{pmatrix}$	$x - a$ (opposite)
$af(x)$	stretch y -axis, s.f. a	$y \times a$
$f(ax)$	stretch x -axis, s.f. $\frac{1}{a}$	$x \div a$ (opposite)
$-f(x)$	reflect in the x -axis	$y \rightarrow -y$
$f(-x)$	reflect in the y -axis	$x \rightarrow -x$

01 Map a Point

Inside changes x the opposite way. Outside changes y the same way. Never both.

WHEN YOU SEE → YOU DO

“state the coordinates of the image of P ”

→ inside → x , outside → y

“ $f(x + 3)$ ”

→ $x - 3$ (opposite)

“ $f(3x)$ ”

→ $x \div 3$

“ P is mapped to P' , find a ”

→ compare the coordinates, solve

“describe fully the transformation”

→ name + vector / scale factor + direction

WORKED MODEL • OCTOBER 2025

$P\left(\frac{3\pi}{2}, 4\right)$ on $y = f(x)$.

1 $f\left(x - \frac{\pi}{6}\right): x + \frac{\pi}{6}$

inside, opposite

2 $-\frac{1}{2}f(x): y \times \left(-\frac{1}{2}\right)$

outside, same

ANSWER $\left(\frac{5\pi}{3}, 4\right), \left(\frac{3\pi}{2}, -2\right)$

IDEA 01

Practise: Map a Point

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Inside or outside?

P1 · TOPIC 7 4 marks

$P(4, -2)$ lies on $y = f(x)$. Find the image of P on (a) $y = f(x) + 5$ (b) $y = f(x - 3)$
(c) $y = 2f(x)$ (d) $y = f(2x)$

Q2 DR ESLAM Describe fully

P1 · TOPIC 7 2 marks

Describe the transformation that maps $y = x^2$ onto $y = (x + 5)^2$.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 7 3 marks

3 The point $P(2,6)$ lies on the curve with equation $y = f(x)$.

State the coordinates of the image of point P on the curves with the following equations:

(i) $y = f(x) + 1$

(ii) $y = -f(x)$

(iii) $y = f\left(\frac{1}{4}x\right)$.

(3 marks)

Q4**PRACTICE**

Practice · Medium

P1 · TOPIC 7**4 marks**

- 1 The point $P(-1, 4)$ lies on the curve with equation $y = f(x)$.

State the coordinates of the image of point P on the curves with the following equations:

(i) $y = f(x) + 3$

(ii) $y = f(x + 3)$

(iii) $y = 3f(x)$

(iv) $y = f(3x)$

(4 marks)

Q5**PRACTICE** Practice · Medium**P1 · TOPIC 7** 4 marks

3 The point $P(3,2)$ lies on the curve with equation $y = f(x)$.

(i) On the graph of $y = f(x) + a$, where a is a constant, the point P is mapped to the point $(3, -5)$. Determine the value of a .

(ii) On the graph of $y = f(x + b)$, where b is a constant, the point P is mapped to the point $(-1, 2)$. Determine the value of b .

(iii) On the graph of $y = cf(x)$, where c is a constant, the point P is mapped to the point $(3, 1)$. Determine the value of c .

(iv) On the graph of $y = f(dx)$, where d is a constant, the point P is mapped to the point $(1, 2)$. Determine the value of d .

(4 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA11 · January 2023 · Q7

P1 · TOPIC 7 10 marks

7. (a) On Diagram 1, sketch a graph of the curve
- C
- with equation

$$y = \frac{6}{x} \quad x \neq 0 \quad (2)$$

The curve C is transformed onto the curve with equation $y = \frac{6}{x-2} \quad x \neq 2$

- (b) Fully describe this transformation.
- (2)

The curve with equation

$$y = \frac{6}{x-2} \quad x \neq 2$$

and the line with equation

$$y = kx + 7 \quad \text{where } k \text{ is a constant}$$

intersect at exactly two points, P and Q .Given that the x coordinate of point P is -4

- (c) find the value of
- k
- ,
- (2)

- (d) find, using algebra, the coordinates of point
- Q
- .

(Solutions relying entirely on calculator technology are not acceptable.)

(4)

Question 7 continued

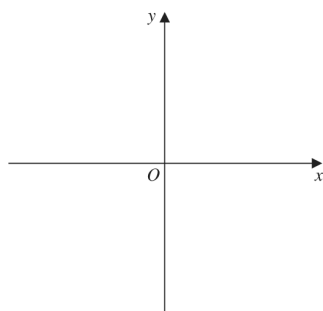
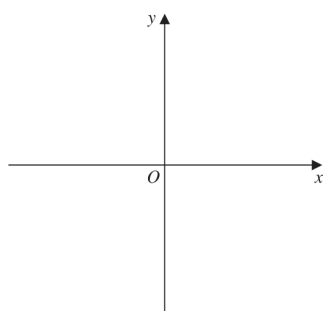


Diagram 1

Only use this copy of Diagram 1 if you need to redraw your graph.



Copy of Diagram 1

(Total for Question 7 is 10 marks)

Q7**PAST PAPER**

Pearson IAL WMA11 · October 2025 · Q9

P1 · TOPIC 7 6 marks

9.

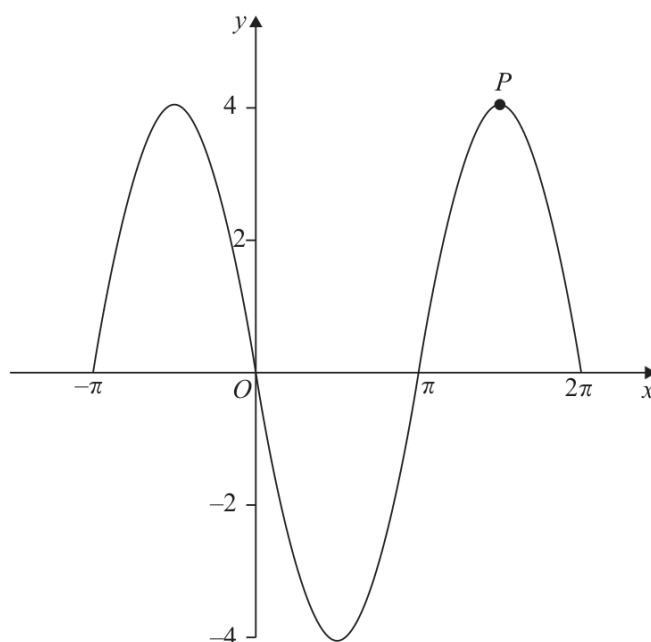
**Figure 3**

Figure 3 shows a sketch of part of the graph of the trigonometric function with equation $y = f(x)$

- (a) Write down an expression for $f(x)$

(2)

The point P lies on $y = f(x)$ and is shown in Figure 3.

- (b) State the coordinates of the point to which P is transformed when the graph of $y = f(x)$ is transformed to the graph with equation

(i) $y = f\left(x - \frac{\pi}{6}\right)$

(2)

(ii) $y = -\frac{1}{2}f(x)$

(2)**(Total for Question 9 is 6 marks)**

02 Sketch the Transformed Curve

List the key points — roots, y-intercept, turning points, asymptotes — move each one, then draw the same shape through them.

WHEN YOU SEE → YOU DO

**“on separate diagrams sketch
 $y = f(x + 3)$ and $y = f(-3x)$ ”**

→ move every key point

“state the equation of the asymptote”

→ move the asymptote too

“touches the x-axis”

→ a touching point stays a touching point

y-intercept after $f(x + a)$

→ it is a new point: $f(a)$

WORKED MODEL • JUNE 2022

Min $P(-1, 0)$, max $Q\left(\frac{3}{2}, 2\right)$, asymptote $y = 1$.

1 $f(x) - 2$: $(-1, -2)$, $\left(\frac{3}{2}, 0\right)$, $y = -1$

down 2

2 $f(-x)$: $(1, 0)$, $\left(-\frac{3}{2}, 2\right)$, $y = 1$

reflect in y-axis

ANSWER same shape, points moved

IDEA 02

Practise: Sketch the Transformed Curve

My questions first, then selected practice, then real past papers.

Q8**DR ESLAM**

Move the key points

P1 · TOPIC 7 3 marks

$y = f(x)$ has a maximum at $(1, 4)$ and an asymptote $y = 2$. State the maximum and the asymptote of $y = f(x + 2) - 1$.

SELECTED PRACTICE

Q9**PRACTICE** Practice · Easy

P1 · TOPIC 7 7 marks

6 (a) The function $f(x)$ is defined as $f(x) = (x - 2)(x - 6)$

Sketch the graph of $y = f(x)$, , showing clearly the coordinates of the points where the graph intersects the coordinate axes and the coordinates of the turning point.

(3 marks)

(b) On separate diagrams sketch the graphs of:

(i) $y = f(x - 4)$,

(ii) $y = f(-x)$.

In each case clearly show the coordinates of the points where the graph intersects the coordinate axes and the coordinates of the turning point.

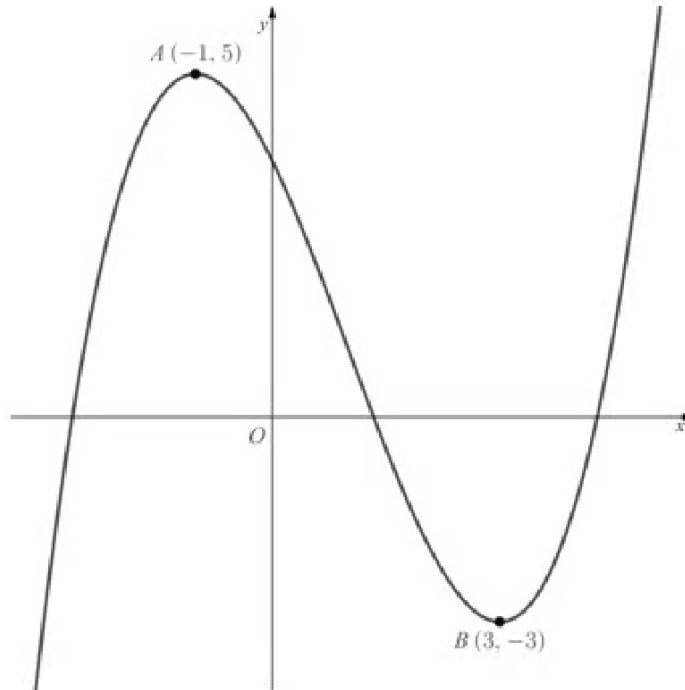
(4 marks)

Q10**PRACTICE**

Practice · Medium

P1 · TOPIC 7 6 marks

- 4 (a)** The diagram below shows the graph of $y = f(x)$. The two marked points $A(-1, 5)$ and $B(3, -3)$ lie on the graph.



In separate diagrams, sketch the curves with equation

(i) $y = f(x - 1)$

(ii) $y = f(x) + 3$

On each diagram, give the coordinates of the images of points A and B under the given transformation.

(4 marks)

- (b)** On the graph of $y = f(x + a)$ the image of one of the two marked points has an x coordinate of 2. Find the two possible values of a .

(2 marks)

PAST PAPERS

Q11 PAST PAPER Pearson IAL WMA11 · June 2022 · Q4

P1 · TOPIC 7 6 marks

4.

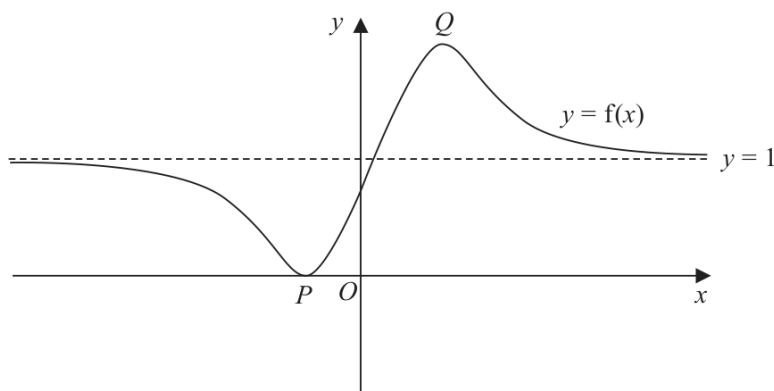


Figure 1

Figure 1 shows a sketch of a curve with equation $y = f(x)$

The curve has a minimum at $P(-1, 0)$ and a maximum at $Q\left(\frac{3}{2}, 2\right)$

The line with equation $y = 1$ is the only asymptote to the curve.

On separate diagrams sketch the curves with equation

(i) $y = f(x) - 2$ (3)

(ii) $y = f(-x)$ (3)

On each sketch you must clearly state

- the coordinates of the maximum and minimum points
- the equation of the asymptote

(Total 6 marks)

Q12**PAST PAPER**

Pearson IAL WMA11 · June 2024 · Q3

P1 · TOPIC 7 6 marks

3.

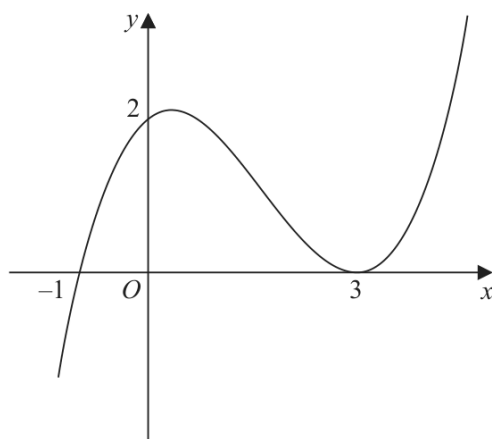
**Figure 1**

Figure 1 shows a sketch of the curve with equation $y = f(x)$.

The curve passes through the points $(-1, 0)$ and $(0, 2)$ and touches the x -axis at the point $(3, 0)$.

On separate diagrams, sketch the curve with equation

(a) $y = f(x + 3)$

(3)

(b) $y = f(-3x)$

(3)

On each diagram, show clearly the coordinates of all the points where the curve cuts or touches the coordinate axes.

(Total for Question 3 is 6 marks)

03 Transformed Curves Meeting

Two transformed curves meet where their equations are equal. Write each one as an equation in x , then solve.

WHEN YOU SEE → YOU DO

“ $y = f(2x)$ meets $y = f(x) + 3$ ”

$$\rightarrow \sqrt{2x} = \sqrt{x} + 3$$

“show that the x coordinate satisfies $\sqrt{x} = \dots$ ”

→ collect $\sqrt{x}$, factorise

“in simplest form”

→ rationalise, then square

“find the range for which $f(x) < 6$ from the graph”

→ read where the curve is below $y = 6$

WORKED MODEL • OCTOBER 2021

$f(x) = \sqrt{x}$; $y = f(2x)$ meets $y = f(x) + 3$.

$$1 \quad \sqrt{2}\sqrt{x} = \sqrt{x} + 3 \Rightarrow \sqrt{x}(\sqrt{2} - 1) = 3$$

collect, factorise

$$2 \quad \sqrt{x} = \frac{3}{\sqrt{2} - 1} = 3(\sqrt{2} + 1)$$

rationalise

ANSWER $Q(27 + 18\sqrt{2}, 6 + 3\sqrt{2})$

IDEA 03

Practise: Transformed Curves Meeting

My questions first, then real past papers.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA11 · October 2021 · Q9

P1 · TOPIC 7 9 marks

9. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

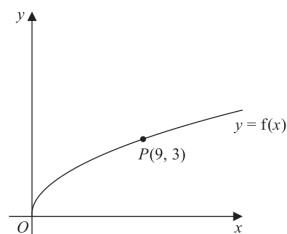


Figure 5

Figure 5 shows a sketch of the curve with equation $y = f(x)$ where

$$f(x) = \sqrt{x} \quad x > 0$$

The point $P(9, 3)$ lies on the curve and is shown in Figure 5.

On the next page there is a copy of Figure 5 called Diagram 1.

(a) On Diagram 1, sketch and clearly label the graphs of

$$y = f(2x) \quad \text{and} \quad y = f(x) + 3$$

Show on each graph the coordinates of the point to which P is transformed.

(3)

The graph of $y = f(2x)$ meets the graph of $y = f(x) + 3$ at the point Q .(b) Show that the x coordinate of Q is the solution of

$$\sqrt{x} = 3(\sqrt{2} + 1)$$

(3)

(c) Hence find, in simplest form, the coordinates of Q .

(3)

Question 9 continued

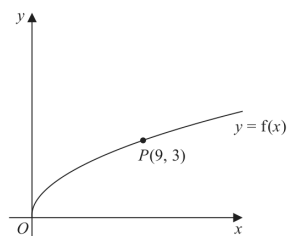
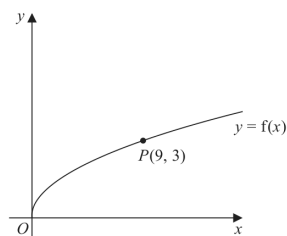


Diagram 1

Turn over for a copy of Diagram 1 if you need to redraw your graphs.

Only use this copy if you need to redraw your graphs.



Copy of Diagram 1

(Total 9 marks)

Q14 PAST PAPER Pearson IAL WMA11 · October 2022 · Q7

P1 · TOPIC 7 8 marks

7.

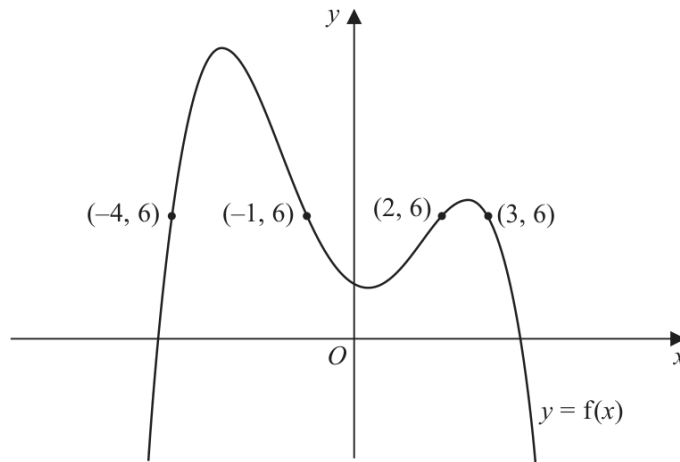
**Figure 1**

Figure 1 shows the curve with equation $y = f(x)$.

The points $P(-4, 6)$, $Q(-1, 6)$, $R(2, 6)$ and $S(3, 6)$ lie on the curve.

(a) Using Figure 1, find the range of values of x for which

$$f(x) < 6 \quad (3)$$

(b) State the largest solution of the equation

$$f(2x) = 6 \quad (1)$$

(c) (i) Sketch the curve with equation $y = f(-x)$.

On your sketch, state the coordinates of the points to which P , Q , R and S are transformed.

(ii) Hence find the set of values of x for which

$$f(-x) \geq 6 \text{ and } x < 0 \quad (4)$$

(Total for Question 7 is 8 marks)

Q15 PAST PAPER Pearson IAL WMA11 · June 2025 · Q9

P1 · TOPIC 7 10 marks

9.

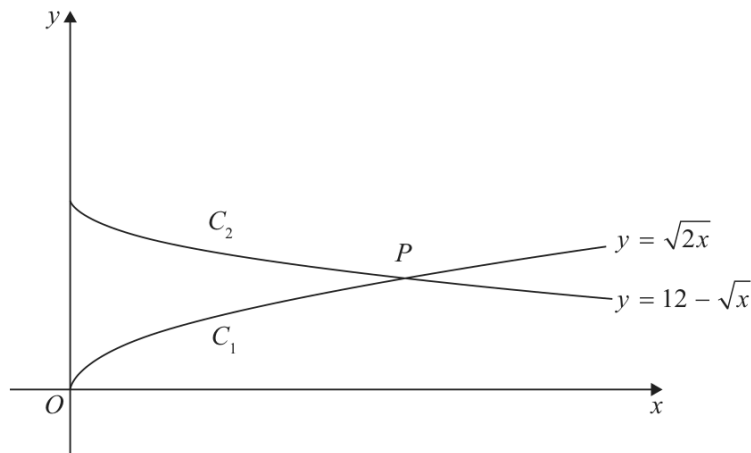


Figure 4

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Figure 4 shows a sketch of

- the graph C_1 with equation $y = \sqrt{2x}$
- the graph C_2 with equation $y = 12 - \sqrt{x}$

(a) Describe fully the single transformation that would transform

- the graph with equation $y = \sqrt{x}$ onto C_1
- the graph with equation $y = -\sqrt{x}$ onto C_2

(4)

The graphs C_1 and C_2 meet at the point P , as shown in Figure 4.

(b) (i) Show that the x coordinate of P is a solution of

$$\sqrt{x} = 12(\sqrt{2} - 1)$$

(ii) Hence find, in simplest form, the exact coordinates of P .

(6)

(Total for Question 9 is 10 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------|--|
| 01 | Map a point | Inside $\rightarrow$ x, opposite. Outside $\rightarrow$ y, same. |
| 02 | Sketch | Move every key point and asymptote; keep the shape. |
| 03 | Meeting | Write both as equations, set equal, solve exactly. |

BOTTOM LINE Inside the bracket: x, the opposite way · outside: y, the way you see $\rightarrow$ move the key points.

Answers

Transformations of Functions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • MAP A POINT

- Q1** Dr Eslam • Inside or outside? (a) (4, 3) (b) (7, -2) (c) (4, -4) (d) (2, -2)
- Q2** Dr Eslam • Describe fully translation by $\begin{pmatrix} -5 \\ 0 \end{pmatrix}$
- Q3** Practice • Easy (i) (2, 7) (ii) (2, -6) (iii) (8, 6)
- Q4** Practice • Medium (i) (-1, 7) (ii) (-4, 4) (iii) (-1, 12) (iv) $\left(-\frac{1}{3}, 4\right)$
- Q5** Practice • Medium $a = -7, b = -4, c = \frac{1}{2}, d = 3$
- Q6** Pearson IAL WMA11 • January 2023 • Q7 (b) translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ (c) $k = 2$ (d) $Q\left(\frac{5}{2}, 12\right)$
- Q7** Pearson IAL WMA11 • October 2025 • Q9 (a) $f(x) = -4 \sin x$ (b)(i) $\left(\frac{5\pi}{3}, 4\right)$ (ii) $\left(\frac{3\pi}{2}, -2\right)$

IDEA 02 • SKETCH THE TRANSFORMED CURVE

- Q8** Dr Eslam • Move the key points maximum (-1, 3), asymptote $y = 1$
- Q9** Practice • Easy (a) roots 2, 6, y-int 12, min (4, -4) (b)(i) roots 6, 10, y-int 60, min (8, -4) (ii) roots -2, -6, y-int 12, min (-4, -4)
- Q10** Practice • Medium (a)(i) A(0, 5), B(4, -3) (ii) A(-1, 8), B(3, 0) (b) $a = -3$ or $a = -1$
- Q11** Pearson IAL WMA11 • June 2022 • Q4 (i) min (-1, -2), max $\left(\frac{3}{2}, 0\right)$, $y = -1$ (ii) min (1, 0), max $\left(-\frac{3}{2}, 2\right)$, $y = 1$
- Q12** Pearson IAL WMA11 • June 2024 • Q3 (a) cuts (-4, 0), passes (-3, 2), touches (0, 0) (b) cuts $\left(\frac{1}{3}, 0\right)$, touches (-1, 0), y-int (0, 2)

IDEA 03 · TRANSFORMED CURVES MEETING

- Q13** Pearson IAL WMA11 · October 2021 · Q9 (a) $(4.5, 3)$ and $(9, 6)$ (c) $Q(27 + 18\sqrt{2}, 6 + 3\sqrt{2})$
- Q14** Pearson IAL WMA11 · October 2022 · Q7 (a) $x < -4, -1 < x < 2, x > 3$ (b) $x = \frac{3}{2}$ (c)(i) $(4, 6), (1, 6), (-2, 6), (-3, 6)$ (ii) $-3 \leq x \leq -2$
- Q15** Pearson IAL WMA11 · June 2025 · Q9 (a)(i) stretch $\parallel$ x-axis s.f. $\frac{1}{2}$ (ii) translation $\begin{pmatrix} 0 \\ 12 \end{pmatrix}$ (b)(ii) $P(432 - 288\sqrt{2}, 24 - 12\sqrt{2})$

Equation of a Straight Line

Every line question needs a gradient and a point.
Then $y - y_1 = m(x - x_1)$ — and perpendicular means
flip and change sign.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Equation of a Straight Line

01 — Gradient + point = line

$ax + by + c = 0$ through A and B

02 — Parallel and perpendicular

$m_1 m_2 = -1$ perpendicular bisector

03 — Intersections, areas, shapes

meet at P area of a triangle parallelogram

04 — Lines with curves and models

tangent parallel to l_1 $V = pt + q$

THE THREE TOOLS

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$y - y_1 = m(x - x_1)$$

$$m_1 m_2 = -1$$

01 Gradient + Point = Line

Find the gradient, pick a point, write $y - y_1 = m(x - x_1)$, then the form they ask for.

WHEN YOU SEE → YOU DO

“passes through A and B”

→ $m = \frac{\Delta y}{\Delta x}$, then one of the points

“in the form $ax + by + c = 0$ where a, b, c are integers”

→ clear fractions, everything on one side

“find the gradient of $2x - 5y + 7 = 0$ ”

→ make y the subject: $m = \frac{2}{5}$

“ $H = at + b$ ”, two data points

→ gradient = change in H over change in t

WORKED MODEL • JANUARY 2026

$A(-1, 8), B(7, 2)$ on L_1 .

1 $m = \frac{2 - 8}{7 - (-1)} = -\frac{3}{4}$

gradient

2 $y - 8 = -\frac{3}{4}(x + 1)$

point A

ANSWER $3x + 4y - 29 = 0$

IDEA 01

Practise: Gradient + Point = Line

My questions first, then real past papers.

Q1 **DR ESLAM** Two points

P1 • TOPIC 8 3 marks

Find the equation of the line through (2, 1) and (4, 7) in the form $y = mx + c$.

PAST PAPERS

Q2 **PAST PAPER** Pearson IAL WMA11 · October 2019 · Q2**P1 · TOPIC 8** **5 marks**

2. A tree was planted in the ground.

Exactly 2 years after it was planted, the height of the tree was 1.85 m.

Exactly 7 years after it was planted, the height of the tree was 3.45 m.

Given that the height, H metres, of the tree, t years after it was planted in the ground, can be modelled by the equation

$$H = at + b$$

where a and b are constants,

- (a) find the value of a and the value of b .

(4)

- (b) State, according to the model, the height of the tree when it was planted.

(1)

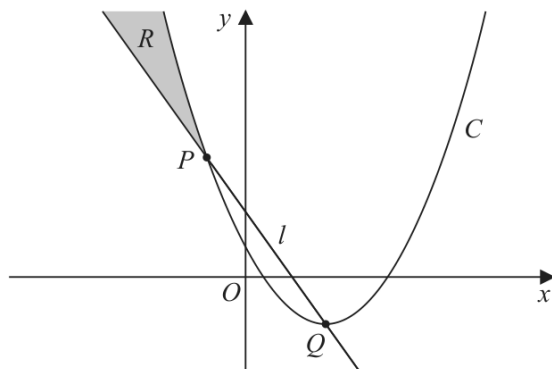
(Total 5 marks)

Q3**PAST PAPER**

Pearson IAL WMA11 · January 2021 · Q4

P1 · TOPIC 8 8 marks

4.

**Figure 2**

The points P and Q , as shown in Figure 2, have coordinates $(-2, 13)$ and $(4, -5)$ respectively.

The straight line l passes through P and Q .

- (a) Find an equation for l , writing your answer in the form $y = mx + c$, where m and c are integers to be found. (3)

The quadratic curve C passes through P and has a minimum point at Q .

- (b) Find an equation for C . (3)

The region R , shown shaded in Figure 2, lies in the second quadrant and is bounded by C and l only.

- (c) Use inequalities to define region R . (2)

(Total 8 marks)

Q4 PAST PAPER Pearson IAL WMA11 · October 2022 · Q3

P1 · TOPIC 8 5 marks

3. The share price of a company is monitored.

Exactly 3 years after monitoring began, the share price was £1.05

Exactly 5 years after monitoring began, the share price was £1.65

The share price, £ V , of the company is modelled by the equation

$$V = pt + q$$

where t is the number of years after monitoring began and p and q are constants.

- (a) Find the value of p and the value of q .

(3)

Exactly T years after monitoring began, the share price was £2.50

- (b) Find the value of T , according to the model, giving your answer to one decimal place.

(2)

(Total for Question 3 is 5 marks)

Q5**PAST PAPER**

Pearson IAL WMA11 · January 2026 · Q4

P1 · TOPIC 8 9 marks

4. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The point $A(-1, 8)$ and the point $B(7, 2)$ lie on the line L_1

- (a) Find an equation for L_1 in the form $ax + by + c = 0$, where a , b and c are integers. (3)

The line L_2

- passes through A and is perpendicular to L_1
- cuts the x -axis at the point C

- (b) Find the coordinates of C . (3)

- (c) Hence find the area of triangle ABC . (3)

(Total for Question 4 is 9 marks)

02 Parallel and Perpendicular

Parallel: same gradient. Perpendicular: flip and change the sign, $m_2 = -\frac{1}{m_1}$.

WHEN YOU SEE → YOU DO

“parallel to l_1 ”

→ same m

“perpendicular to l_1 ”

→ $m_2 = -\frac{1}{m_1}$

“prove angle $PQR = 90^\circ$ ”

→ $m_{PQ} \times m_{QR} = -1$

“perpendicular bisector”, “through the midpoint of AB ”

→ midpoint $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

“shortest distance from the line to the origin”

→ perpendicular from O , then $\sqrt{x^2 + y^2}$

WORKED MODEL • JUNE 2025

$l_1 : x - 2y + 25 = 0$; l_2 through O , perpendicular.

1 $m_1 = \frac{1}{2} \Rightarrow m_2 = -2: y = -2x$

perpendicular

2 $x - 2(-2x) + 25 = 0 \Rightarrow P(-5, 10)$

intersect

ANSWER distance $OP = \sqrt{25 + 100} = 5\sqrt{5}$

IDEA 02

Practise: Parallel and Perpendicular

My questions first, then real past papers.

Q6 DR ESLAM Perpendicular

P1 • TOPIC 8 3 marks

Find the line through (2, 1) perpendicular to $y = 3x - 5$, in the form $ax + by + c = 0$.

PAST PAPERS

Q7 PAST PAPER Pearson IAL WMA11 · January 2022 · Q8

P1 · TOPIC 8 9 marks

8. The line l_1 has equation

$$2x - 5y + 7 = 0$$

- (a) Find the gradient of l_1 (1)

Given that

- the point A has coordinates $(6, -2)$
 - the line l_2 passes through A and is perpendicular to l_1
- (b) find the equation of l_2 giving your answer in the form $y = mx + c$, where m and c are constants to be found. (3)

The lines l_1 and l_2 intersect at the point M .

- (c) Using algebra and showing all your working, find the coordinates of M .

(Solutions relying on calculator technology are not acceptable.) (3)

Given that the diagonals of a square $ABCD$ meet at M ,

- (d) find the coordinates of the point C . (2)

(Total 9 marks)

Q8 **PAST PAPER** Pearson IAL WMA11 · January 2023 · Q2**P1 · TOPIC 8** **5 marks**

2. The points P , Q and R have coordinates $(-3, 7)$, $(9, 11)$ and $(12, 2)$ respectively.

(a) Prove that angle $PQR = 90^\circ$

(3)

Given that the point S is such that $PQRS$ forms a rectangle,

(b) find the coordinates of S .

(2)

(Total for Question 2 is 5 marks)

Q9**PAST PAPER**

Pearson IAL WMA11 · October 2024 · Q1

P1 · TOPIC 8 6 marks

1. The line l_1 passes through the point $A(-5, 20)$ and the point $B(3, -4)$.

- (a) Find an equation for l_1 giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

The line l_2 is perpendicular to l_1 and passes through the midpoint of AB

- (b) Find an equation for l_2 giving your answer in the form $px + qy + r = 0$, where p , q and r are integers.

(3)**(Total for Question 1 is 6 marks)**

Q10**PAST PAPER**

Pearson IAL WMA11 · January 2025 · Q2

P1 · TOPIC 8 5 marks**2.**

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Given that

- the point A has coordinates $(-2\sqrt{3}, 5)$
 - the point B has coordinates $(7\sqrt{3}, 8)$
 - the straight line l_1 passes through A and B
- (a) show that the gradient of l_1 is $p\sqrt{3}$, where p is a rational constant to be found.
You must show each step of your working.

(2)The straight line l_2 is perpendicular to l_1 and passes through A .

- (b) Find the equation of l_2 , giving your answer in the form $y = mx + c$, where m and c are constants.

(3)**(Total for Question 2 is 5 marks)**

Q11 **PAST PAPER** Pearson IAL WMA11 · June 2025 · Q5**P1 · TOPIC 8 7 marks**

5. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The line l_1 has equation

$$x - 2y + 25 = 0$$

The line l_2 passes through the origin and is perpendicular to l_1

- (a) Find an equation for l_2 (2)

The lines l_1 and l_2 intersect at the point P .

- (b) Use algebra to find the coordinates of P . (3)

- (c) Hence find the shortest distance from l_1 to the origin.

Write your answer as a fully simplified surd. (2)

(Total for Question 5 is 7 marks)

03 Intersections, Areas, Shapes

Two lines meet: solve simultaneously. Areas: find the vertices, then base $\times$ height, or split into triangles.

WHEN YOU SEE $\rightarrow$ YOU DO

“the lines meet at P ”

$\rightarrow$ solve the two equations

“cuts the x -axis”

$\rightarrow y = 0$ · “the y -axis” $\rightarrow x = 0$

“area of triangle OPQ ”, “area of MPN ”

$\rightarrow$ sketch, base on an axis, height = the other coordinate

“ $PQRS$ is a parallelogram / rectangle / square”

$\rightarrow S = P + R - Q$ · diagonals meet at the midpoint

“both possible pairs for C ”, “length $BC = 5\sqrt{5}$ ”

$\rightarrow$ distance formula $\rightarrow$ quadratic $\rightarrow$ two answers

WORKED MODEL · JUNE 2026

$l_1 : y = 10 - 3x$; $l_2 \perp l_1$ meets it at $x = 4$.

1 $P(4, -2)$, $m_2 = \frac{1}{3}$: $x - 3y - 10 = 0$

perpendicular through P

2 $M(0, 10)$, $N(0, -\frac{10}{3})$

y -axis

ANSWER area = $\frac{1}{2} \cdot \frac{40}{3} \cdot 4 = \frac{80}{3}$

IDEA 03

Practise: Intersections, Areas, Shapes

My questions first, then real past papers.

Q12 **DR ESLAM** Where they meet

P1 · TOPIC 8 2 marks

Find where $y = 2x + 1$ meets $x + y = 7$.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA11 · January 2020 · Q6

P1 · TOPIC 8 8 marks

6. The line l_1 has equation $3x - 4y + 20 = 0$

The line l_2 cuts the x -axis at $R(8,0)$ and is parallel to l_1

- (a) Find the equation of l_2 , writing your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(3)

The line l_1 cuts the x -axis at P and the y -axis at Q .

Given that $PQRS$ is a parallelogram, find

- (b) the area of $PQRS$,

(3)

- (c) the coordinates of S .

(2)

(Total 8 marks)

Q14 PAST PAPER Pearson IAL WMA11 · October 2021 · Q5

P1 · TOPIC 8 8 marks

5. The line l_1 has equation $3y - 2x = 30$

The line l_2 passes through the point $A(24, 0)$ and is perpendicular to l_1

Lines l_1 and l_2 meet at the point P

- (a) Find, using algebra and showing your working, the coordinates of P . (5)

Given that l_1 meets the x -axis at the point B ,

- (b) find the area of triangle BPA . (3)

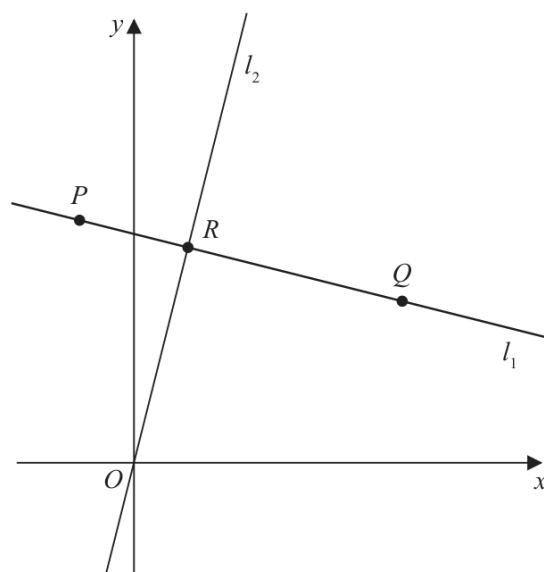
(Total 8 marks)

Q15**PAST PAPER**

Pearson IAL WMA11 · January 2024 · Q5

P1 · TOPIC 8 10 marks

5.

**Figure 1****In this question you must show all stages of your working.****Solutions relying on calculator technology are not acceptable.**

The straight line l_1 , shown in Figure 1, passes through the points $P(-2, 9)$ and $Q(10, 6)$.

- (a) Find the equation of l_1 , giving your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

The straight line l_2 passes through the origin O and is perpendicular to l_1 .

The lines l_1 and l_2 meet at the point R as shown in Figure 1.

- (b) Find the coordinates of R

(4)

- (c) Find the exact area of triangle OPQ .

(3)**(Total for Question 5 is 10 marks)**

Q16 PAST PAPER Pearson IAL WMA11 · June 2024 · Q9

P1 · TOPIC 8 9 marks

9. Given that

- the point A has coordinates $(4, 2)$
 - the point B has coordinates $(15, 7)$
 - the line l_1 passes through A and B
- (a) find an equation for l_1 , giving your answer in the form $px + qy + r = 0$ where p , q and r are integers to be found.

(3)

The line l_2 passes through A and is parallel to the x -axis.

The point C lies on l_2 so that the length of BC is $5\sqrt{5}$

- (b) Find both possible pairs of coordinates of the point C .

(4)

- (c) Hence find the minimum possible area of triangle ABC .

(2)

(Total for Question 9 is 9 marks)

Q17 **PAST PAPER** Pearson IAL WMA11 · June 2026 · Q2**P1 · TOPIC 8** **6 marks**

2. The line l_1 has equation $y = 10 - 3x$

The line l_2

- is perpendicular to l_1
- intersects l_1 at the point P with x coordinate 4

(a) Find an equation for l_2 writing the answer in the form $ax + by + c = 0$, where a , b and c are integers.

(3)

Given that

- line l_1 cuts the y -axis at the point M
- line l_2 cuts the y -axis at the point N

(b) find the area of triangle MPN , making your method clear.

(3)

Question 2 continued

(Total for Question 2 is 6 marks)

04 Lines with Curves and Models

Lines join quadratics: the turning point gives the curve, the line through two points gives the region, inequalities define it.

WHEN YOU SEE → YOU DO

“maximum point $(-2, 12)$, cuts at -5 ”

→ $f(x) = 12 - k(x + 2)^2$, the root gives k

“a line touches C and is parallel to l_1 ”

→ $\frac{dy}{dx}$ = that gradient, or discriminant = 0

“use inequalities to define R ”

→ one inequality per edge

“ $V = pt + q$... find T ”

→ two data points → p, q → substitute

WORKED MODEL • JUNE 2022

Max $(-2, 12)$, cuts at $x = -5$.

1 $f(x) = 12 - k(x + 2)^2$

turning point form

2 $0 = 12 - 9k \Rightarrow k = \frac{4}{3}$

root -5

ANSWER $f(x) = 12 - \frac{4}{3}(x + 2)^2$

IDEA 04

Practise: Lines with Curves and Models

My questions first, then real past papers.

PAST PAPERS

Q18 PAST PAPER Pearson IAL WMA11 · October 2021 · Q3

P1 · TOPIC 8 6 marks

3. (i) Solve

$$\frac{3}{x} > 4$$

(3)

(ii)

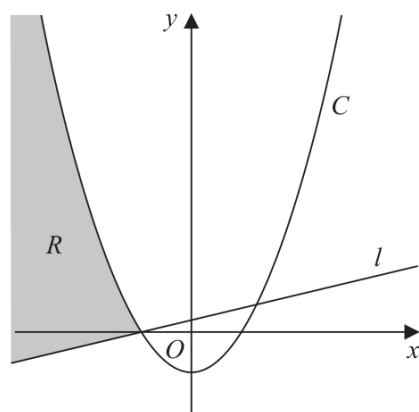


Figure 1

Figure 1 shows a sketch of the curve C and the straight line l .

The infinite region R , shown shaded in Figure 1, lies in quadrants 2 and 3 and is bounded by C and l only.

Given that

- l has a gradient of 3
- C has equation $y = 2x^2 - 50$
- C and l intersect on the negative x -axis

use inequalities to define the region R .

(3)

(Total 6 marks)

Q19**PAST PAPER**

Pearson IAL WMA11 · October 2021 · Q8

P1 · TOPIC 8 10 marks

8.

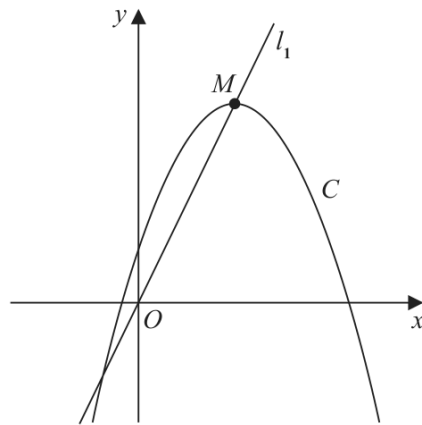
**Figure 4**

Figure 4 shows a sketch of the curve C with equation

$$y = 4 + 12x - 3x^2$$

The point M is the maximum turning point on C .

- (a) (i) Write $4 + 12x - 3x^2$ in the form

$$a + b(x + c)^2$$

where a , b and c are constants to be found.

- (ii) Hence, or otherwise, state the coordinates of M .

(5)

The line l_1 passes through O and M , as shown in Figure 4.

A line l_2 touches C and is parallel to l_1

- (b) Find an equation for l_2

(5)**(Total 10 marks)**

Q20 PAST PAPER Pearson IAL WMA11 · June 2022 · Q5

P1 · TOPIC 8 9 marks

5. The curve C has equation $y = f(x)$

Given that

- $f(x)$ is a quadratic expression
- the maximum turning point on C has coordinates $(-2, 12)$
- C cuts the negative x -axis at -5

- (a) find $f(x)$

(4)

The line l_1 has equation $y = \frac{4}{5}x$

Given that the line l_2 is perpendicular to l_1 and passes through $(-5, 0)$

- (b) find an equation for l_2 , writing your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

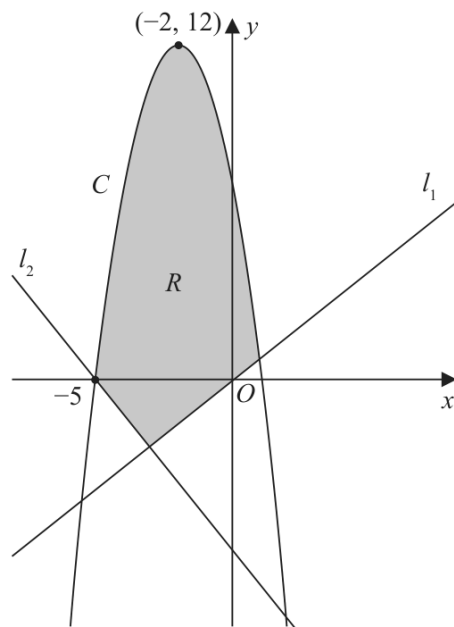


Figure 2

Figure 2 shows a sketch of the curve C and the lines l_1 and l_2

- (c) Define the region R , shown shaded in Figure 2, using inequalities.

(2)

(Total 9 marks)

10.

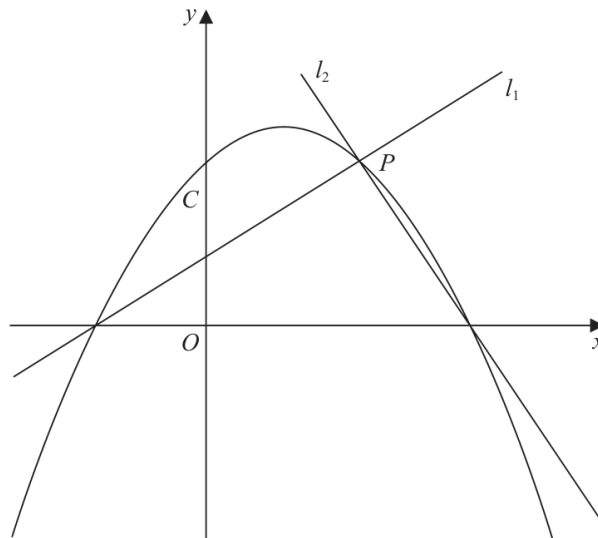


Figure 5

Figure 5 shows a sketch of the quadratic curve C with equation

$$y = -\frac{1}{4}(x+2)(x-b) \quad \text{where } b \text{ is a positive constant}$$

The line l_1 also shown in Figure 5,

- has gradient $\frac{1}{2}$
- intersects C on the negative x -axis and at the point P

(a) (i) Write down an equation for l_1

(1)

(ii) Find, in terms of b , the coordinates of P

(3)

Given that the line l_2 is perpendicular to l_1 and intersects C on the positive x -axis,

(b) find, in terms of b , an equation for l_2

(2)

Given also that l_2 intersects C at the point P

(c) show that another equation for l_2 is

$$y = -2x + \frac{5b}{2} - 4$$

(2)

(d) Hence, or otherwise, find the value of b

(Total for Question 10 is 10 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 The line** $m = \frac{\Delta y}{\Delta x}$, $y - y_1 = m(x - x_1)$, then the asked form.
- 02 $\parallel$ and $\perp$** Same $m \cdot m_2 = -\frac{1}{m_1}$. Midpoint for bisectors.
- 03 Intersections & areas** Solve together. Sketch, base on an axis. $S = P + R - Q$.
- 04 With curves** Turning point form + a root. Regions: one inequality per edge.

BOTTOM LINE Gradient + a point → the line. Perpendicular → flip and change the sign.

Answers

Equation of a Straight Line • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • GRADIENT + POINT = LINE

- Q1** Dr Eslam • Two points $m = 3: y = 3x - 5$
- Q2** Pearson IAL WMA11 • October 2019 • Q2 (a) $a = 0.32, b = 1.21$ (b) 1.21 m
- Q3** Pearson IAL WMA11 • January 2021 • Q4 (a) $y = 7 - 3x$ (b) $y = \frac{1}{2}(x - 4)^2 - 5$ (c) $y \geq 7 - 3x, y \leq \frac{1}{2}(x - 4)^2 - 5, x \leq 0$
- Q4** Pearson IAL WMA11 • October 2022 • Q3 (a) $p = 0.3, q = 0.15$ (b) $T = 7.8$
- Q5** Pearson IAL WMA11 • January 2026 • Q4 (a) $3x + 4y - 29 = 0$ (b) $C(-7, 0)$ (c) 50

IDEA 02 • PARALLEL AND PERPENDICULAR

- Q6** Dr Eslam • Perpendicular $m = -\frac{1}{3}: x + 3y - 5 = 0$
- Q7** Pearson IAL WMA11 • January 2022 • Q8 (a) $\frac{2}{5}$ (b) $y = -\frac{5}{2}x + 13$ (c) $M(4, 3)$ (d) $C(2, 8)$
- Q8** Pearson IAL WMA11 • January 2023 • Q2 (a) $m_{PQ} = \frac{1}{3}, m_{QR} = -3$, product -1 (b) $S(0, -2)$
- Q9** Pearson IAL WMA11 • October 2024 • Q1 (a) $y = -\frac{24}{5}x - 4$ (b) $5x - 24y + 187 = 0$
- Q10** Pearson IAL WMA11 • January 2025 • Q2 (a) $p = \frac{1}{9}$ (b) $y = -3\sqrt{3}x - 13$
- Q11** Pearson IAL WMA11 • June 2025 • Q5 (a) $y = -2x$ (b) $P(-5, 10)$ (c) $5\sqrt{5}$

IDEA 03 · INTERSECTIONS, AREAS, SHAPES

- Q12** Dr Eslam · Where they meet (2, 5)
- Q13** Pearson IAL WMA11 · January 2020 · Q6 (a) $3x - 4y - 24 = 0$ (b) $\frac{220}{3}$ (c) $S\left(\frac{4}{3}, -5\right)$
- Q14** Pearson IAL WMA11 · October 2021 · Q5 (a) $P(12, 18)$ (b) 351
- Q15** Pearson IAL WMA11 · January 2024 · Q5 (a) $y = -\frac{1}{4}x + \frac{17}{2}$ (b) $R(2, 8)$ (c) 51
- Q16** Pearson IAL WMA11 · June 2024 · Q9 (a) $5x - 11y + 2 = 0$ (b) $(25, 2), (5, 2)$ (c) $\frac{5}{2}$
- Q17** Pearson IAL WMA11 · June 2026 · Q2 (a) $x - 3y - 10 = 0$ (b) $\frac{80}{3}$

IDEA 04 · LINES WITH CURVES AND MODELS

- Q18** Pearson IAL WMA11 · October 2021 · Q3 (i) $0 < x < \frac{3}{4}$ (ii) $3x + 15 \leq y \leq 2x^2 - 50, x \leq -5$
- Q19** Pearson IAL WMA11 · October 2021 · Q8 (a) $16 - 3(x - 2)^2; M(2, 16)$ (b) $y = 8x + \frac{16}{3}$
- Q20** Pearson IAL WMA11 · June 2022 · Q5 (a) $f(x) = 12 - \frac{4}{3}(x + 2)^2$ (b) $y = -\frac{5}{4}x - \frac{25}{4}$
- Q21** Pearson IAL WMA11 · June 2023 · Q10 (a)(i) $y = \frac{1}{2}x + 1$ (ii) $P\left(b - 2, \frac{b}{2}\right)$ (b) $y = -2x + 2b$
(d) $b = 8$

DR ESLAM AHMED
eliteigcse.com

Basic Trigonometry

Any triangle, three rules. Look at what you are given — it tells you which rule to use.

THE TOPIC ON ONE MAP

Basic Trigonometry

01 — Sine rule and the ambiguous case

an opposite pair

obtuse: $180^\circ - \theta$

two triangles

02 — Cosine rule

two sides + angle

three sides

03 — Area $\frac{1}{2}ab \sin C$

area given

work backwards

04 — Exact values, multi-step shapes

no calculator

shared side

Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

a side-angle **pair** is known**Cosine rule**

$$a^2 = b^2 + c^2 - 2bc \cos A$$

two sides & the angle between • or three sides

Area

$$\frac{1}{2}ab \sin C$$

two sides & the angle between

01 Sine Rule and the Ambiguous Case

Use it when you know one side **and its opposite angle**. For an angle, $\sin$ gives two answers: α and $180^\circ - \alpha$.

WHEN YOU SEE → YOU DO

“use the sine rule”, a side and its opposite angle

$$\rightarrow \frac{a}{\sin A} = \frac{b}{\sin B}$$

“angle is obtuse”, “the longest side”, “ $\theta > 90^\circ$ ”

$$\rightarrow \text{take } 180^\circ - \alpha$$

“two triangles”, “the other possible triangle”

$$\rightarrow \text{both } \alpha \text{ and } 180^\circ - \alpha$$

“to 4 decimal places” then “hence”

$$\rightarrow \text{keep the exact sine, then invert}$$

WORKED MODEL • JUNE 2022

$AB = 21$, $BC = 13$, $\angle BAC = 25^\circ$, $\angle ACB = x^\circ$; AB is the longest side.

$$1 \quad \frac{\sin x}{21} = \frac{\sin 25^\circ}{13} \Rightarrow \sin x = 0.6827$$

sine rule

$$2 \quad x = 43.05^\circ \text{ or } 136.95^\circ$$

two answers

ANSWER $x = 136.95$ (longest side $\Rightarrow$ largest angle)

IDEA 01

Practise: Sine Rule and the Ambiguous Case

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Side from the sine rule

P1 • TOPIC 9 2 marks

In triangle ABC , $BC = 10$ cm, $\angle A = 70^\circ$, $\angle B = 40^\circ$. Find AC .

AC = _____

Q2 DR ESLAM Two possible angles

P1 • TOPIC 9 3 marks

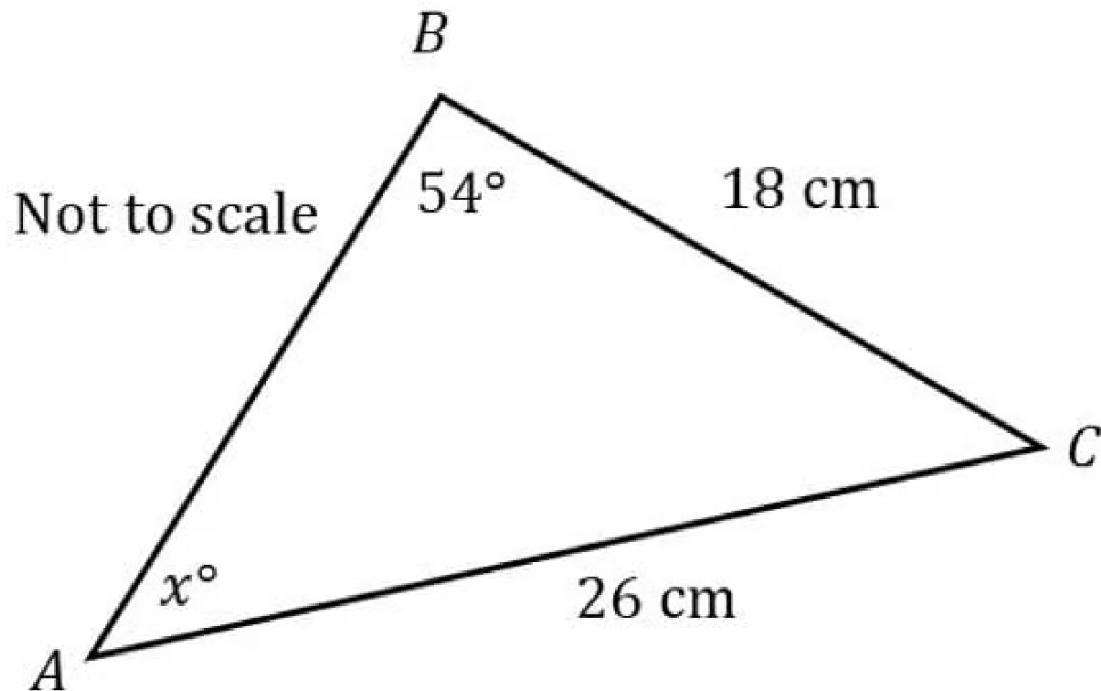
$AB = 8$, $BC = 6$, $\angle A = 40^\circ$. Find the two possible values of $\angle C$.

SELECTED PRACTICE

Q3 PRACTICE Practice · Easy

P1 · TOPIC 9 3 marks

6



x is an acute angle. Use the sine rule to calculate the angle x° . Give your answer to the nearest degree.

(3 marks)

Q4**PRACTICE**

Practice • Medium

P1 • TOPIC 9**4 marks**

5 Two triangles both have sides $AB = 5.3$ cm, $BC = 6.4$ cm and angle $ACB = 38^\circ$.

(i) Show that the angle BAC for one of the triangles is 132° , to 3 s.f.

(ii) Find the angle ABC for the other triangle.

(4 marks)**Q5****PRACTICE**

Practice • Hard

P1 • TOPIC 9**4 marks**

5 Two triangles both have sides $AB = 2.8$ cm, $BC = 3.5$ cm and angle $ACB = 43^\circ$.

Find the largest angle in each of the triangles.

(4 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA11 · June 2022 · Q2

P1 · TOPIC 9 5 marks

2. In the triangle ABC ,

- $AB = 21$ cm
- $BC = 13$ cm
- angle $BAC = 25^\circ$
- angle $ACB = x^\circ$

(a) Use the sine rule to find the value of $\sin x^\circ$, giving your answer to 4 decimal places.
(2)

Given also that AB is the longest side of the triangle,

(b) find the value of x , giving your answer to 2 decimal places.
(3)

(Total 5 marks)

02 Cosine Rule

Two sides and the angle between them → the third side. Three sides → any angle (the largest is opposite the longest side).

WHEN YOU SEE → YOU DO

“find the length of BD ”, two sides and the angle between

$$\longrightarrow a^2 = b^2 + c^2 - 2bc \cos A$$

three sides, “find the largest angle”

→ angle opposite the longest side

“sides in the ratio 2:4:5”

→ use 2, 4, 5 for the angle

“show that $p^2 + q^2 - pq = 8$ ”

$$\longrightarrow \text{cosine rule with } \cos 60^\circ = \frac{1}{2}$$

bearings

→ draw north lines, find the angle inside the triangle

WORKED MODEL • OCTOBER 2025

Sides in the ratio 2 : 4 : 5; area 140 cm^2 .

$$1 \quad \cos \theta = \frac{4 + 16 - 25}{2(2)(4)} = -\frac{5}{16}$$

largest angle

$$2 \quad \theta = 108.2^\circ$$

invert

$$3 \quad \frac{1}{2}(2k)(4k) \sin 108.2^\circ = 140 \Rightarrow k = 6.07$$

area with $2k, 4k$

ANSWER 108.2° , shortest side 12.1 cm

IDEA 02

Practise: Cosine Rule

My questions first, then selected practice, then real past papers.

Q7 DR ESLAM Cosine rule, exact

P1 · TOPIC 9 2 marks

$AB = 8$, $AC = 5$, $\angle A = 60^\circ$. Find BC exactly.

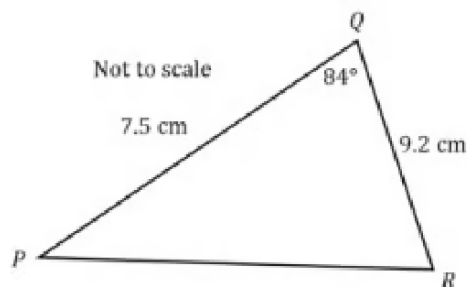
$BC =$ _____

SELECTED PRACTICE**Q8** PRACTICE Practice · Easy

P1 · TOPIC 9 3 marks

- 5 PQR is a triangle with angle $PQR = 84^\circ$ and side lengths $PQ = 7.5$ cm and $QR = 9.2$ cm.

Use cosine rule to calculate the length of PR .



(3 marks)

Q9**PRACTICE**

Practice · Medium

P1 · TOPIC 9 4 marks

- 1 (a) A triangle ABC has side lengths $AB = 5$ cm, $BC = 4$ cm and $AC = 7$ cm.

Calculate the size of the angle BAC , giving your answer to the nearest degree.

(2 marks)

- (b) A triangle PQR has side lengths $PQ = 12$ cm, $QR = 8$ cm and angle $PQR = 60^\circ$.

Calculate the length of RP , giving your answer to three significant figures.

(2 marks)**Q10****PRACTICE**

Practice · Medium

P1 · TOPIC 9 3 marks

- 7 Three telephone masts are used to triangulate locations. Mast B is due west of Mast A . Mast C is 0.6 km from Mast A and 0.7 km from Mast B on a bearing of 136° .

Show that the bearing of Mast C from Mast A is 213° .

(3 marks)

PAST PAPERS

Q11 PAST PAPER Pearson IAL WMA11 · October 2019 · Q4

P1 · TOPIC 9 5 marks

4. A parallelogram $ABCD$ has area 40 cm^2

Given that AB has length 10 cm , BC has length 6 cm and angle DAB is obtuse, find

(a) the size of angle DAB , in degrees, to 2 decimal places, (3)

(b) the length of diagonal BD , in cm , to one decimal place. (2)

(Total 5 marks)

Q12**PAST PAPER**

Pearson IAL WMA11 · October 2025 · Q3

P1 · TOPIC 9 5 marks

3. The sides of a triangle are in the ratio 2:4:5

(a) Find the size of the **largest** angle of the triangle, in degrees, to one decimal place.

(2)

Given that the area of the triangle is 140cm^2

(b) find the length of the **shortest** side, in cm, to one decimal place.

(3)

(Total for Question 3 is 5 marks)

03 Area $\frac{1}{2}ab \sin C$ — and Working Backwards

Area with the angle **between** the two sides. Given the area, the same formula finds $\sin C$ — which has two angles.

WHEN YOU SEE → YOU DO

“the area is 100 cm^2 , find $\sin \theta$ ”

$$\rightarrow \frac{1}{2}ab \sin \theta = 100$$

“given $\theta > 90^\circ$ ” after $\sin \theta$

$\rightarrow \cos \theta$ is negative in the cosine rule

parallelogram

$\rightarrow \text{area} = ab \sin \theta$ (two triangles)

“show that one possible value is 48.1° ”

$\rightarrow$ the other is $180^\circ - 48.1^\circ$

WORKED MODEL • JANUARY 2024

$AB = 15$, $AC = 25$, area 100 , $\theta > 90^\circ$.

$$1 \quad \frac{1}{2}(15)(25) \sin \theta = 100 \Rightarrow \sin \theta = \frac{8}{15}$$

area formula

$$2 \quad \cos \theta = -\sqrt{1 - \frac{64}{225}}$$

obtuse: negative

$$3 \quad BC^2 = 225 + 625 + 2(15)(25)(0.8433)$$

cosine rule

ANSWER $BC = 38.5 \text{ cm}$

IDEA 03

Practise: Area and Working Backwards

My questions first, then selected practice, then real past papers.

Q13 DR ESAM Area

P1 • TOPIC 9 2 marks

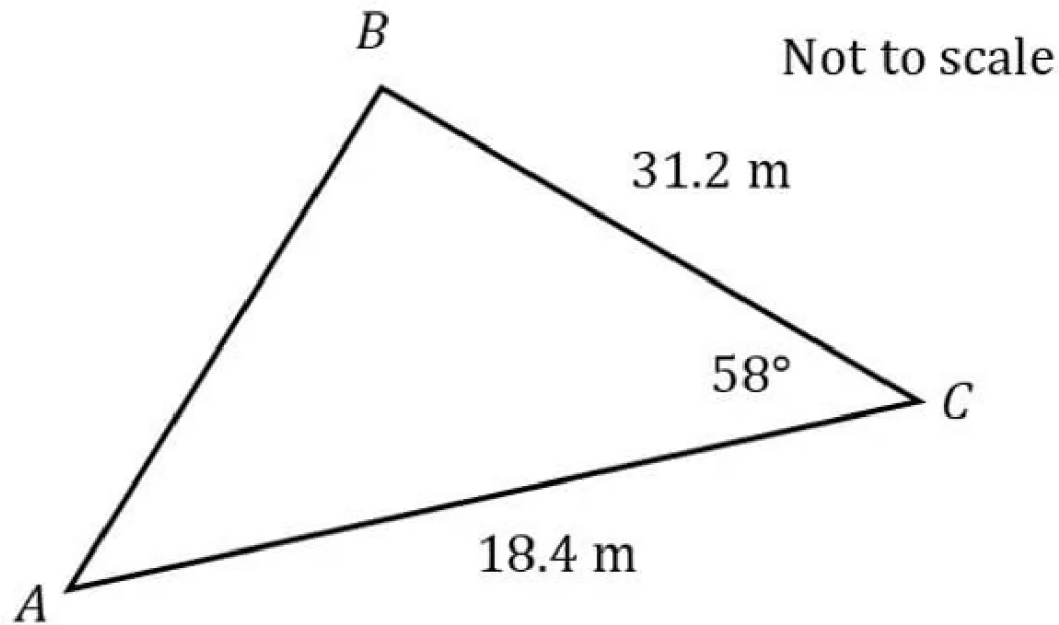
Find the area of a triangle with sides 6 cm and 9 cm and included angle 150° .

SELECTED PRACTICE

Q14 PRACTICE Practice · Easy

P1 · TOPIC 9 3 marks

7 A triangular field is shown in the diagram below. Calculate the area of the field, give your answer to the nearest square metre.



(3 marks)

Q15**PRACTICE**

Practice · Easy

P1 · TOPIC 9 3 marks

- 9 The area of a triangle WXY is 75 cm^2 . $XY = 14.3 \text{ cm}$ and angle $WXY = 104^\circ$.

Using the formula $A = \frac{1}{2} ab \sin C$, calculate the length of WX . Give your answer to three significant figures.

(3 marks)

Q16**PRACTICE**

Practice · Medium

P1 · TOPIC 9 8 marks

- 8 (a)** A triangular field has an area of 10m^2 . The field has two side lengths of 3.2 m and 8.4 m which have an angle θ between them.

Show that one possible value for θ is 48.1° to three significant figures.

(3 marks)

- (b)** Show that there is another possible value that works for θ . You must show all your working.

(2 marks)

- (c)** Using $\theta = 48.1^\circ$. Calculate the perimeter of the field. Giving your answer to three significant figures.

(3 marks)

PAST PAPERS

Q17 PAST PAPER Pearson IAL WMA11 · January 2024 · Q2

P1 · TOPIC 9 5 marks

2. The triangle ABC is such that

- $AB = 15$ cm
- $AC = 25$ cm
- angle $BAC = \theta^\circ$
- area triangle $ABC = 100$ cm²

(a) Find the value of $\sin \theta^\circ$

(2)

Given that $\theta > 90$ (b) find the length of BC , in cm, to 3 significant figures.

(3)

(Total for Question 2 is 5 marks)

Q18 **PAST PAPER** Pearson IAL WMA11 · June 2026 · Q3**P1 · TOPIC 9** **6 marks**

3. A triangular plot of land, ABC , has area 60 m^2

Given

- $AB = 15 \text{ m}$
- $AC = 10 \text{ m}$
- angle $BAC = \theta^\circ$

(a) show that $\sin \theta^\circ = \frac{4}{5}$ (1)

Given also that θ° is the **largest** angle in the triangle,

(b) find the length of the side BC , in metres, to 3 significant figures, (3)

(c) find the size of the **smallest** angle in the triangle, in degrees, to one decimal place. (2)

Question 3 continued

(Total for Question 3 is 6 marks)

04 Exact Values and Multi-Step Shapes

“Show all stages”, “exact” — use $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$ and keep surds. Frames and plots join two triangles along a shared side.

WHEN YOU SEE → YOU DO

“solutions relying on calculator ... not acceptable”

→ exact values, algebra

“AC is 2 metres longer than AB”

→ $q = p + 2$, substitute

two triangles sharing a side

→ solve the first, carry the shared side

	30°	45°	60°
sin	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
cos	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$

WORKED MODEL • JUNE 2021

$$BC = 2\sqrt{2}, \angle A = 60^\circ, AB = p, AC = q = p + 2.$$

$$1 \quad 8 = p^2 + q^2 - 2pq \cdot \frac{1}{2}$$

cosine rule, exact

$$2 \quad p^2 + 2p - 4 = 0 \Rightarrow p = -1 + \sqrt{5}$$

substitute $q = p + 2$

$$\text{ANSWER} \quad p = \sqrt{5} - 1, q = \sqrt{5} + 1, \text{ area} = \frac{1}{2}pq \cdot \frac{\sqrt{3}}{2} = \sqrt{3}$$

IDEA 04

Practise: Exact Values and Multi-Step Shapes

My questions first, then real past papers.

Q19**DR ESLAM**

Exact and algebraic

P1 · TOPIC 9

4 marks

$AB = x$, $AC = x + 1$, $BC = \sqrt{7}$, $\angle A = 60^\circ$. Find x .

$x =$ _____

PAST PAPERS

Q20 PAST PAPER Pearson IAL WMA11 · June 2021 · Q3

P1 · TOPIC 9 9 marks

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

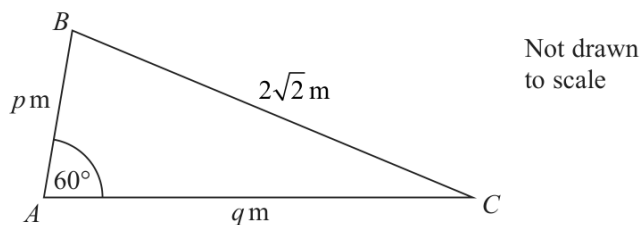


Figure 1

Figure 1 shows the plan view of a flower bed.
The flowerbed is in the shape of a triangle ABC with

- $AB = p$ metres
- $AC = q$ metres
- $BC = 2\sqrt{2}$ metres
- angle $BAC = 60^\circ$

(a) Show that

$$p^2 + q^2 - pq = 8 \quad (2)$$

Given that side AC is 2 metres longer than side AB , use algebra to find

- (b) (i) the exact value of p ,
- (ii) the exact value of q . (5)

Using the answers to part (b),

- (c) calculate the exact area of the flower bed. (2)

(Total 9 marks)

Q21 PAST PAPER Pearson IAL WMA11 • October 2023 • Q5

P1 • TOPIC 9 7 marks

5.

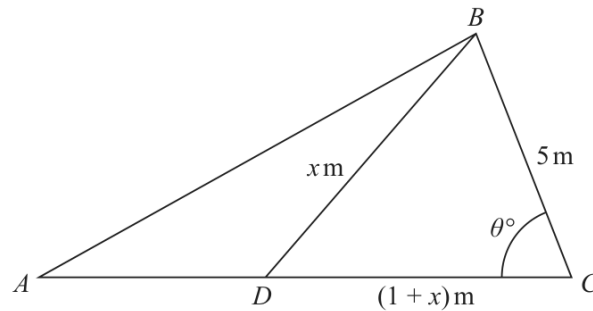
Diagram NO
accurately dr**Figure 2**

Figure 2 shows the plan view of a frame for a flat roof.

The shape of the frame consists of triangle ABD joined to triangle BCD .

Given that

- $BD = x$ m
- $CD = (1 + x)$ m
- $BC = 5$ m
- angle $BCD = \theta^\circ$

(a) show that $\cos \theta^\circ = \frac{13 + x}{5 + 5x}$

(2)

Given also that

- $x = 2\sqrt{3}$
- angle $BAC = 30^\circ$
- ADC is a straight line

(b) find the area of triangle ABC , giving your answer, in m^2 , to one decimal place.

(5)

(Total for Question 5 is 7 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------------|---|
| 01 | Sine rule | A side with its opposite angle. Angle: α or $180^\circ - \alpha$. |
| 02 | Cosine rule | Two sides + included angle, or three sides.
Largest angle opposite longest side. |
| 03 | Area | $\frac{1}{2}ab \sin C$. Given area $\rightarrow \sin C \rightarrow$ two possible angles. |
| 04 | Exact step / multi- | $\sin 60 = \frac{\sqrt{3}}{2}, \cos 60 = \frac{1}{2}$. Shared side carries across. |

BOTTOM LINE Pair known $\rightarrow$ sine rule • included angle or three sides $\rightarrow$ cosine rule • always check the obtuse answer.

Answers

Basic Trigonometry • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • SINE RULE AND THE AMBIGUOUS CASE

- Q1** Dr Eslam • Side from the sine rule

$$\frac{10 \sin 40^\circ}{\sin 70^\circ} = 6.84 \text{ cm}$$
- Q2** Dr Eslam • Two possible angles

$$\sin C = \frac{8 \sin 40^\circ}{6} = 0.857: C = 59.0^\circ \text{ or } 121.0^\circ$$
- Q3** Practice • Easy

$$x = 34^\circ$$
- Q4** Practice • Medium
 (i) $\sin A = \frac{6.4 \sin 38^\circ}{5.3}: 132^\circ$ (ii) $\angle ABC = 94.0^\circ$
- Q5** Practice • Hard

$$\sin A = \frac{3.5 \sin 43^\circ}{2.8} = 0.853: A = 58.5^\circ \text{ or } 121.5^\circ;$$

 largest angles 78.5° and 121.5°
- Q6** Pearson IAL WMA11 • June 2022 • Q2
 (a) 0.6827 (b) $x = 136.95$

IDEA 02 • COSINE RULE

- Q7** Dr Eslam • Cosine rule, exact

$$\sqrt{64 + 25 - 40} = 7$$
- Q8** Practice • Easy

$$PR = 11.2 \text{ cm}$$
- Q9** Practice • Medium
 (a) 34° (b) 10.6 cm
- Q10** Practice • Medium
 bearing 213° (shown)
- Q11** Pearson IAL WMA11 • October 2019 • Q4
 (a) 138.19° (b) 15.0 cm
- Q12** Pearson IAL WMA11 • October 2025 • Q3
 (a) 108.2° (b) 12.1 cm

IDEA 03 • AREA AND WORKING BACKWARDS

- Q13** Dr Eslam • Area 13.5 cm^2
- Q14** Practice • Easy 243 m^2
- Q15** Practice • Easy $WX = 10.8 \text{ cm}$
- Q16** Practice • Medium (b) 131.9° also works (c) 18.3 m
- Q17** Pearson IAL WMA11 • January 2024 • Q2 (a) $\frac{8}{15}$ (b) 38.5 cm
- Q18** Pearson IAL WMA11 • June 2026 • Q3 (b) 22.5 m (c) 20.9°

IDEA 04 • EXACT VALUES AND MULTI-STEP SHAPES

- Q19** Dr Eslam • Exact and algebraic $7 = x^2 + (x+1)^2 - x(x+1) \Rightarrow x^2 + x - 6 = 0: x = 2$
- Q20** Pearson IAL WMA11 • June 2021 • Q3 (b) $p = \sqrt{5} - 1, q = \sqrt{5} + 1$ (c) $\sqrt{3} \text{ m}^2$
- Q21** Pearson IAL WMA11 • October 2023 • Q5 (b) 16.1 m^2

DR ESLAM AHMED
eliteigcse.com

Radian Measure

Two formulas, $s = r\theta$ and $A = \frac{1}{2}r^2\theta$. Every question is a shape built from sectors and triangles — break it into pieces you know.

THE TOPIC ON ONE MAP

Radian Measure

01 — Radians, arcs and sectors

$s = r\theta$

$A = \frac{1}{2}r^2\theta$

in terms of π

02 — Triangles and segments

segment = sector – triangle

obtuse angle

03 — Composite shapes

total area

perimeter

04 — Equations in r and θ

area and perimeter given

quadratic in r

ONLY IN RADIANS

$$\text{arc } s = r\theta \quad \text{sector } A = \frac{1}{2}r^2\theta \quad \text{triangle}$$

$$\frac{1}{2}ab \sin C$$

$\pi \text{ rad} = 180^\circ$ • calculator in RAD mode

01 Radians, Arcs and Sectors

$\pi = 180^\circ$. With θ in radians, the arc and the sector are one multiplication each.

WHEN YOU SEE → YOU DO

“exact value”, “in terms of π ”	→ keep π and surds
“perimeter of the sector”	→ $2r + r\theta$ — two radii plus the arc
“area of the sector is 15, find r ”	→ $\frac{1}{2}r^2\theta = 15$
a major (reflex) sector	→ angle = $2\pi - \theta$

WORKED MODEL • OCTOBER 2019

Sector, $\theta = 1.25$, area 15 cm^2 .

1 $\frac{1}{2}r^2(1.25) = 15 \Rightarrow r^2 = 24 \Rightarrow r = 2\sqrt{6}$

area formula

2 $P = 2r + r\theta = 4\sqrt{6} + 2.5\sqrt{6}$

perimeter

ANSWER $r = 2\sqrt{6}$, $P = \frac{13}{2}\sqrt{6}$

× **TRAP** Calculator in DEG mode gives wrong sines. Check: $\sin 1.2 = 0.932$ in RAD.

IDEA 01

Practise: Radians, Arcs and Sectors

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Arc and sector

P1 · TOPIC 10 3 marks

A sector has radius 6 cm and angle $\frac{2\pi}{3}$. Find, in terms of π , its arc length, area and perimeter.

Q2 DR ESLAM Find θ

P1 · TOPIC 10 2 marks

A sector of radius 5 cm has area 20 cm^2 . Find the angle in radians.

 $\theta =$ _____

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 10

5 marks

1 Without using a calculator, write down:

(i) 60° *in radians*,

(ii) $\sin 45^\circ$

(iii) $\tan \pi$

(iv) $\frac{5\pi}{6}$ *radians in degrees*

(iv) $\cos 0$

(5 marks)

Q4**PRACTICE**

Practice · Easy

P1 · TOPIC 10**4 marks**

- 2 A sector of a circle, OPQ , is such that it has radius 4 cm and the angle at its centre, O , is $\frac{\pi}{4}$ radians.

(i) Find the length of the arc PQ .

(ii) Find the area of the sector OPQ .

(4 marks)

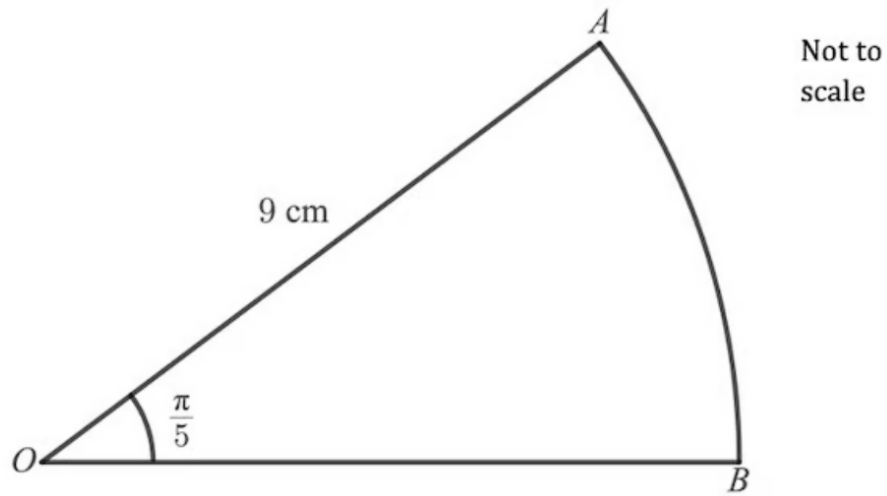
Q5**PRACTICE**

Practice · Medium

P1 · TOPIC 10

5 marks

3 The sector of a circle is shown below.



Find the area and the perimeter of the sector.

(5 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA11 · October 2019 · Q1

P1 · TOPIC 10 5 marks

1.

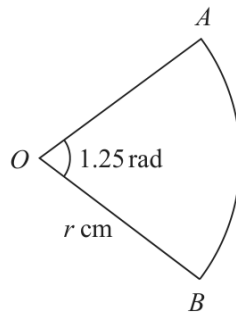
**Figure 1**

Figure 1 shows a sector AOB of a circle with centre O and radius r cm.

The angle AOB is 1.25 radians.

Given that the area of the sector AOB is 15 cm^2

- (a) find the exact value of r , (2)
- (b) find the exact length of the perimeter of the sector. Write your answer in simplest form. (3)

(Total 5 marks)

Q7**PAST PAPER**

Pearson IAL WMA11 · June 2022 · Q8

P1 · TOPIC 10 10 marks

8.

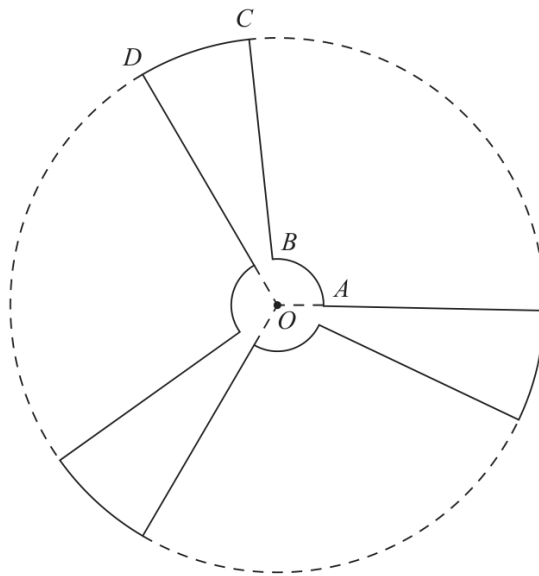
**Figure 3**

Figure 3 shows a sketch of the outline of the face of a ceiling fan viewed from below.

The fan consists of three identical sections congruent to $OABCD$, shown in Figure 3, where

- $OABO$ is a sector of a circle with centre O and radius 9 cm
- $OBCDO$ is a sector of a circle with centre O and radius 84 cm
- angle $AOD = \frac{2\pi}{3}$ radians

Given that the length of the arc AB is 15 cm,

- (a) show that the length of the arc CD is 35.9 cm to one decimal place. (3)

The face of the fan is modelled to be a flat surface.

Find, according to the model,

- (b) the perimeter of the face of the fan, giving your answer to the nearest cm, (2)
- (c) the surface area of the face of the fan.

Give your answer to 3 significant figures and make your units clear. (5)

(Total 10 marks)

Q8 PAST PAPER Pearson IAL WMA11 · October 2024 · Q5

P1 · TOPIC 10 9 marks

5. A plot of land OAB is in the shape of a sector of a circle with centre O .

Given

- $OA = OB = 5 \text{ km}$
- angle $AOB = 1.2 \text{ radians}$

(a) find the perimeter of the plot of land.

(2)

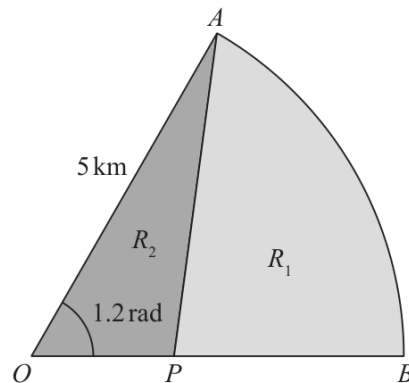


Figure 2

Diagram **NO**
accurately dr

A point P lies on OB such that the line AP divides the plot of land into two regions R_1 and R_2 as shown in Figure 2.

Given that

$$\text{area of } R_1 = 3 \times \text{area of } R_2$$

(b) show that the area of $R_2 = 3.75 \text{ km}^2$

(3)

(c) Find the length of AP , giving your answer to the nearest 100 m.

(4)

(Total for Question 5 is 9 marks)

02 Triangles and Segments

Most shapes add or subtract a triangle. Find its missing side or angle with the sine or cosine rule.

WHEN YOU SEE → YOU DO

“segment”, “shaded region between chord and arc”

$$\longrightarrow \text{sector} - \text{triangle} = \frac{1}{2}r^2(\theta - \sin \theta)$$

two sides and the angle between

$$\longrightarrow \text{cosine rule for the third side} \cdot \text{area } \frac{1}{2}ab \sin C$$

a side opposite a known angle

$$\longrightarrow \text{sine rule}$$

“angle OAB is obtuse”, “ $> \frac{\pi}{2}$ ”

$$\longrightarrow \text{sine rule gives the acute one: use } \pi - \alpha$$

“show that angle AOB is 1.231 radians”

$$\longrightarrow \text{give 4 s.f. or more before rounding}$$

WORKED MODEL • JANUARY 2022

Semicircle $r = 3$, $\angle BOA = 0.7$, $BD \parallel AE$.

$$1 \quad \angle BOD = \pi - 2(0.7) = 1.742$$

symmetry

$$2 \quad \text{segment} = \frac{1}{2}(9)(1.742) - \frac{1}{2}(9) \sin 1.742$$

sector - triangle

$$3 \quad BD^2 = 9 + 9 - 18 \cos 1.742 \Rightarrow BD = 4.59$$

cosine rule

$$\text{ANSWER} \quad \text{area } 3.4 \text{ m}^2, \text{ perimeter } 3(1.742) + 4.59 = 9.8 \text{ m}$$

IDEA 02

Practise: Triangles and Segments

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Segment

P1 · TOPIC 10

3 marks

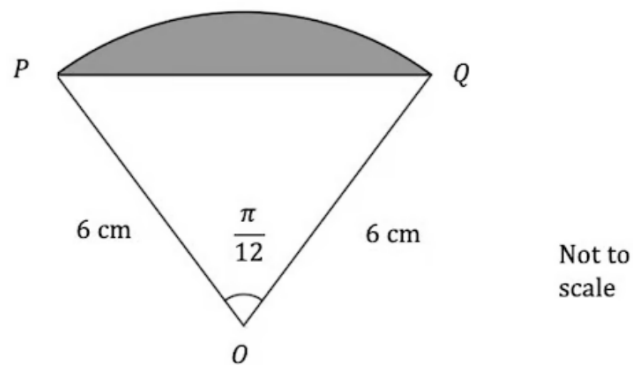
A sector has $r = 4$ cm and $\theta = 1.2$. Find the area of the segment cut off by the chord.

SELECTED PRACTICE

Q10 PRACTICE Practice · Easy

P1 · TOPIC 10 6 marks

- 7 (a) The diagram below shows the sector of a circle, OPQ , radius 6 cm and centre angle $\frac{\pi}{12}$ radians.



Use the formula $\frac{1}{2}ab \sin C$ to find the area of the triangle OPQ .

(2 marks)

- (b) Find the area of the sector OPQ .

(2 marks)

- (c) Hence or otherwise find the area of the shaded segment.

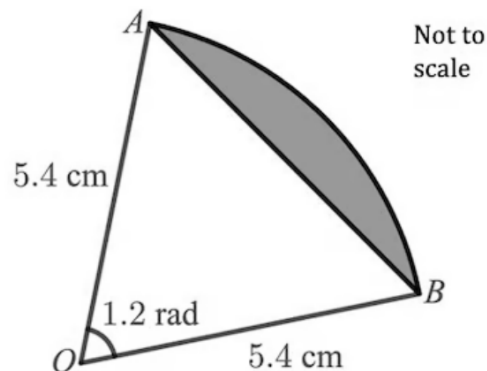
(2 marks)

Q11**PRACTICE**

Practice • Medium

P1 • TOPIC 10 8 marks

5 (a) The diagram below shows the sector of a circle .



- (i) Find the area of the sector OAB , giving your answer to 3 significant figures.
- (ii) Find the area of the triangle OAB , giving your answer to 3 significant figures.
- (iii) Find the area of the shaded segment, giving your answer to 3 significant figures.

(5 marks)

(b) (i) Find the length of the arc AB .

- (ii) Find the perimeter of the sector OAB .

(3 marks)

PAST PAPERS

Q12 PAST PAPER Pearson IAL WMA11 • January 2021 • Q5

P1 • TOPIC 10 9 marks

5.

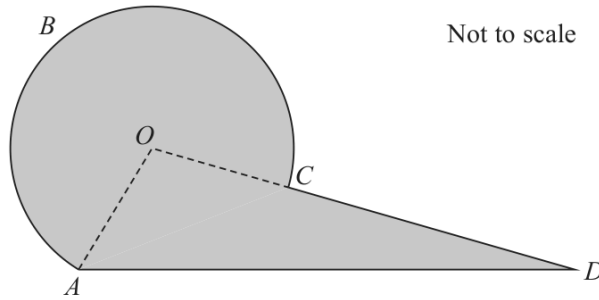


Figure 3

Figure 3 shows the plan view of a viewing platform at a tourist site.

The shape of the viewing platform consists of a sector $ABCOA$ of a circle, centre O , joined to a triangle AOD .

Given that

- $OA = OC = 6$ m
- $AD = 14$ m
- angle $ADC = 0.43$ radians
- angle AOD is an obtuse angle
- OCD is a straight line

find

- the size of angle AOD , in radians, to 3 decimal places, (3)
- the length of arc ABC , in metres, to one decimal place, (2)
- the total area of the viewing platform, in m^2 , to one decimal place. (4)

(Total 9 marks)

Q13 PAST PAPER Pearson IAL WMA11 • October 2021 • Q7

P1 • TOPIC 10 10 marks

7.

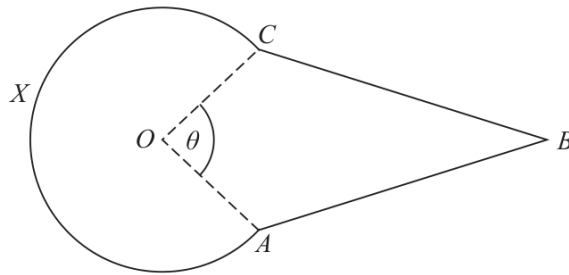
**Figure 3**

Figure 3 shows the design for a sign at a bird sanctuary.

The design consists of a kite $OABC$ joined to a sector $OCXA$ of a circle centre O .

In the design

- $OA = OC = 0.6$ m
- $AB = CB = 1.4$ m
- Angle $OAB = \text{Angle } OCB = 2$ radians
- Angle $AOC = \theta$ radians, as shown in Figure 3

Making your method clear,

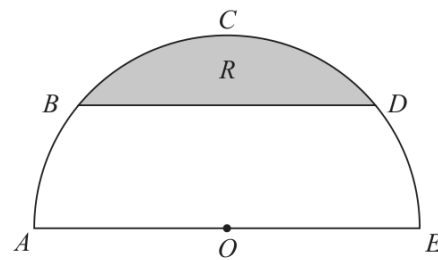
- show that $\theta = 1.64$ radians to 3 significant figures, (4)
- find the perimeter of the sign, in metres to 2 significant figures, (2)
- find the area of the sign, in m^2 to 2 significant figures. (4)

(Total 10 marks)

Q14 PAST PAPER Pearson IAL WMA11 • January 2022 • Q5

P1 • TOPIC 10 7 marks

5.



Not to scale

Figure 2

Figure 2 shows a plan view of a semicircular garden $ABCDEOA$

The semicircle has

- centre O
- diameter AOE
- radius 3 m

The straight line BD is parallel to AE and angle BOA is 0.7 radians.

- (a) Show that, to 4 significant figures, angle BOD is 1.742 radians.

(1)

The flowerbed R , shown shaded in Figure 2, is bounded by BD and the arc BCD .

- (b) Find the area of the flowerbed, giving your answer in square metres to one decimal place.

(3)

- (c) Find the perimeter of the flowerbed, giving your answer in metres to one decimal place.

(3)

(Total 7 marks)

8.

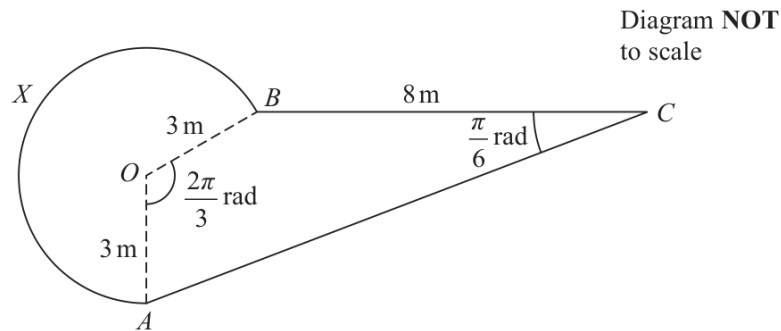


Figure 2

Figure 2 shows the plan view of a design for a pond.

The design consists of a sector $AOBX$ of a circle centre O joined to a quadrilateral $AOBC$.

- $BC = 8$ m
- $OA = OB = 3$ m
- angle AOB is $\frac{2\pi}{3}$ radians
- angle BCA is $\frac{\pi}{6}$ radians

(a) Calculate (i) the exact area of the sector $AOBX$,

(ii) the exact perimeter of the sector $AOBX$.

(5)

(b) Calculate the exact area of the triangle AOB .

(2)

(c) Show that the length AB is $3\sqrt{3}$ m.

(2)

(d) Find the total surface area of the pond. Give your answer in m^2 correct to 2 significant figures.

(5)

(Total for Question 8 is 14 marks)

Q16 PAST PAPER Pearson IAL WMA11 · January 2023 · Q6

P1 · TOPIC 10 10 marks

6.

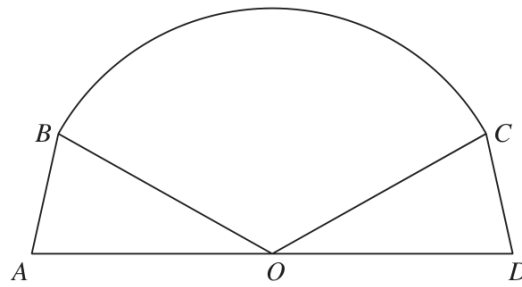
Diagram **NO**
accurately dr**Figure 1**

Figure 1 shows the plan view for the design of a stage.

The design consists of a sector BOC of a circle, with centre O , joined to two congruent triangles OAB and ODC .

Given that

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40 \text{ m}^2$
- AOD is a straight line of length 12.5 m

(a) find the radius of the sector, giving your answer, in m , to 2 decimal places,

(2)

(b) find the size of angle AOB , in radians, to 2 decimal places.

(1)

Hence find

(c) the total area of the stage, giving your answer, in m^2 , to one decimal place,

(3)

(d) the total perimeter of the stage, giving your answer, in m , to one decimal place.

(4)

(Total for Question 6 is 10 marks)

Q17 PAST PAPER Pearson IAL WMA11 • June 2024 • Q5

P1 • TOPIC 10 7 marks

5.

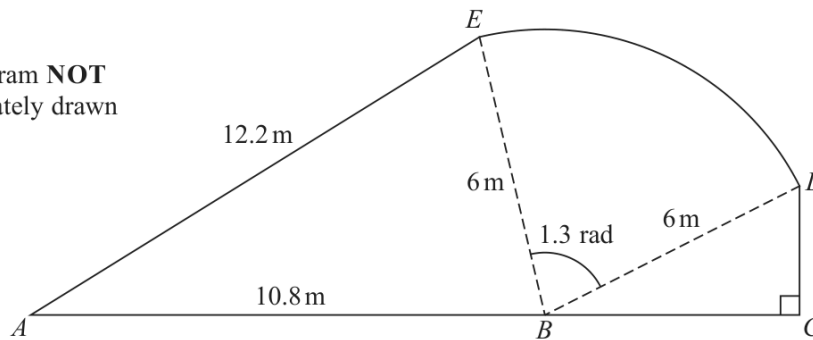
Diagram **NOT**
accurately drawn**Figure 2**

Figure 2 shows the plan view of a garden.

The shape of the garden $ABCDEA$ consists of a triangle ABE and a right-angled triangle BCD joined to a sector BDE of a circle with radius 6 m and centre B .

The points A , B and C lie on a straight line with $AB = 10.8$ m

Angle $BCD = \frac{\pi}{2}$ radians, angle $EBD = 1.3$ radians and $AE = 12.2$ m

- (a) Find the area of the sector BDE , giving your answer in m^2 (2)
- (b) Find the size of angle ABE , giving your answer in radians to 2 decimal places. (2)
- (c) Find the area of the garden, giving your answer in m^2 to 3 significant figures. (3)

(Total for Question 5 is 7 marks)

Q18 PAST PAPER Pearson IAL WMA11 · January 2025 · Q8

P1 · TOPIC 10 13 marks

8.

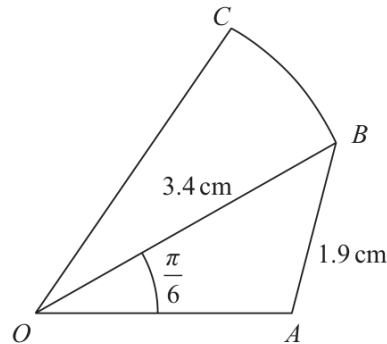
**Figure 1**

Figure 1 shows a sketch of a design for a badge.

The design consists of a triangle OAB joined to a sector OBC of a circle with centre O

In the design

- $OB = 3.4$ cm
- $AB = 1.9$ cm
- angle $AOB = \frac{\pi}{6}$ radians
- angle $OAB > \frac{\pi}{2}$ radians

Making your method clear,

(a) find the size of angle OAB , giving your answer in radians to 4 significant figures, (3)

(b) find the area of triangle OAB , in cm^2 , giving your answer to 3 significant figures. (2)

Given that the ratio of the area of sector OBC to the area of triangle OAB is 3 : 2

(c) show that angle BOC is 0.462 radians to 3 significant figures. (3)

(d) Hence find the perimeter of the badge, in cm, to the nearest integer. (5)

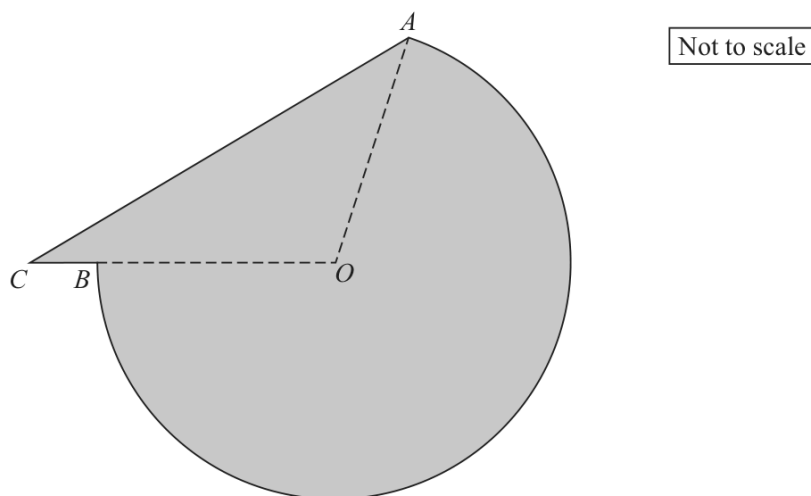
(Total for Question 8 is 13 marks)

Q19**PAST PAPER**

Pearson IAL WMA11 • June 2025 • Q6

P1 • TOPIC 10 9 marks

6.

**Figure 2**

The shaded area in Figure 2 shows the plan view of a helicopter landing pad.

The area consists of the major sector AOB of a circle centre O joined to a triangle AOC .

Given that

- $AO = OB = 15$ m
- $BC = 2$ m
- CBO is a straight line
- angle $ACO = 0.6$ radians

(a) show that angle COA is 1.847 radians to 3 decimal places.

(3)

(b) Find the total area of the helicopter landing pad.
Give your answer in m^2 to 3 significant figures.

(3)

(c) Find the perimeter of the helicopter landing pad.
Give your answer in metres to 3 significant figures.

(3)

(Total for Question 6 is 9 marks)

Q20**PAST PAPER**

Pearson IAL WMA11 · January 2026 · Q5

P1 · TOPIC 10**9 marks**

5.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

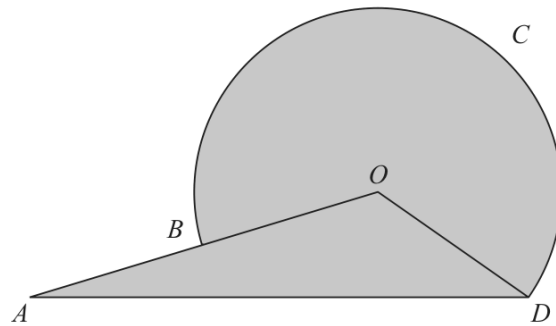


Diagram NOT
to scale

Figure 2

Figure 2 shows the logo for the tourist board of an island.

The logo consists of a sector $BODCB$ of a circle centre O joined to a triangle AOD .

Given that

- ABO is a straight line
- $OD = 3$ cm
- $AD = 8$ cm
- angle $AOD = 2.5$ radians

(a) show that angle ODA is 0.415 radians to 3 significant figures.

(3)

(b) Find the total area of the logo, in cm^2 , to one decimal place.

(3)

(c) Find the perimeter of the logo, $ABCD A$, in cm, to one decimal place.

(3)

(Total for Question 5 is 9 marks)

03 Composite Shapes: Break It Up

Gardens, logos, platforms, pools. Label every piece: which sectors, which triangles, which straight edges.

WHEN YOU SEE → YOU DO

- | | |
|---------------------------|---|
| "total area of the ..." | → add the pieces (subtract any hole) |
| "perimeter of the ..." | → walk around the outside only — arcs and straight edges |
| "DA:AO = 3:5", "PQ = 2.8" | → write every radius in terms of one letter |
| "to 3 s.f.", "nearest m" | → keep full accuracy until the last line |

1

Name the pieces

Sector(s), triangle(s), rectangle(s). Mark every radius.

2

Missing lengths and angles

Sine rule, cosine rule, or $\theta = 2\pi - \dots$.

3

Add areas • walk the perimeter

Only the outside edges count for the perimeter.

WORKED MODEL • JANUARY 2024

Sector DOC ($\angle 0.8$, arc 9) joined to major sector $AOBEA$; $DA : AO = 3 : 5$.

1 $OD = \frac{8}{5}AO$, $0.8 \cdot \frac{8}{5}AO = 9 \Rightarrow AO = 7.03$

arc gives the radius

2 $P = 9 + 2(11.25 - 7.03) + 7.03(2\pi - 0.8)$

walk the outside

ANSWER $P = 56.0 \text{ m}$, $A = 186 \text{ m}^2$

IDEA 03

Practise: Composite Shapes

My questions first, then real past papers.

Q21 **DR ESLAM** Walk the outside

P1 · TOPIC 10 3 marks

A shape is a semicircle of radius 3 cm on top of a 6×4 cm rectangle. Find its perimeter and area.

PAST PAPERS

Q22 PAST PAPER Pearson IAL WMA11 • January 2020 • Q4

P1 • TOPIC 10 9 marks

4.

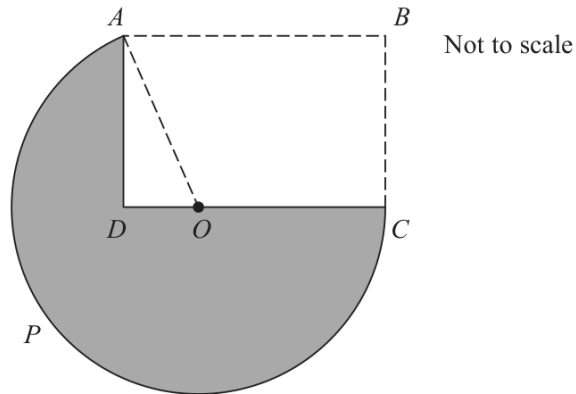


Figure 2

Figure 2 shows the plan view of a house $ABCD$ and a lawn $APCDA$.

$ABCD$ is a rectangle with $AB = 16$ m.

$APCOA$ is a sector of a circle centre O with radius 12 m.

The point O lies on the line DC , as shown in Figure 2.

(a) Show that the size of angle AOD is 1.231 radians to 3 decimal places.

(2)

The lawn $APCDA$ is shown shaded in Figure 2.

(b) Find the area of the lawn, in m^2 , to one decimal place.

(4)

(c) Find the perimeter of the lawn, in metres, to one decimal place.

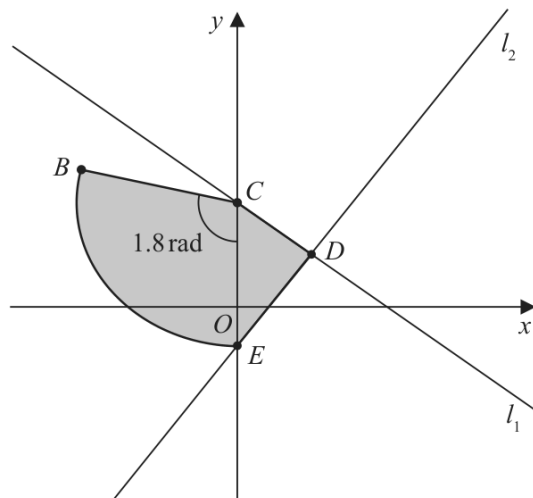
(3)

(Total 9 marks)

Q23 PAST PAPER Pearson IAL WMA11 · June 2021 · Q7

P1 · TOPIC 10 10 marks

7.

**Figure 3**

The line l_1 has equation $4y + 3x = 48$

The line l_1 cuts the y -axis at the point C , as shown in Figure 3.

(a) State the y coordinate of C .

(1)

The point $D(8, 6)$ lies on l_1

The line l_2 passes through D and is perpendicular to l_1

The line l_2 cuts the y -axis at the point E as shown in Figure 3.

(b) Show that the y coordinate of E is $-\frac{14}{3}$

(3)

A sector BCE of a circle with centre C is also shown in Figure 3.

Given that angle BCE is 1.8 radians,

(c) find the length of arc BE .

(3)

The region $CBED$, shown shaded in Figure 3, consists of the sector BCE joined to the triangle CDE .

(d) Calculate the exact area of the region $CBED$.

(3)**(Total 10 marks)**

Q24 PAST PAPER Pearson IAL WMA11 · June 2023 · Q5

P1 · TOPIC 10 10 marks

5.

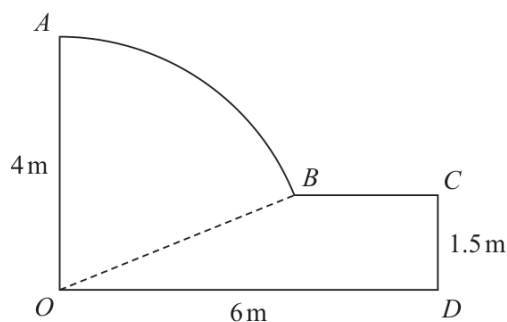
**Figure 1**

Figure 1 shows the plan for a garden.

In the plan

- OA and CD are perpendicular to OD
- AB is an arc of the circle with centre O and radius 4 metres
- BC is parallel to OD
- OD is 6 metres, OA is 4 metres and CD is 1.5 metres

(a) Show that angle AOB is 1.186 radians to 4 significant figures.

(2)

(b) Find the perimeter of the garden, giving your answer in metres to 3 significant figures.

(4)

(c) Find the area of the garden, giving your answer in square metres to 3 significant figures.

(4)

(Total for Question 5 is 10 marks)

Q25 **PAST PAPER** Pearson IAL WMA11 · January 2024 · Q8

P1 · TOPIC 10 **9 marks**

8.

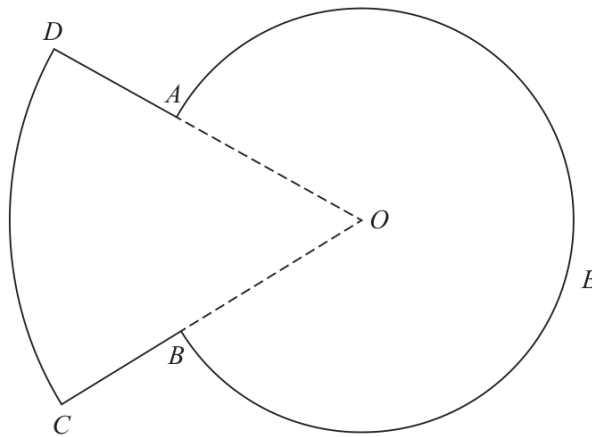


Figure 3

Figure 3 shows a sketch of the plan view of a platform.

The plan view of the platform consists of a sector DOC of a circle centre O joined to a sector $AOBEA$ of a different circle, also with centre O .

Given that

- angle $AOB = 0.8$ radians
- arc length $CD = 9$ m
- $DA : AO = 3 : 5$

(a) show that $AO = 7.03$ m to 3 significant figures.

(3)

(b) Find the perimeter of the platform, in m, to 3 significant figures.

(3)

(c) Find the total area of the platform, giving your answer in m^2 to the nearest whole number.

(3)

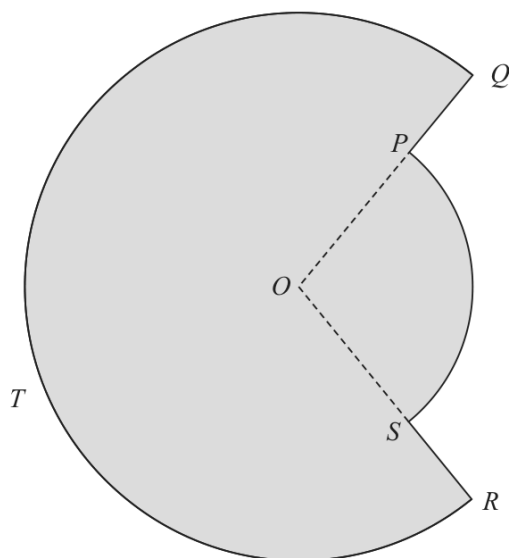
(Total for Question 8 is 9 marks)

Q26**PAST PAPER**

Pearson IAL WMA11 · October 2025 · Q7

P1 · TOPIC 10 9 marks

7.

**Figure 2**

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 2 shows the plan view of a design for a swimming pool.

The design consists of a sector POS of a circle centre O joined to a major sector $QORTQ$ of a different circle, also with centre O .

Given that

- angle POS is 1.65 radians
- the area of sector POS is 30 m^2
- $PQ = 2.8 \text{ m}$

(a) show that, to 3 significant figures, $OQ = 8.83 \text{ m}$

(3)

(b) Find the total surface area of the swimming pool in m^2 to the nearest integer.

(3)

(c) Find the total perimeter of the swimming pool in metres to 2 significant figures.

(3)

(Total for Question 7 is 9 marks)

04 Equations in r and θ

Two facts (area, perimeter, arc) give two equations. Solve them simultaneously — often a quadratic.

WHEN YOU SEE → YOU DO

“the area is 1 and the perimeter is 8”

$$\rightarrow \frac{1}{2}r^2\theta = 1, 2r + r\theta = 8$$

“show that $5x^2 + 12x - 1500 = 0$ ”

→ area of the big sector – small sector = given

“exact value ... in the form $a + b\sqrt{c}$ ”

→ quadratic formula, no decimals

WORKED MODEL • JUNE 2026

Area 1, perimeter 8, $r > 2$.

$$1 \quad r\theta = 8 - 2r \Rightarrow \frac{1}{2}r(8 - 2r) = 1$$

substitute

$$2 \quad r^2 - 4r + 1 = 0 \Rightarrow r = 2 \pm \sqrt{3}$$

quadratic

ANSWER $r = 2 + \sqrt{3}, \quad \theta = \frac{2}{r^2} = 14 - 8\sqrt{3}$

IDEA 04

Practise: Equations in r and θ

My questions first, then selected practice, then real past papers.

Q27 DR ESAM Two facts, two equations

P1 · TOPIC 10 4 marks

A sector has perimeter 20 cm and area 16 cm^2 . Find the two possible radii.

SELECTED PRACTICE**Q28** PRACTICE Practice · Easy

P1 · TOPIC 10 2 marks

6 The arc length of the sector of a circle is 5 cm .

The radius is twice the length of the arc.

Find the angle at the centre of the circle.

(2 marks)

Q29**PRACTICE**

Practice · Easy

P1 · TOPIC 10**4 marks**

8 The area of a sector is $8\pi \text{ cm}^2$.

The arc length of the sector is $4\pi \text{ cm}$.

Find the radius and the angle at the centre of the circle.

(4 marks)**Q30****PRACTICE**

Practice · Hard

P1 · TOPIC 10**4 marks**

3 A sector of a circle, OPQ , is such that it has radius a cm and the angle at its centre, O , is $a^2 \pi$ radians.

Given that the area of the sector OPQ in cm^2 is three times the length of the arc PQ in cm, find the value of a .

(4 marks)

PAST PAPERS

Q31 PAST PAPER Pearson IAL WMA11 • October 2023 • Q9

P1 • TOPIC 10 7 marks

9.

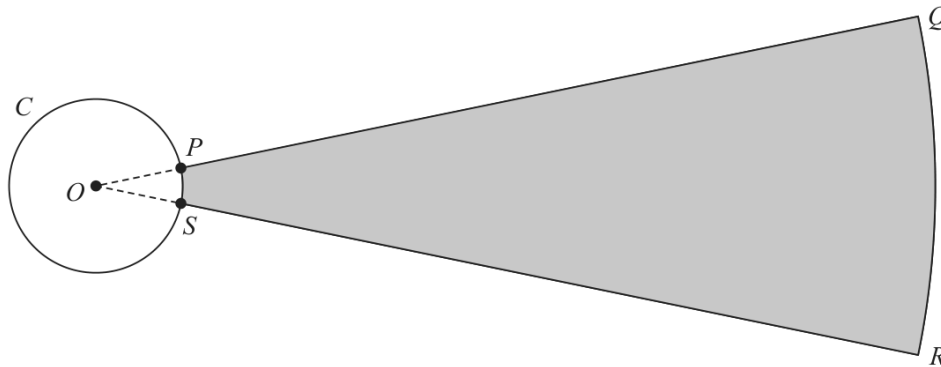
Diagram NO
accurately dr

Figure 3

Figure 3 shows the plan view of the area being used for a ball-throwing competition.

Competitors must stand within the circle C and throw a ball as far as possible into the target area, $PQRS$, shown shaded in Figure 3.

Given that

- circle C has centre O
- P and S are points on C
- $OPQRSO$ is a sector of a circle with centre O
- the length of arc PS is 0.72 m
- the size of angle POS is 0.6 radians

(a) show that $OP = 1.2$ m

(1)

Given also that

- the target area, $PQRS$, is 90 m^2
- length $PQ = x$ metres

(b) show that

$$5x^2 + 12x - 1500 = 0 \quad (3)$$

(c) Hence calculate the total perimeter of the target area, $PQRS$, giving your answer to the nearest metre.

(3)

(Total for Question 9 is 7 marks)

Q32 PAST PAPER Pearson IAL WMA11 • June 2026 • Q7

P1 • TOPIC 10 7 marks

7.

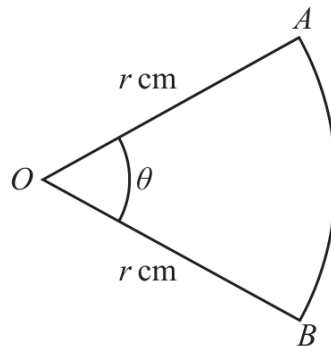
Diagram **NOT**
accurately drawn**Figure 3**

Figure 3 shows a sector AOB of a circle, centre O .

The radius OA of the sector is r cm and angle AOB is θ radians.

Given that

- the **area** of sector AOB is 1 cm^2
- the **perimeter** of sector AOB is 8 cm

(a) write down two different equations involving r and θ .

(2)

Given that $r > 2$

(b) solve the two simultaneous equations in part (a) to find the exact value of r and the exact value of θ .

Write each answer in the form $a + b\sqrt{c}$ where a , b and c are integers.

(Solutions relying on calculator technology are not acceptable.)

(5)

Question 7 continued**(Total for Question 7 is 7 marks)**

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 Arc & sector** $s = r\theta$, $A = \frac{1}{2}r^2\theta$. Major sector: $2\pi - \theta$. RAD mode.
- 02 Triangles** $\frac{1}{2}ab \sin C$, sine rule, cosine rule. Segment = $\frac{1}{2}r^2(\theta - \sin \theta)$.
- 03 Composite** Name the pieces → missing lengths → add areas, walk the outside.
- 04 Equations** Two facts → two equations in r, θ → substitute.

BOTTOM LINE Radians only → break the shape into sectors and triangles → full accuracy to the last line.

Answers

Radian Measure • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • RADIANS, ARCS AND SECTORS

- Q1** Dr Eslam • Arc and sector arc 4π ; area 12π ; perimeter $12 + 4\pi$
- Q2** Dr Eslam • Find θ $\theta = 1.6$
- Q3** Practice • Easy (i) $\frac{\pi}{3}$ (ii) $\frac{\sqrt{2}}{2}$ (iii) 0 (iv) 150° (v) 1
- Q4** Practice • Easy (i) π cm (ii) 2π cm²
- Q5** Practice • Medium area $\frac{81\pi}{10} \approx 25.4$ cm²; perimeter $18 + \frac{9\pi}{5} \approx 23.7$ cm
- Q6** Pearson IAL WMA11 • October 2019 • Q1 (a) $r = 2\sqrt{6}$ (b) $\frac{13}{2}\sqrt{6}$ cm
- Q7** Pearson IAL WMA11 • June 2022 • Q8 (b) 603 cm (c) 4730 cm² (0.473 m²)
- Q8** Pearson IAL WMA11 • October 2024 • Q5 (a) 16 km (c) $AP = 4.7$ km

IDEA 02 • TRIANGLES AND SEGMENTS

- Q9** Dr Eslam • Segment $\frac{1}{2}(16)(1.2 - \sin 1.2) = 2.14 \text{ cm}^2$
- Q10** Practice • Easy (a) 4.66 cm^2 (b) $1.5\pi = 4.71 \text{ cm}^2$ (c) 0.0536 cm^2
- Q11** Practice • Medium (a)(i) 17.5 (ii) 13.6 (iii) 3.91 cm^2 (b)(i) 6.48 cm (ii) 17.3 cm
- Q12** Pearson IAL WMA11 • January 2021 • Q5 (a) 1.805 (b) 26.9 m (c) 113.7 m^2
- Q13** Pearson IAL WMA11 • October 2021 • Q7 (b) 5.6 m (c) 1.6 m^2
- Q14** Pearson IAL WMA11 • January 2022 • Q5 (b) 3.4 m^2 (c) 9.8 m
- Q15** Pearson IAL WMA11 • October 2022 • Q8 (a) 6π , $6 + 4\pi$ (b) $\frac{9\sqrt{3}}{4}$ (d) 43 m^2
- Q16** Pearson IAL WMA11 • January 2023 • Q6 (a) 5.77 (b) 0.37 (c) 53.1 m^2 (d) 30.9 m
- Q17** Pearson IAL WMA11 • June 2024 • Q5 (a) 23.4 m^2 (b) 1.54 (c) 60.9 m^2
- Q18** Pearson IAL WMA11 • January 2025 • Q8 (a) 2.034 (b) 1.78 cm^2 (d) 9 cm
- Q19** Pearson IAL WMA11 • June 2025 • Q6 (b) 622 m^2 (c) 94.1 m
- Q20** Pearson IAL WMA11 • January 2026 • Q5 (b) 21.9 cm^2 (c) 21.7 cm

IDEA 03 • COMPOSITE SHAPES

- Q21** Dr Eslam • Walk the outside $P = 6 + 8 + 3\pi \approx 23.4 \text{ cm}$; $A = 24 + 4.5\pi \approx 38.1 \text{ cm}^2$
- Q22** Pearson IAL WMA11 • January 2020 • Q4 (a) $\cos^{-1} \frac{4}{12} = 1.231$ (b) 292.2 m^2 (c) 79.8 m
- Q23** Pearson IAL WMA11 • June 2021 • Q7 (a) 12 (c) 30 (d) $\frac{950}{3}$
- Q24** Pearson IAL WMA11 • June 2023 • Q5 (b) 18.5 m (c) 15.7 m^2
- Q25** Pearson IAL WMA11 • January 2024 • Q8 (b) 56.0 m (c) 186 m^2
- Q26** Pearson IAL WMA11 • October 2025 • Q7 (b) 211 m^2 (c) 57 m

IDEA 04 • EQUATIONS IN r AND θ

- Q27** Dr Eslam • Two facts, two equations $r\theta = 20 - 2r$, $\frac{1}{2}r(20 - 2r) = 16 \Rightarrow r^2 - 10r + 16 = 0$:
 $r = 2$ or 8
- Q28** Practice • Easy $r = 10$, $\theta = 0.5$
- Q29** Practice • Easy $r = 4 \text{ cm}$, $\theta = \pi$
- Q30** Practice • Hard $a = 6$
- Q31** Pearson IAL WMA11 • October 2023 • Q9 (c) $x = 16.2$; perimeter 43 m
- Q32** Pearson IAL WMA11 • June 2026 • Q7 (a) $\frac{1}{2}r^2\theta = 1$, $2r + r\theta = 8$ (b) $r = 2 + \sqrt{3}$, $\theta = 14 - 8\sqrt{3}$

DR ESLAM AHMED
eliteigcse.com

Trigonometric Functions

Three graphs, four moves, one symmetry. Every question in this topic is read off a sketch — draw it first.

THE TOPIC ON ONE MAP

Trigonometric Functions

01 — The three graphs

$\sin$, $\cos$, $\tan$ period asymptote

02 — Transformations

$a \sin x$ $\sin bx$ $\sin(x + c)$ $+k$

03 — Solving by symmetry

all values in the range in terms of α

04 — Counting solutions

two graphs a huge interval

05 — Modelling with trig

$30 \cos(24t)^\circ$ dips

01 The Three Graphs

Learn the five key points of each graph. Every other trig graph is one of these, moved.

WHEN YOU SEE → YOU DO

“sketch ... showing the coordinates where it meets the axes”

→ zeros, y-intercept, max, min

“state the period”

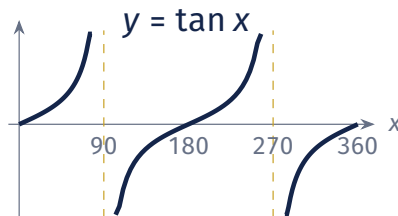
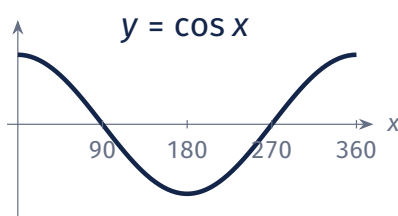
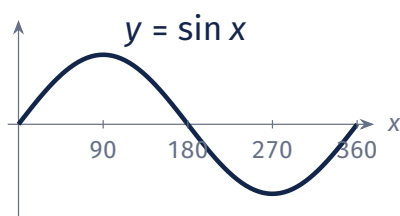
→ sin, cos: 360° (2π) · tan: 180° (π)

“asymptote” to $y = \tan x$

→ $x = 90^\circ + 180^\circ k$ ($\frac{\pi}{2} + k\pi$)

“radians”

→ $\pi = 180^\circ$ — label the axis in multiples of $\frac{\pi}{2}$



x	0	90	180	270	360
sin x	0	1	0	-1	0
cos x	1	0	-1	0	1
tan x	0	–	0	–	0

IDEA 01

Practise: The Three Graphs

My questions first, then selected practice.

Q1 DR ESLAM Five key points

P1 • TOPIC 11 3 marks

Sketch $y = \cos x$ for $0 \leq x \leq 360^\circ$, labelling every point where it meets the axes, the maximum and the minimum.

Q2 DR ESLAM Periods and asymptotes

P1 • TOPIC 11 2 marks

State the period of $y = \tan x$ and the equations of its asymptotes for $0 \leq x \leq 360^\circ$.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 11 6 marks

1 On separate diagrams sketch the graphs of:

(i) $y = \sin x \quad -180^\circ \leq x \leq 180^\circ$

(ii) $y = \cos x \quad 0^\circ \leq x \leq 360^\circ$

(iii) $y = \tan x \quad -180^\circ \leq x \leq 180^\circ$

(6 marks)

Q4**PRACTICE**

Practice · Medium

P1 · TOPIC 11 3 marks

1 (i) Sketch the graph of $y = \cos \theta$ in the interval $0^\circ \leq \theta \leq 360^\circ$. The sketch must include coordinates of all points where the graph meets the coordinate axes.

(ii) Write down all the values of θ for which $\cos \theta = 0$ for $0^\circ \leq \theta \leq 360^\circ$.

(3 marks)**Q5****PRACTICE**

Practice · Hard

P1 · TOPIC 11 3 marks

3 (i) Sketch the graph of $y = \sin \theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$. The sketch must include coordinates of all points where the graph meets the coordinate axes.

(ii) Given that $\sin 60^\circ = \frac{\sqrt{3}}{2}$, use your graph to find all values of θ in the given interval for which $\sin \theta = -\frac{\sqrt{3}}{2}$.

(3 marks)

02 Transformations: Read a, b, c, k

$y = a \sin(bx + c) + k$ is $y = \sin x$ moved. Each letter does exactly one thing — and you can read it back off a graph.

WHEN YOU SEE → YOU DO

$a \sin x$, “**maximum value of y is ...**”

→ stretch $\parallel y$: $\max = a$, $\min = -a$

$\sin bx$

→ stretch $\parallel x$, s.f. $\frac{1}{b}$; period $\frac{360}{b}$

$\sin(x + c)$

→ translate $\begin{pmatrix} -c \\ 0 \end{pmatrix}$ — inside moves the opposite way

$+k$, “**has a maximum value of 10**”

→ translate up k : new $\max = a + k$

$-\cos x$ / $-\sin x$

→ reflect in the x -axis

“ **P is a minimum point ... find the coordinates of Q** ”

→ count along whole periods

WORKED MODEL • JUNE 2024

$y = 12 \sin x$; $C_2 : y = 12 \sin x + k$ has maximum 3. Find the minimum with the smallest positive x .

1 $12 + k = 3 \Rightarrow k = -9$

outside: y moves

2 min of $\sin x$ first at $x = \frac{3\pi}{2}$

inside unchanged

3 $y = -12 - 9$

$\min = -a + k$

ANSWER $\left(\frac{3\pi}{2}, -21\right)$

IDEA 02

Practise: Transformations

My questions first, then selected practice, then real past papers.

Q6 DR ESLAM Read a and k

P1 • TOPIC 11 2 marks

$y = 4 \cos x + k$ has a maximum value of 7. Find k and the minimum value.

Q7 DR ESLAM Move a point

P1 • TOPIC 11 3 marks

$(90^\circ, 1)$ is the maximum of $y = \sin x$. Write down the maximum point of (a) $y = 3 \sin x$ (b) $y = \sin 2x$ (c) $y = \sin(x - 30^\circ)$

Q8 DR ESLAM Period from b

P1 • TOPIC 11 2 marks

State the period of (a) $y = \cos 3x$ (b) $y = \sin \frac{x}{2}$ (in degrees).

SELECTED PRACTICE

Q9**PRACTICE** Practice · Easy**P1 · TOPIC 11** 3 marks

- 4 The point P has coordinates $(90^\circ, 1)$ and lies on the graph of $y = \sin x$, where $0^\circ \leq x \leq 180^\circ$.

Write down the coordinates of the image of point P under the following transformations:

(i) $y = f(x) + 2$

(ii) $y = f(3x)$

(iii) $y = f(x + 30^\circ)$

(3 marks)

Q10**PRACTICE**

Practice · Hard

P1 · TOPIC 11 4 marks

4 (i) Sketch the graph of $y = \tan\left(\theta - \frac{\pi}{4}\right)$ in the interval $-2\pi \leq \theta \leq 2\pi$.

(ii) Write down all the values of θ for which $\tan\left(\theta - \frac{\pi}{4}\right) = 1$ in the given interval.

(4 marks)

Q11**PRACTICE**

Practice · Very Hard

P1 · TOPIC 11**6 marks**

10 A function $f(x) = \cos px$, $0 \leq x \leq 2\pi$, first crosses the x -axis at $\frac{\pi}{10}$.

(i) Determine the value of p and sketch the graph of $y = f(x)$.

(ii) State the period of $f(x)$.

(6 marks)

PAST PAPERS

Q12 PAST PAPER Pearson IAL WMA11 · October 2019 · Q9

P1 · TOPIC 11 4 marks

9.

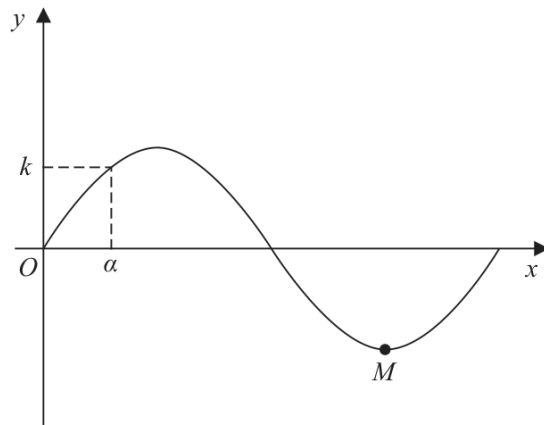


Figure 5

Figure 5 shows a sketch of part of the curve C with equation $y = \sin\left(\frac{x}{12}\right)$, where x is measured in radians. The point M shown in Figure 5 is a minimum point on C .

- (a) State the period of C . (1)
- (b) State the coordinates of M . (1)

The smallest positive solution of the equation $\sin\left(\frac{x}{12}\right) = k$, where k is a constant, is α .

Find, in terms of α ,

- (c) (i) the negative solution of the equation $\sin\left(\frac{x}{12}\right) = k$ that is closest to zero,
 (ii) the smallest positive solution of the equation $\cos\left(\frac{x}{12}\right) = k$. (2)

(Total 4 marks)

Q13 PAST PAPER Pearson IAL WMA11 · January 2020 · Q7

P1 · TOPIC 11 5 marks

7.

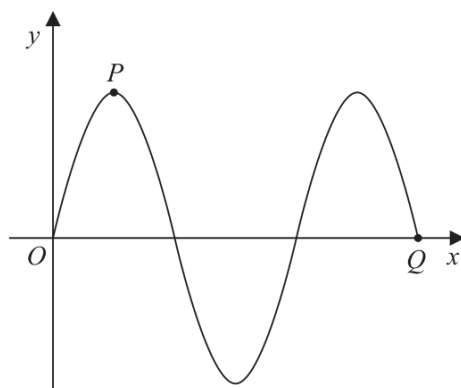
**Figure 3**

Figure 3 shows part of the curve C_1 with equation $y = 3 \sin x$, where x is measured in degrees.

The point P and the point Q lie on C_1 and are shown in Figure 3.

(a) State

- (i) the coordinates of P ,
- (ii) the coordinates of Q .

(3)

A different curve C_2 has equation $y = 3 \sin x + k$, where k is a constant.

The curve C_2 has a maximum y value of 10

The point R is the minimum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R .

(2)**(Total 5 marks)**

Q14 PAST PAPER Pearson IAL WMA11 · January 2021 · Q3

P1 · TOPIC 11 5 marks

3.

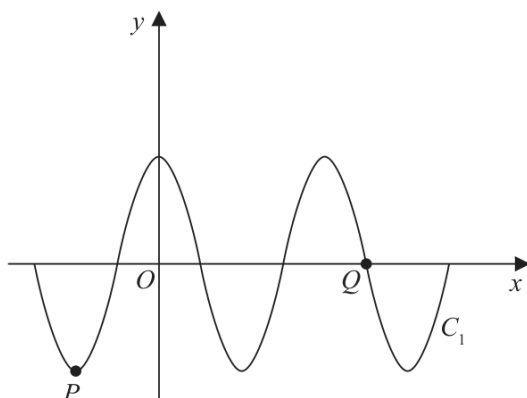
**Figure 1**

Figure 1 shows a sketch of part of the curve C_1 with equation $y = 4 \cos x^\circ$

The point P and the point Q lie on C_1 and are shown in Figure 1.

(a) State

(i) the coordinates of P ,

(ii) the coordinates of Q .

(3)

The curve C_2 has equation $y = 4 \cos x^\circ + k$, where k is a constant.

Curve C_2 has a minimum y value of -1

The point R is the maximum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R .

(2)**(Total 5 marks)**

Q15**PAST PAPER**

Pearson IAL WMA11 · October 2021 · Q4

P1 · TOPIC 11 5 marks

4.

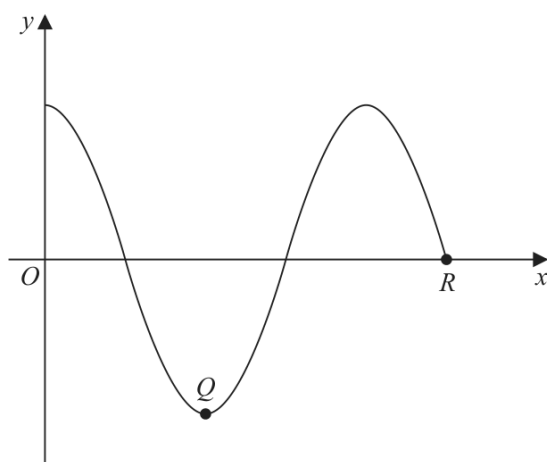
**Figure 2**

Figure 2 shows a sketch of the curve with equation $y = f(x)$, where

$$f(x) = \cos 2x^\circ \quad 0 \leq x \leq k$$

The point Q and the point $R(k, 0)$ lie on the curve and are shown in Figure 2.

(a) State

- (i) the coordinates of Q ,
- (ii) the value of k .

(3)

(b) Given that there are exactly two solutions to the equation

$$\cos 2x^\circ = p \quad \text{in the region } 0 \leq x \leq k$$

find the range of possible values for p .

(2)**(Total 5 marks)**

Q16**PAST PAPER**

Pearson IAL WMA11 · January 2022 · Q9

P1 · TOPIC 11 4 marks

9.

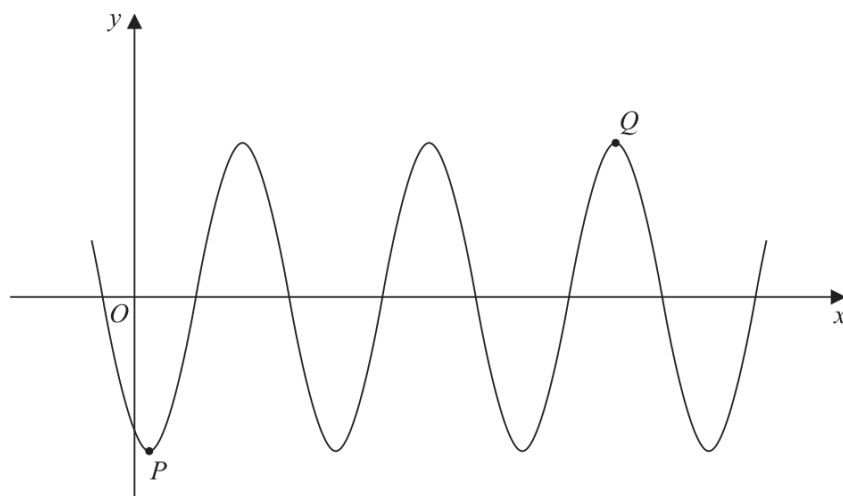
**Figure 4**

Figure 4 shows part of the curve with equation

$$y = A \cos(x - 30)^\circ$$

where A is a constant.

The point P is a minimum point on the curve and has coordinates $(30, -3)$ as shown in Figure 4.

(a) Write down the value of A .

(1)

The point Q is shown in Figure 4 and is a maximum point.

(b) Find the coordinates of Q .

(3)**(Total 4 marks)**

9. (i)

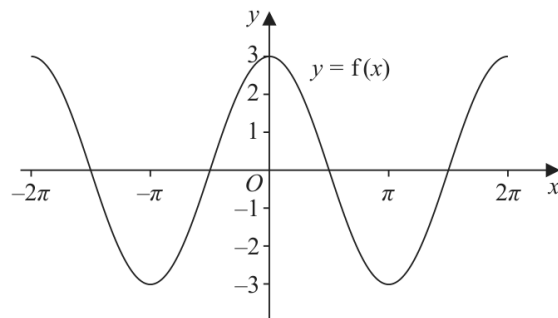


Figure 3

Figure 3 shows part of the graph of the trigonometric function with equation $y = f(x)$

(a) Write down an expression for $f(x)$

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = f\left(x + \frac{\pi}{4}\right)$

Show clearly the coordinates of all the points where the curve intersects the coordinate axes.

(3)

(ii)

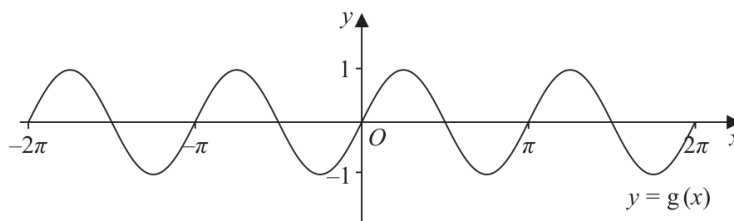


Figure 4

Figure 4 shows part of the graph of the trigonometric function with equation $y = g(x)$

(a) Write down an expression for $g(x)$

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = g(x) - 2$

Show clearly the coordinates of the y intercept.

(2)

(Total for Question 9 is 9 marks)

10.

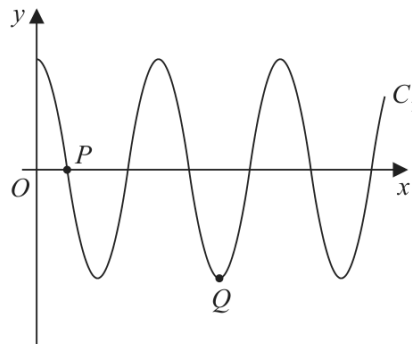


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 3 \cos\left(\frac{x}{n}\right)^\circ \quad x \geq 0$$

where n is a constant.

The curve C_1 cuts the positive x -axis for the first time at point $P(270, 0)$, as shown in Figure 4.

(a) (i) State the value of n

(ii) State the period of C_1

(2)

The point Q , shown in Figure 4, is a minimum point of C_1

(b) State the coordinates of Q .

(2)

The curve C_2 has equation $y = 2 \sin x^\circ + k$, where k is a constant.

The point $R\left(a, \frac{12}{5}\right)$ and the point $S\left(-a, -\frac{3}{5}\right)$, both lie on C_2

Given that a is a constant less than 90

(c) find the value of k .

(2)

(Total for Question 10 is 6 marks)

Q19**PAST PAPER**

Pearson IAL WMA11 · January 2024 · Q6

P1 · TOPIC 11 6 marks

6.

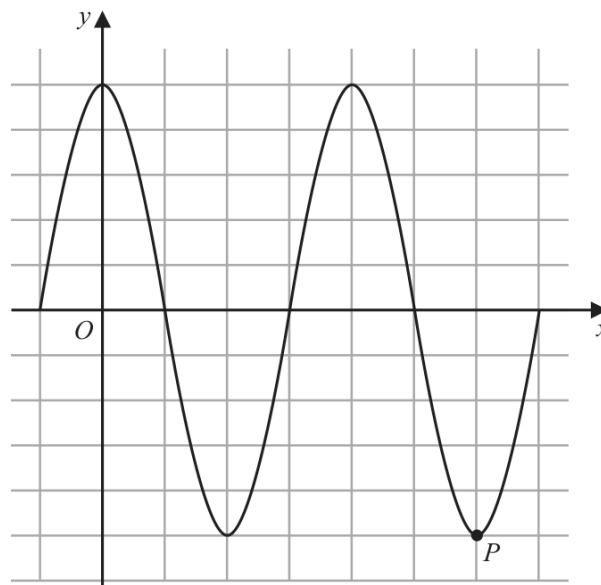
**Figure 2**

Figure 2 shows a plot of part of the curve C_1 with equation

$$y = 5 \cos x$$

with x being measured in degrees.

The point P , shown in Figure 2, is a minimum point on C_1

(a) State the coordinates of P

(2)

The point Q lies on a different curve C_2

Given that point Q

- is a maximum point on the curve
- is the maximum point with the **smallest** x coordinate, $x > 0$

(b) find the coordinates of Q when

(i) C_2 has equation $y = 5 \cos x - 2$

(ii) C_2 has equation $y = -5 \cos x$

(4)

(Total for Question 6 is 6 marks)

Q20 PAST PAPER Pearson IAL WMA11 · June 2024 · Q11

P1 · TOPIC 11 6 marks

11.

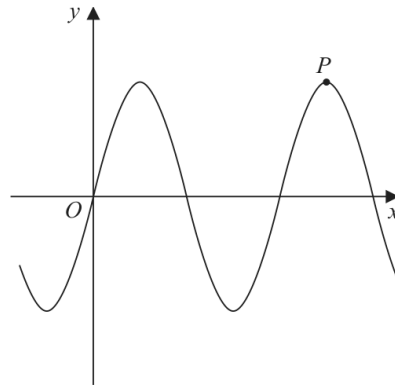


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 12 \sin x$$

where x is measured in radians.

The point P shown in Figure 4 is a maximum point on C_1 .

(a) Find the coordinates of P .

(2)

The curve C_2 has equation

$$y = 12 \sin x + k$$

where k is a constant.

Given that the **maximum** value of y on C_2 is 3

(b) find the coordinates of the **minimum** point on C_2 which has the **smallest** positive x coordinate.

(2)

The curve C_3 has equation

$$y = 12 \sin(x + B)$$

where B is a positive constant.

Given that $\left(\frac{\pi}{4}, A\right)$, where A is a constant, is the **minimum** point on C_3 which has the **smallest** positive x coordinate,

(c) find

(i) the value of A ,

(ii) the smallest possible value of B .

(2)

(Total for Question 11 is 6 marks)

TOTAL FOR PAPER IS 75 MARKS

4.

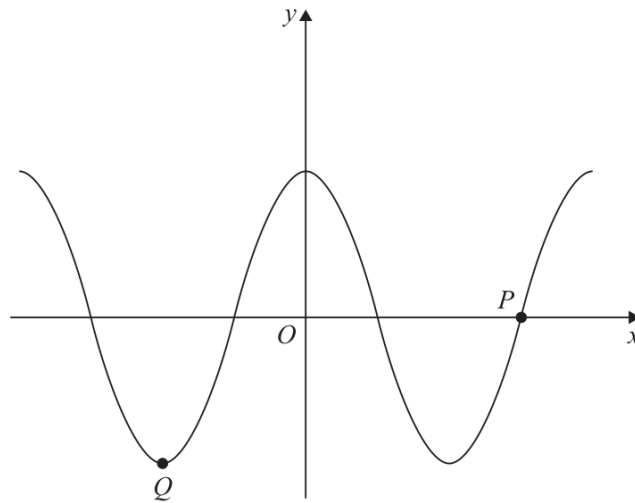


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = 4 \cos x$$

where x is measured in degrees.

The points P and Q lie on the curve and are shown in Figure 1.

- (a) State the coordinates of P . (1)
- (b) State the coordinates of Q . (2)
- (c) State the **number** of solutions of the equation
 - (i) $4 \cos x = 3$ in the interval $0 < x \leq 18\,000^\circ$
 - (ii) $5 + 4 \cos x = 1$ in the interval $-720^\circ < x \leq 720^\circ$
 - (iii) $4 \cos x - 3 = 1$ in the interval $-1080^\circ \leq x \leq 1080^\circ$ (3)

(Total for Question 4 is 6 marks)

03 Solving by Symmetry

Your calculator gives one answer. The graph gives all the others.

WHEN YOU SEE → YOU DO

“given that $\sin 60^\circ = \dots$, find all values ...”

→ use the symmetry of the graph

“in terms of α ” / “in terms of p ”

→ write the other solutions from α

$\sin 3x = \dots$ for $0 \leq x \leq 2\pi$

→ solve for $3x$ in $0 \leq 3x \leq 6\pi$, then divide

$$\sin x = k \quad \alpha \quad 180^\circ - \alpha$$

$$\cos x = k \quad \alpha \quad 360^\circ - \alpha \text{ (or } -\alpha)$$

$$\tan x = k \quad \alpha \quad \alpha + 180^\circ$$

Then add or subtract whole periods to stay in the interval.

WORKED MODEL • JUNE 2022

$\sin \alpha = p$. Find, in terms of α , the solutions of $\sin 2x = p$ for $0 < x < 180^\circ$.

1 $2x = \alpha$ or $2x = 180^\circ - \alpha$

symmetry of sin

2 range of $2x$: $0 < 2x < 360^\circ$

no more in range

ANSWER $x = \frac{\alpha}{2}$ or $x = 90^\circ - \frac{\alpha}{2}$

IDEA 03

Practise: Solving by Symmetry

My questions first, then selected practice, then real past papers.

Q22 DR ESLAM Symmetry of sine

P1 • TOPIC 11 2 marks

Given $\sin 30^\circ = 0.5$, find all solutions of $\sin x = 0.5$ for $0 \leq x \leq 720^\circ$.

Q23 DR ESLAM Inside a bracket

P1 • TOPIC 11 3 marks

Solve $\cos 2x = 0$ for $0 \leq x \leq 360^\circ$.

Work with $2x$ in $0 \leq 2x \leq 720^\circ$ first.

SELECTED PRACTICE

Q24

PRACTICE

Practice · Easy

P1 · TOPIC 11 2 marks

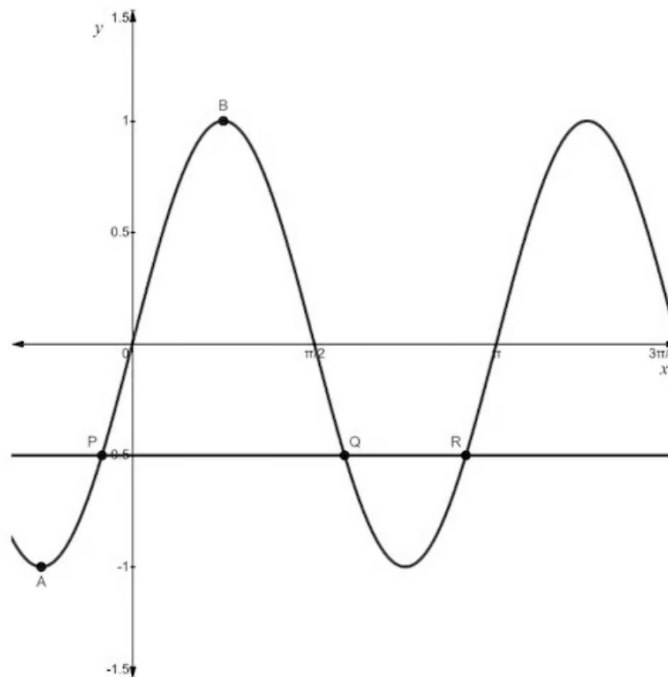
- 5 Write down the values for which $\cos x = \frac{1}{2}$, for $0 \leq x \leq 2\pi$.

(2 marks)

Q25 PRACTICE Practice • Medium

P1 • TOPIC 11 3 marks

- 8 (a) The graph below shows the curve with equation $y = \sin 2x$ in the interval $-\frac{\pi}{3} \leq x \leq \frac{3\pi}{2}$.



Point A has coordinates $\left(-\frac{\pi}{4}, -1\right)$ and is the minimum point closest to the origin.

Point B is the maximum point closest to the origin. State the coordinates of B .

(1 mark)

- (b) A straight line with equation $y = -\frac{1}{2}$ meets the graph of $y = \sin 2x$ at the three points P , Q and R , as shown in the diagram.

Given that point P has coordinates $\left(-\frac{\pi}{12}, -\frac{1}{2}\right)$, use graph symmetries to determine the coordinates of Q and R .

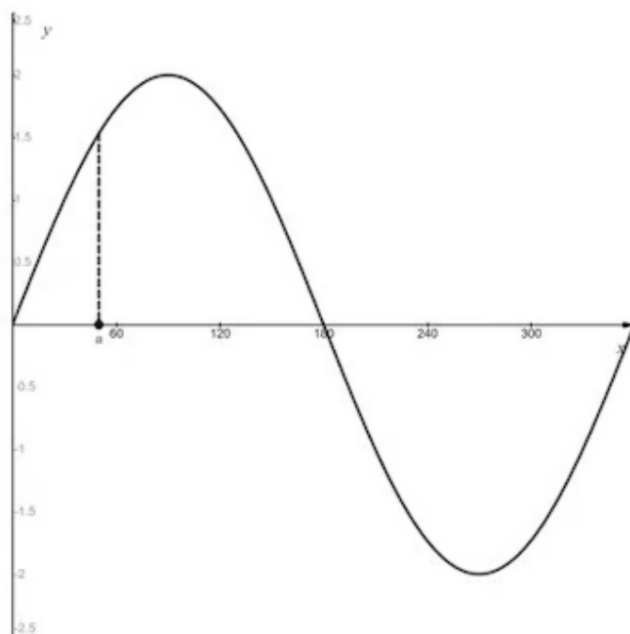
(2 marks)

Q26**PRACTICE**

Practice · Very Hard

P1 · TOPIC 11 2 marks

- 9 The graph below shows the curve with equation $y = 2 \sin \theta$, in the interval $0^\circ \leq \theta \leq 360^\circ$. One value of θ has been labelled ($\theta = a^\circ$).



Use the graph, along with the symmetry properties of the sine function, to verify that

$$2 \sin a = 2 \sin(180^\circ - a) = -2 \sin(180^\circ + a) = -2 \sin(360^\circ - a).$$

(2 marks)

PAST PAPERS

Q27 PAST PAPER Pearson IAL WMA11 · June 2022 · Q9

P1 · TOPIC 11 8 marks

9.

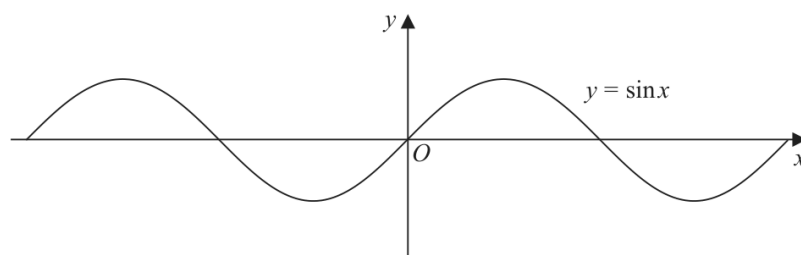


Figure 4

Figure 4 shows part of the graph of the curve with equation $y = \sin x$

Given that $\sin \alpha = p$, where $0 < \alpha < 90^\circ$

(a) state, in terms of p , the value of

(i) $2 \sin(180^\circ - \alpha)$

(ii) $\sin(\alpha - 180^\circ)$

(iii) $3 + \sin(180^\circ + \alpha)$

(3)

A copy of Figure 4, labelled Diagram 1, is shown on page 27.

On Diagram 1,

(b) sketch the graph of $y = \sin 2x$

(2)

(c) Hence find, in terms of α , the x coordinates of any points in the interval $0 < x < 180^\circ$ where

$$\sin 2x = p$$

(3)

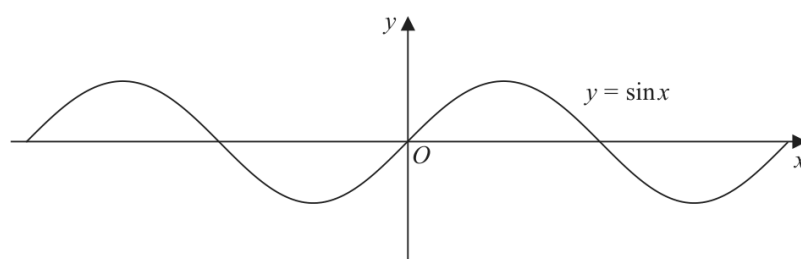


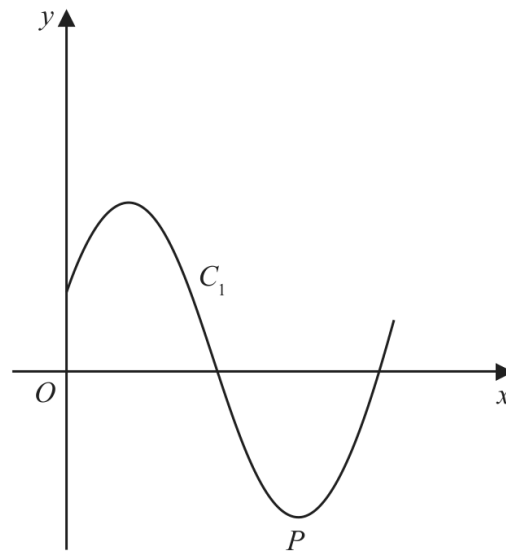
Diagram 1

(Total 8 marks)

Q28 PAST PAPER Pearson IAL WMA11 · June 2026 · Q9

P1 · TOPIC 11 5 marks

9.

**Figure 4**

The curve C_1 has equation $y = \sin\left(x + \frac{\pi}{8}\right)$

Figure 4 shows a sketch of C_1 in the interval $0 \leq x < 2\pi$

The point P is a minimum turning point on C_1 and is shown in Figure 4.

(a) State the x coordinate of P .

(1)

(b) State the number of roots of the equation

$$\sin\left(x + \frac{\pi}{8}\right) = \sin\left(\frac{\pi}{8}\right)$$

in the interval $0 \leq x < 101\pi$, giving a reason for your answer.

(2)

The curve C_2 has equation $y = -5\sin\left(x + \frac{\pi}{8}\right)$

The point Q is the minimum turning point on C_2 in the interval $0 \leq x < 2\pi$

(c) State the coordinates of Q .

(2)

Question 9 continued**(Total for Question 9 is 5 marks)**

04 Counting Solutions: Two Graphs

“How many solutions” is never solved — it is **counted**. Draw both sides as graphs and count the crossings.

WHEN YOU SEE → YOU DO

“state the number of solutions/roots” → sketch both sides, count intersections

“giving a reason” → “the graphs intersect n times”

a huge interval ($-100\pi \leq x \leq 100\pi$) → count in one period, multiply, fix the ends

$\tan x = \text{bounded curve}$ → one crossing on every branch

1

Rearrange

Put the equation as *graph you have* = *graph you can draw*.

2

Count one period

Mark the crossings in one period.

3

Scale up, check the ends

Multiply by the number of periods; look carefully at the first and last part-period.

WORKED MODEL • JANUARY 2023

$$\tan x = \pi x \text{ in } -100\pi < x < 100\pi$$

1 central branch: $x = 0$ and one each side

3 roots

2 every other full branch: the line crosses once

99 each side

ANSWER $3 + 2 \times 99 = 201$

IDEA 04

Practise: Counting Solutions

My questions first, then selected practice, then real past papers.

Q29 **DR ESLAM** Count, do not solve**P1 • TOPIC 11** 2 marks

By sketching $y = \sin x$ and $y = \frac{x}{360}$ for $0 \leq x \leq 360^\circ$, state the number of solutions of $\sin x = \frac{x}{360}$.

SELECTED PRACTICE**Q30** **PRACTICE** Practice • Medium**P1 • TOPIC 11** 5 marks

- 6 (i) On the same set of axes, sketch the graphs of $y = \cos \theta$ and $y = \cos 3\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$, giving the coordinates of all points of intersection with the coordinate axes.
- (ii) State the coordinates of any points where the graphs intersect.

(5 marks)

Q31**PRACTICE**

Practice · Hard

P1 · TOPIC 11 8 marks

- 12 (i) On the same set of axes, sketch the graphs of $y = \tan \frac{1}{4} \theta$ and $y = \cos(\theta + 120^\circ)$ in the interval $0^\circ \leq \theta \leq 270^\circ$. Show clearly the coordinates of all points of intersection with the coordinate axes.
- (ii) Deduce the number of solutions to the equation $\cos(\theta + 120^\circ) - \tan \frac{1}{4} \theta = 0$, in the interval $0^\circ \leq \theta \leq 270^\circ$.

(8 marks)

Q32**PRACTICE**

Practice · Very Hard

P1 · TOPIC 11 7 marks

- 4 (a) On the same set of axes, sketch the graphs of $y = \tan \frac{1}{2}\theta$ and $y = \tan\left(\theta - \frac{\pi}{6}\right)$ in the interval $-2\pi \leq \theta \leq 2\pi$. Label the coordinates of points of intersection with the coordinate axes.

(4 marks)

- (b) Within the interval $-2\pi \leq \theta \leq 2\pi$, determine the coordinates of the two points where $\tan \frac{1}{2}\theta = \tan\left(\theta - \frac{\pi}{6}\right)$. Give your answer in surd form.

(3 marks)

PAST PAPERS

Q33 PAST PAPER Pearson IAL WMA11 • June 2021 • Q9

P1 • TOPIC 11 7 marks

9.

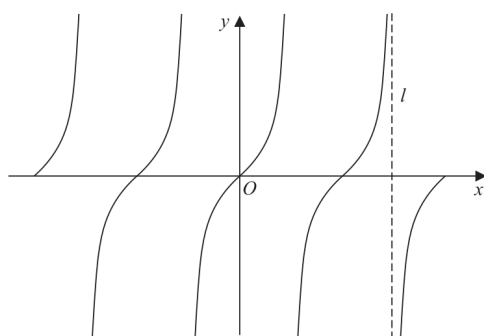


Figure 4

Figure 4 shows a sketch of the curve with equation

$$y = \tan x \quad -2\pi \leq x \leq 2\pi$$

The line l , shown in Figure 4, is an asymptote to $y = \tan x$

(a) State an equation for l .

(1)

A copy of Figure 4, labelled Diagram 1, is shown on the next page.

(b) (i) On Diagram 1, sketch the curve with equation

$$y = \frac{1}{x} + 1 \quad -2\pi \leq x \leq 2\pi$$

stating the equation of the horizontal asymptote of this curve.

(ii) Hence, **giving a reason**, state the number of solutions of the equation

$$\tan x = \frac{1}{x} + 1$$

in the region $-2\pi \leq x \leq 2\pi$

(4)

(c) State the number of solutions of the equation $\tan x = \frac{1}{x} + 1$ in the region

(i) $0 \leq x \leq 40\pi$

(ii) $-10\pi \leq x \leq \frac{5}{2}\pi$

(2)

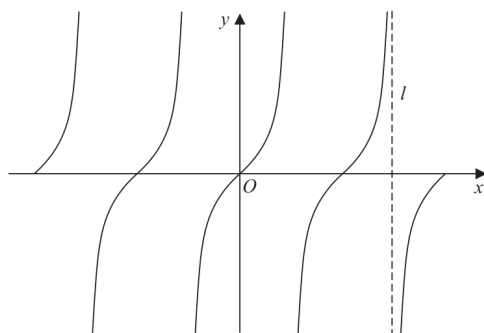


Diagram 1

(Total 7 marks)

Q34 **PAST PAPER** Pearson IAL WMA11 · January 2023 · Q9

P1 · TOPIC 11 **4 marks**

9.

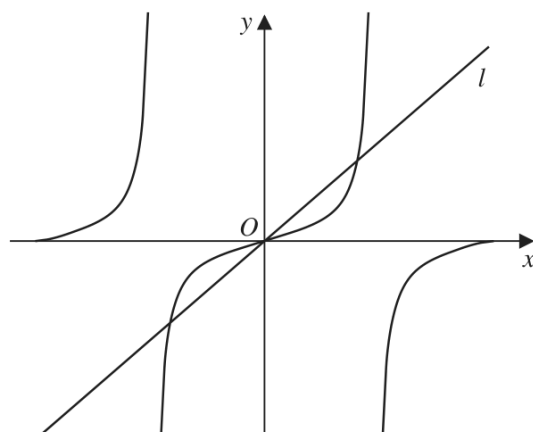


Figure 3

Figure 3 shows a sketch of

- the curve with equation $y = \tan x$
- the straight line l with equation $y = \pi x$

in the interval $-\pi < x < \pi$

(a) State the period of $\tan x$

(1)

(b) Write down the number of roots of the equation

(i) $\tan x = (\pi + 2)x$ in the interval $-\pi < x < \pi$

(1)

(ii) $\tan x = \pi x$ in the interval $-2\pi < x < 2\pi$

(1)

(iii) $\tan x = \pi x$ in the interval $-100\pi < x < 100\pi$

(1)

(Total for Question 9 is 4 marks)

Q35 PAST PAPER Pearson IAL WMA11 · October 2024 · Q7

P1 · TOPIC 11 8 marks

7.

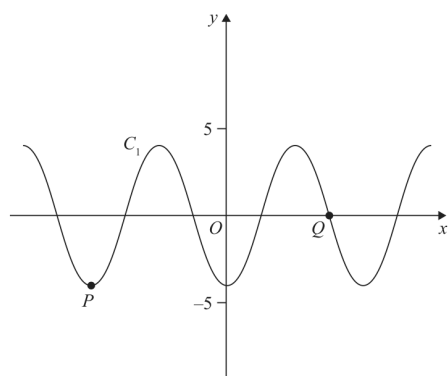
**Figure 3**

Figure 3 shows a plot of part of the curve C_1 with equation

$$y = -4 \cos x$$

where x is measured in radians.

Points P and Q lie on the curve and are shown in Figure 3.

(a) State

(i) the coordinates of P

(ii) the coordinates of Q

(3)

The curve C_2 has equation $y = -4 \cos x + k$ where x is measured in radians and k is a constant.

Given that C_2 has a maximum y value of 11

(b) (i) state the value of k

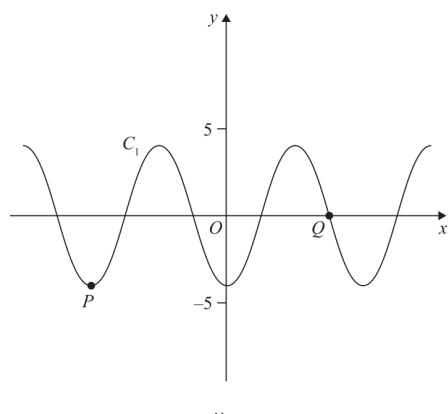
(ii) state the coordinates of the minimum point on C_2 with the smallest positive x coordinate.

(3)

On the opposite page there is a copy of Figure 3 labelled Diagram 1.

(c) Using Diagram 1, state the number of solutions of the equation

$$-4 \cos x = 5 - \frac{10}{\pi}x$$



(Total for Question 7 is 8 marks)

Q36

PAST PAPER

Pearson IAL WMA11 • January 2026 • Q9

P1 • TOPIC 11 9 marks

9.

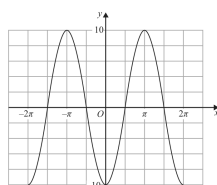


Figure 4

Figure 4 shows a sketch of part of the graph of the trigonometric function with equation $y = f(x)$.

(a) Write down an expression for $f(x)$.

(2)

Copies of Figure 4 (labelled Diagram 1 and Diagram 2) can be found on the following pages.

(b) (i) On Diagram 1 sketch a graph of the curve with equation

$$y = 10 - x^2$$

(ii) Hence find the **number** of solutions of the equation

$$f(x) = 10 - x^2 \quad \text{in the interval } -100\pi \leq x \leq 100\pi$$

(3)

(c) (i) On Diagram 2 sketch a graph of the curve with equation

$$y = \tan x \quad -2\pi \leq x \leq 2\pi$$

(ii) Hence find the **number** of solutions of the equation

$$f(x) = \tan x \quad \text{in the interval } -100\pi \leq x \leq 100\pi$$

giving a reason for your answer.

(4)

(a)

(b)(i)

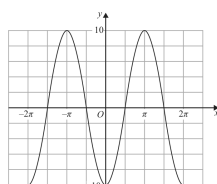
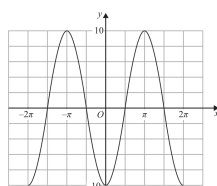


Diagram 1



Copy of Diagram 1

Only use this copy if you need to redraw your graph.

(b)(ii)

(c)(i)

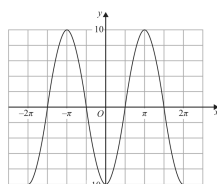
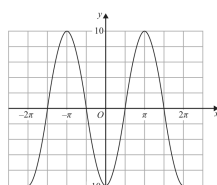


Diagram 2



Copy of Diagram 2

Only use this copy if you need to redraw your graph.

(c)(ii)

(Total for Question 9 is 9 marks)

05 Modelling with Trig

Tides, rollercoasters, dips in the ground. Same graph — the question is about its period and its max/min.

WHEN YOU SEE → YOU DO

“ $y = 30 \cos(24t)$ ”

→ period = $\frac{360}{24} = 15$ s

“how many dips / falls”

→ count minima in the interval

“at or above halfway”

→ solve $\sin(\dots) \geq \frac{1}{2}$ with the graph

WORKED MODEL • TIDAL ESTUARY

$y = \sin\left(\frac{\pi}{8}t\right)$; when is $y \geq \frac{1}{2}$?

1 $\frac{\pi}{8}t = \frac{\pi}{6}$ or $\frac{5\pi}{6}$

symmetry

2 $t = \frac{4}{3}$ to $t = \frac{20}{3}$ hours

solve

ANSWER about 3:30 pm to 8:30 pm

IDEA 05

Practise: Modelling with Trig

My questions first, then selected practice.

Q37 DR ESLAM Period of a model

P1 • TOPIC 11 2 marks

The height of a wave is $h = 2 \sin(30t)$ metres, t in seconds. Find the period and the greatest height.

SELECTED PRACTICE

Q38

PRACTICE

Practice · Medium

P1 · TOPIC 11 6 marks

- 11** A section of a new rollercoaster has a series of rises and falls. The vertical displacement of the rollercoaster carriage, y , measured in metres relative to a fixed reference height, can be modelled using the function $y = 30 \cos(24t)^\circ$, where t is the time in seconds.
- (i) Sketch the function for the interval $0 \leq t \leq 30$.
- (ii) How many times will the rollercoaster carriage fall during the 30 seconds?
- (iii) How long does the model suggest it will take for the rollercoaster carriage to reach the bottom of the first fall?

(6 marks)

Q39**PRACTICE**

Practice · Hard

P1 · TOPIC 11 5 marks

- 10** Changes in the depth of water in a small tidal estuary relative to a fixed reference depth can be modelled using the function $y = \left(\sin \frac{\pi}{8} t \right)$, where y is measured in metres, the angle is measured in radians, and t is the time in hours.

(i) Sketch the function for the interval $0 \leq t \leq 8$.

(ii) If $t = 0$ represents 2pm, during what times, to the nearest half hour, will the estuary be at or above the halfway point between $y = 0$ and its maximum depth?

(5 marks)

Q40**PRACTICE**

Practice · Hard

P1 · TOPIC 11 5 marks

11 A series of dips and mounds caused by underground mining has a cross-section which can be modelled using the function $y = 4\cos(18x)^\circ$, where x and y are respectively the horizontal and vertical displacements, in metres, from a fixed origin point.

(i) Sketch the function for the interval $0 \leq x \leq 40$ and state the periodicity of the model.

(ii) How many dips are in this model in the given interval?

(5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------|--|
| 01 | Three graphs | Five key points each. Period $360^\circ / 360^\circ / 180^\circ$. |
| 02 | $a \sin(bx + c) + k$ | a height $\cdot b$ squash (period = $\frac{360}{b}$) $\cdot c$ opposite shift $\cdot k$ up. |
| 03 | Symmetry | $\sin: 180 - \alpha \cdot \cos: 360 - \alpha \cdot \tan: \alpha + 180$. |
| 04 | Counting | Two graphs $\rightarrow$ count crossings $\rightarrow$ per period $\rightarrow$ the ends. |
| 05 | Models | Period and max/min answer the story. |

BOTTOM LINE Sketch first $\rightarrow$ move the key points $\rightarrow$ one answer from the calculator, the rest from symmetry $\rightarrow$ count, do not solve.

Answers

Trigonometric Functions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • THE THREE GRAPHS

- Q1** Dr Eslam • Five key points $(0, 1), (90, 0), (180, -1), (270, 0), (360, 1)$
- Q2** Dr Eslam • Periods and asymptotes $180^\circ; x = 90^\circ, x = 270^\circ$
- Q3** Practice • Easy
- (i) zeros $-180, 0, 180$; max $(90, 1)$, min $(-90, -1)$ (ii) $(0, 1), (90, 0), (180, -1), (270, 0), (360, 1)$ (iii) zeros $-180, 0, 180$; asymptotes $x = \pm 90$
- Q4** Practice • Medium (i) $(0, 1), (90, 0), (270, 0)$ (ii) $90^\circ, 270^\circ$
- Q5** Practice • Hard (ii) $-60^\circ, -120^\circ$

IDEA 02 • TRANSFORMATIONS

- Q6** Dr Eslam • Read a and k $k = 3$; minimum -1
- Q7** Dr Eslam • Move a point (a) $(90, 3)$ (b) $(45, 1)$ (c) $(120, 1)$
- Q8** Dr Eslam • Period from b (a) 120° (b) 720°
- Q9** Practice • Easy (i) $(90, 3)$ (ii) $(30, 1)$ (iii) $(60, 1)$
- Q10** Practice • Hard (ii) $-\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$
- Q11** Practice • Very Hard (i) $p = 5$ (ii) period $\frac{2\pi}{5}$
- Q12** Pearson IAL WMA11 • October 2019 • Q9 (a) 24π (b) $(18\pi, -1)$ (c)(i) $-12\pi - \alpha$ (ii) $6\pi - \alpha$
- Q13** Pearson IAL WMA11 • January 2020 • Q7 (a) $P(90, 3), Q(360, 0)$ (b) $R(270, 4)$
- Q14** Pearson IAL WMA11 • January 2021 • Q3 (a) $P(-180, -4), Q(450, 0)$ (b) $R(360, 7)$
- Q15** Pearson IAL WMA11 • October 2021 • Q4 (a) $Q(90, -1), k = 225$ (b) $-1 < p < 0$ or $p = 1$
- Q16** Pearson IAL WMA11 • January 2022 • Q9 (a) $A = -3$ (b) $Q(930, 3)$
- Q17** Pearson IAL WMA11 • June 2023 • Q9 (i) $f(x) = 3 \cos x$; y -int $\frac{3\sqrt{2}}{2}$ (ii) $g(x) = \sin 2x$; y -int -2
- Q18** Pearson IAL WMA11 • October 2023 • Q10 (a) $n = 3$, period 1080° (b) $(540, -3)$ (c) $k = 0.9$
- Q19** Pearson IAL WMA11 • January 2024 • Q6 (a) $(540, -5)$ (b)(i) $(360, 3)$ (ii) $(180, 5)$
- Q20** Pearson IAL WMA11 • June 2024 • Q11 (a) $(\frac{5\pi}{2}, 12)$ (b) $(\frac{3\pi}{2}, -21)$ (c) $A = -12, B = \frac{5\pi}{4}$
- Q21** Pearson IAL WMA11 • June 2025 • Q4 (a) $(270, 0)$ (b) $(-180, -4)$ (c) $100, 4, 7$

IDEA 03 • SOLVING BY SYMMETRY

- Q22** Dr Eslam • Symmetry of sine 30, 150, 390, 510
- Q23** Dr Eslam • Inside a bracket 45, 135, 225, 315
- Q24** Practice • Easy $\frac{\pi}{3}, \frac{5\pi}{3}$
- Q25** Practice • Medium (a) $B(\frac{\pi}{4}, 1)$ (b) $Q(\frac{7\pi}{12}, -\frac{1}{2}), R(\frac{11\pi}{12}, -\frac{1}{2})$
- Q26** Practice • Very Hard verified by the symmetry of sin
- Q27** Pearson IAL WMA11 • June 2022 • Q9 (a) $2p, -p, 3 - p$ (c) $\frac{\alpha}{2}, 90^\circ - \frac{\alpha}{2}$
- Q28** Pearson IAL WMA11 • June 2026 • Q9 (a) $\frac{11\pi}{8}$ (b) 102 (c) $Q(\frac{3\pi}{8}, -5)$

IDEA 04 • COUNTING SOLUTIONS

- Q29** Dr Eslam • Count, do not solve 2 (at $x = 0$ and once in $(90, 180)$)
- Q30** Practice • Medium (ii) $(-180, -1), (-90, 0), (0, 1), (90, 0), (180, -1)$
- Q31** Practice • Hard (ii) no solutions
- Q32** Practice • Very Hard (b) $(\frac{\pi}{3}, \frac{\sqrt{3}}{3}), (-\frac{5\pi}{3}, \frac{\sqrt{3}}{3})$
- Q33** Pearson IAL WMA11 • June 2021 • Q9 (a) $x = \frac{3\pi}{2}$ (b) asymptote $y = 1$; 5 solutions
(c)(i) 40 (ii) 14
- Q34** Pearson IAL WMA11 • January 2023 • Q9 (a) π (b) 3, 5, 201
- Q35** Pearson IAL WMA11 • October 2024 • Q7 (a) $P(-2\pi, -4), Q(\frac{3\pi}{2}, 0)$ (b) $k = 7$; $(2\pi, 3)$ (c) 1 root
- Q36** Pearson IAL WMA11 • January 2026 • Q9 (a) $f(x) = -10 \cos x$ (b)(ii) 2 (c)(ii) 200

IDEA 05 • MODELLING WITH TRIG

- Q37** Dr Eslam • Period of a model period 12 s; greatest height 2 m
- Q38** Practice • Medium (i) period 15 s, y from 30 to -30 (ii) 2 (iii) 7.5 s
- Q39** Practice • Hard (ii) about 3:30 pm to 8:30 pm
- Q40** Practice • Hard (i) period 20 m (ii) 2 dips

Differentiation

Rewrite as powers of x , bring the power down, subtract one. Then the gradient answers everything: tangents, normals, stationary points.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Differentiation

01 Rewrite first

$$\sqrt{x}, \frac{1}{x^2}$$

$$Ax^p + Bx^q$$

02 Using the gradient

gradient at $x = 2$

stationary point

03 Tangents and normals

tangent

normal

$$ax + by + c = 0$$

04 How many stationary points?

$$f'(x) = 0$$

$$b^2 - 4ac$$

05 First principles

chord PQ

$$2 + h$$

THE ONE RULE

$$y = ax^n \implies \frac{dy}{dx} = anx^{n-1}$$

numbers alone disappear · $\frac{dy}{dx}$ = the gradient of the curve = the gradient of the tangent

01 Rewrite First, Then Differentiate

You can only differentiate a sum of powers of x . Brackets, fractions and roots must be rewritten first.

WHEN YOU SEE → YOU DO

$$\sqrt{x}, \sqrt[3]{x}, \frac{1}{x^2}$$

$$\longrightarrow x^{\frac{1}{2}}, x^{\frac{1}{3}}, x^{-2}$$

$$\text{brackets } (2x - 1)^2(x + 1)$$

→ expand first

$$\text{a fraction } \frac{4x^2 + 9}{2\sqrt{x}}$$

→ split: each term ÷ the bottom

$$\text{"in the form } Ax^p + Bx^q\text{"}$$

→ rewrite, then read A, p, B, q

$$\frac{d^2y}{dx^2} \text{ / } f''(x)$$

→ differentiate twice

WORKED MODEL • A FRACTION WITH A ROOT

$$y = \frac{4x^2 + 9}{2\sqrt{x}}$$

$$1 \quad y = 2x^{\frac{3}{2}} + \frac{9}{2}x^{-\frac{1}{2}}$$

split and rewrite

$$2 \quad \frac{dy}{dx} = 3x^{\frac{1}{2}} - \frac{9}{4}x^{-\frac{3}{2}}$$

power down, minus one

$$\text{ANSWER } 3x^{\frac{1}{2}} - \frac{9}{4}x^{-\frac{3}{2}}$$

× **TRAP** $\frac{1}{2\sqrt{x}}$ is $\frac{1}{2}x^{-\frac{1}{2}}$, not $2x^{-\frac{1}{2}}$. The number stays on the bottom.

IDEA 01

Practise: Rewrite First, Then Differentiate

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Roots become powers

P1 · TOPIC 12 3 marks

Differentiate (a) $\sqrt{x}$ (b) $\frac{4}{x^3}$ (c) $\frac{6}{\sqrt[3]{x}}$

Q2 DR ESLAM Split the fraction

P1 · TOPIC 12 3 marks

Find $\frac{dy}{dx}$ when $y = \frac{x^3 - 4x}{\sqrt{x}}$.

Divide each term by $x^{\frac{1}{2}}$ first.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 12 3 marks

1 Differentiate

(i) $5x$,

(ii) $2x^3$,

(iii) $x^{\frac{1}{2}}$.

(3 marks)

Q4**PRACTICE**

Practice · Medium

P1 · TOPIC 12 4 marks

11 (a) Given that $y = 2x^3 - 8\sqrt{x}$, find

$$\frac{dy}{dx}$$

(2 marks)

(b) $\frac{d^2y}{dx^2}$

(2 marks)

Q5**PRACTICE**

Practice · Very Hard

P1 · TOPIC 12 3 marks

7 A curve is described by the equation $\frac{\sqrt{y}}{-1 + \sqrt{x}} = \frac{1}{x}$, $x > 1$. Find $\frac{dy}{dx}$.

(3 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA11 · January 2023 · Q1

P1 · TOPIC 12 5 marks

1. A curve C has equation

$$y = 2 + 10x^{\frac{1}{2}} - 2x^{\frac{3}{2}} \quad x > 0$$

(a) Find $\frac{dy}{dx}$ giving your answer in simplest form.

(3)

(b) Hence find the exact value of the gradient of the tangent to C at the point where $x = 2$ giving your answer in simplest form.*(Solutions relying on calculator technology are not acceptable.)*

(2)

(Total for Question 1 is 5 marks)

Q7**PAST PAPER**

Pearson IAL WMA11 · June 2023 · Q4

P1 · TOPIC 12 7 marks

4. In this question you must show all stages of your working.

(a) Write

$$y = \frac{5x^2 + \sqrt{x^3}}{\sqrt[3]{8x}}$$

in the form

$$y = Ax^p + Bx^q$$

where A , B , p and q are constants to be found.

(4)

(b) Hence find $\frac{dy}{dx}$ giving each coefficient in simplest form.

(3)

(Total for Question 4 is 7 marks)

Q8 **PAST PAPER** Pearson IAL WMA11 · October 2023 · Q1**P1 · TOPIC 12** **5 marks**

1. Given that

$$y = 5x^3 + \frac{3}{x^2} - 7x \quad x > 0$$

find, in simplest form,

(a) $\frac{dy}{dx}$ (3)

(b) $\frac{d^2y}{dx^2}$ (2)

(Total for Question 1 is 5 marks)

02 Using the Gradient

$\frac{dy}{dx}$ is a gradient machine: put x in, get the gradient out — or set it equal to a number and solve for x .

WHEN YOU SEE → YOU DO

“find the gradient at $x = 2$ ”

→ substitute into $\frac{dy}{dx}$

“the gradient is 4” / “at P and Q the gradient is 2”

→ $\frac{dy}{dx} = 4 \rightarrow$ solve for x

“stationary point”, “ $\frac{dy}{dx} = 0$ ”, “turning point”

→ solve $\frac{dy}{dx} = 0$, then find y

“the gradient is less than -18 ”

→ solve the inequality $\frac{dy}{dx} < -18$

“ a, b, c ” with gradients at two points

→ one equation per point → simultaneous

WORKED MODEL · STATIONARY POINT

$y = 2x^3 - 9x^2 + 12x$ — where is the gradient zero?

1 $\frac{dy}{dx} = 6x^2 - 18x + 12 = 0$

differentiate, = 0

2 $x^2 - 3x + 2 = 0 \Rightarrow x = 1, 2$

solve (calculator first)

ANSWER (1, 5) and (2, 4)

DISGUISED QUADRATIC $12x^4 + 5x^2 - 2 = 0$ or $4x - 5\sqrt{x} - 44 = 0$: let $u = x^2$ or $u = \sqrt{x}$, solve, then go back.

IDEA 02

Practise: Using the Gradient

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Gradient given

P1 · TOPIC 12**3 marks**

Find the point on $y = x^2 - 6x + 1$ where the gradient is 4.

Q10**DR ESLAM**

Stationary points

P1 · TOPIC 12**4 marks**

Find the coordinates of the stationary points of $y = x^3 - 3x + 2$.

SELECTED PRACTICE

Q11**PRACTICE**

Practice · Easy

P1 · TOPIC 12 4 marks

2 (a) Write down the gradient of the line with equation $y = k$, where k is a constant.

(1 mark)

(b) Find the gradient at the point where $x = 8$ for the following functions

(i) $f(x) = 3x^2$,

(ii) $f(x) = 4x^3 - 2x$,

(iii) $f(x) = 3x^{\frac{1}{3}}$.

(3 marks)

Q12**PRACTICE**

Practice · Medium

P1 · TOPIC 12 4 marks

7 (a) A curve is described by the equation $\frac{y}{x-3} = x^2 + 1$.

Make y the subject of the equation.

(1 mark)

(b) Hence find $\frac{dy}{dx}$.

(1 mark)

(c) Find the coordinates of the point on the curve where the gradient is -2 .

(2 marks)

Q13**PRACTICE**

Practice · Very Hard

P1 · TOPIC 12 5 marks

- 8 The curve with equation $y = ax^2 + bx + c$ passes through the point $(-1, 4)$. At the point $(2, 7)$ the gradient of the curve is 7. Find the values of a , b and c .

(5 marks)

PAST PAPERS

Q14 PAST PAPER Pearson IAL WMA11 · January 2020 · Q9

P1 · TOPIC 12 6 marks

9. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$y = \frac{4x^2 + 9}{2\sqrt{x}} \quad x > 0$$

Find the x coordinate of the point on the curve at which $\frac{dy}{dx} = 0$

(6)

(Total 6 marks)

Q15**PAST PAPER**

Pearson IAL WMA11 · January 2021 · Q7

P1 · TOPIC 12 9 marks

7. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = 2x - 3\sqrt{x} - 5 \quad x > 0$$

(a) Solve the equation

$$f(x) = 9$$

(4)

(b) Solve the equation

$$f''(x) = 6$$

(5)**(Total 9 marks)**

Q16**PAST PAPER**

Pearson IAL WMA11 · October 2021 · Q2

P1 · TOPIC 12 5 marks

2. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$y = 3x^5 + 4x^3 - x + 5$$

The points P and Q lie on the curve.

The gradient of the curve at both point P and point Q is 2

Find the x coordinates of P and Q .

(5)

(Total 5 marks)

Q17 PAST PAPER Pearson IAL WMA11 · January 2022 · Q7

P1 · TOPIC 12 11 marks

7.

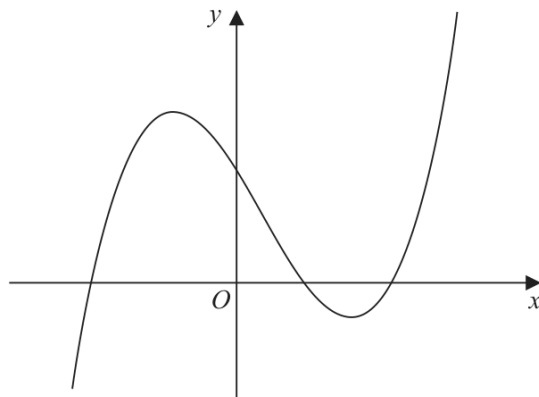
**Figure 3**

Figure 3 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (x + 4)(x - 2)(2x - 9)$$

Given that the curve with equation $y = f(x) - p$ passes through the point with coordinates $(0, 50)$

- (a) find the value of the constant p . (2)

Given that the curve with equation $y = f(x + q)$ passes through the origin,

- (b) write down the possible values of the constant q . (2)

- (c) Find $f'(x)$. (4)

- (d) Hence find the range of values of x for which the gradient of the curve with equation $y = f(x)$ is less than -18 (3)

(Total 11 marks)

Q18 PAST PAPER Pearson IAL WMA11 · June 2024 · Q7

P1 · TOPIC 12 8 marks

7. The curve C has equation $y = f(x)$ where

$$f(x) = 2x^3 - kx^2 + 14x + 24$$

and k is a constant.

(a) Find, in simplest form,

(i) $f'(x)$

(ii) $f''(x)$

(3)

The curve with equation $y = f'(x)$ intersects the curve with equation $y = f''(x)$ at the points A and B .

Given that the x coordinate of A is 5

(b) find the value of k .

(2)

(c) Hence find the coordinates of B .

(3)

(Total for Question 7 is 8 marks)

Q19**PAST PAPER**

Pearson IAL WMA11 · October 2024 · Q3

P1 · TOPIC 12 9 marks

3. In this question you must show all stages of your working.

$$f(x) = \frac{(x+5)^2}{\sqrt{x}} \quad x > 0$$

(a) Find $\int f(x) \, dx$

(4)

(b) (i) Show that when $f'(x) = 0$

$$3x^2 + 10x - 25 = 0$$

(ii) Hence state the value of x for which

$$f'(x) = 0$$

(5)**(Total for Question 3 is 9 marks)**

9.

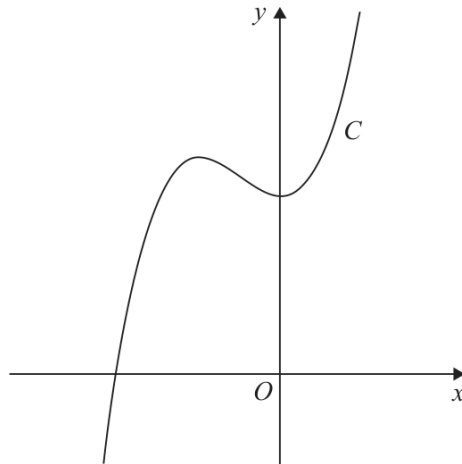
**Figure 4**

Figure 4 shows a sketch of the curve C with equation $y = f(x)$, where

$$f(x) = (x + 5)(3x^2 - 4x + 20)$$

(a) Deduce the range of values of x for which $f(x) \geq 0$ (1)

(b) Find $f'(x)$ giving your answer in simplest form. (3)

The point $R(-4, 84)$ lies on C .

Given that the tangent to C at the point P is parallel to the tangent to C at the point R

(c) find the x coordinate of P . (4)

(d) Find the point to which R is transformed when the curve with equation $y = f(x)$ is transformed to the curve with equation,

- (i) $y = f(x - 3)$
- (ii) $y = 4f(x)$ (2)

(Total for Question 9 is 10 marks)

Q21 **PAST PAPER** Pearson IAL WMA11 · June 2025 · Q2**P1 · TOPIC 12** **5 marks**

2. The curve C has equation

$$y = 2x^{\frac{5}{2}} - 4x + 3$$

- (a) Find $\frac{dy}{dx}$ writing your answer in simplest form.

(2)

The point P lies on C .

Given that

- the x coordinate of P is 2^k where k is a constant
- the gradient of C at the point P is 16

- (b) find the value of k .

(3)

(Total for Question 2 is 5 marks)

Q22**PAST PAPER**

Pearson IAL WMA11 · June 2026 · Q4

P1 · TOPIC 12 10 marks

4. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

$$f(x) = x^3 - 2x^2\sqrt{x} + 14 \quad x > 0$$

- (a) Solve the equation

$$f'(x) = 0 \quad (5)$$

- (b) (i) Show that the equation

$$f''(x) = 66$$

can be written as

$$4x - 5x^{\frac{1}{2}} - 44 = 0$$

- (ii) Hence, using algebra, show that $x = 16$ is the only solution of the equation

$$f''(x) = 66 \quad (5)$$

Question 4 continued

(Total for Question 4 is 10 marks)

03 Tangents and Normals

The tangent has the gradient of the curve. The normal is perpendicular: flip and change sign.

WHEN YOU SEE → YOU DO

“tangent to C at P ”

$$\rightarrow m = \frac{dy}{dx} \text{ at } P, \text{ then } y - y_1 = m(x - x_1)$$

“normal to C at P ”

$$\rightarrow m = -\frac{1}{dy/dx}$$

“in the form $ax + by + c = 0$ where a, b, c are integers”

→ clear fractions, all on one side

“the tangent/normal meets C again”

→ curve = line, solve (you already know one root)

“parallel to the tangent at R ”

→ same gradient: $\frac{dy}{dx} = \text{that number}$

WORKED MODEL · JUNE 2021

Normal to $y = \frac{x^2}{3} + \frac{4}{\sqrt{x}} + \frac{8}{3x} - 5$ at $P(4, 3)$.

$$1 \quad \frac{dy}{dx} = \frac{2}{3}x - 2x^{-\frac{3}{2}} - \frac{8}{3}x^{-2}$$

rewrite, differentiate

$$2 \quad \text{at } x = 4: \frac{8}{3} - \frac{1}{4} - \frac{1}{6} = \frac{9}{4}$$

gradient of tangent

$$3 \quad m_N = -\frac{4}{9}: y - 3 = -\frac{4}{9}(x - 4)$$

normal

$$\text{ANSWER } 4x + 9y - 43 = 0$$

IDEA 03

Practise: Tangents and Normals

My questions first, then selected practice, then real past papers.

Q23 DR ESLAM Tangent

P1 · TOPIC 12 3 marks

Find the equation of the tangent to $y = x^3 - 2x$ at $x = 2$.

Q24 DR ESLAM Normal, integer form

P1 · TOPIC 12 4 marks

Find the normal to $y = \frac{8}{x}$ at $(2, 4)$ in the form $ax + by + c = 0$.

SELECTED PRACTICE

Q25

PRACTICE

Practice · Easy

P1 · TOPIC 12 5 marks

10 (i) Find the gradient of the tangent at the point $(2, 3)$ on the graph of $y = 2x^3 - 3x^2 - 1$.

(ii) Hence find the equation of the tangent at the point $(2, 3)$.

(5 marks)**Q26**

PRACTICE

Practice · Hard

P1 · TOPIC 12 5 marks

9 The curve C has equation $y = 3x^2 - 6x + \sqrt{2x}$. The point $P(2, 2)$ lies on C .

Find an equation of the tangent to C at P .

(5 marks)

Q27**PRACTICE**

Practice · Very Hard

P1 · TOPIC 12 11 marks

10 (a) A curve is described by the equation $y = f(x)$, where

$$f(x) = \frac{1}{\sqrt{x}}, x > 0$$

P is the point on the curve such that the normal to the curve at P also passes through the origin.

Find the coordinates of point P . Give your answer in the form $(2^a, 2^b)$, where a and b are rational numbers to be found.

(6 marks)

(b) Write down the equation of the normal to the curve at P .

(1 mark)

(c) Show that an equation of the tangent to the curve at P is

$$\left(2^{\frac{1}{3}}\right)x + \left(2^{\frac{5}{6}}\right)y = 3$$

(4 marks)

PAST PAPERS

Q28 **PAST PAPER** Pearson IAL WMA11 · October 2019 · Q5**P1 · TOPIC 12** **7 marks**

5. A curve has equation

$$y = \frac{x^3}{6} + 4\sqrt{x} - 15 \quad x \geq 0$$

- (a) Find $\frac{dy}{dx}$, giving the answer in simplest form.

(3)

The point $P\left(4, \frac{11}{3}\right)$ lies on the curve.

- (b) Find the equation of the normal to the curve at P . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(4)**(Total 7 marks)**

Q29 **PAST PAPER** Pearson IAL WMA11 · January 2021 · Q1**P1 · TOPIC 12** **8 marks**

1. A curve has equation

$$y = 2x^3 - 5x^2 - \frac{3}{2x} + 7 \quad x > 0$$

- (a) Find, in simplest form, $\frac{dy}{dx}$ (3)

The point P lies on the curve and has x coordinate $\frac{1}{2}$

- (b) Find an equation of the normal to the curve at P , writing your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found. (5)

(Total 8 marks)

Q30 **PAST PAPER** Pearson IAL WMA11 · June 2021 · Q1**P1 · TOPIC 12** **8 marks**

1. The curve C has equation

$$y = \frac{x^2}{3} + \frac{4}{\sqrt{x}} + \frac{8}{3x} - 5 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(4)

The point $P(4, 3)$ lies on C .

- (b) Find the equation of the normal to C at the point P . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(4)**(Total 8 marks)**

Q31 PAST PAPER Pearson IAL WMA11 · June 2022 · Q10**P1 · TOPIC 12 12 marks**

10.

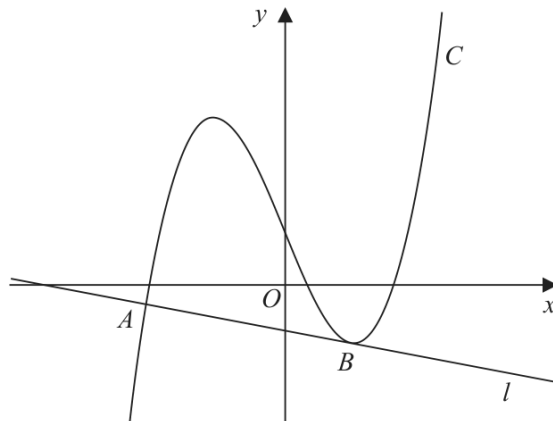
**Figure 5**

Figure 5 shows a sketch of the curve C with equation

$$y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$$

where k is a constant.

- (a) Find $\frac{dy}{dx}$ (2)

The line l , shown in Figure 5, is the normal to C at the point A with x coordinate $-\frac{7}{2}$

Given that l is also a tangent to C at the point B ,

- (b) show that the x coordinate of the point B is a solution of the equation

$$12x^2 + 4x - 33 = 0 \quad (4)$$

- (c) Hence find the x coordinate of B , justifying your answer. (2)

Given that the y intercept of l is -1

- (d) find the value of k . (4)

(Total 12 marks)

Q32**PAST PAPER**

Pearson IAL WMA11 · October 2022 · Q1

P1 · TOPIC 12 6 marks

1. The curve C has equation

$$y = \frac{x^3}{4} - x^2 + \frac{17}{x} \quad x > 0$$

- (a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(3)

The point $R\left(2, \frac{13}{2}\right)$ lies on C .

- (b) Find the equation of the tangent to C at the point R . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

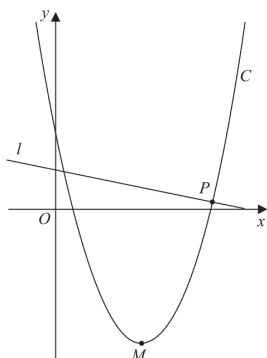
(3)**(Total for Question 1 is 6 marks)**

Q33**PAST PAPER**

Pearson IAL WMA11 · October 2022 · Q9

P1 · TOPIC 12 14 marks

9.

**Figure 3**Figure 3 shows a sketch of the curve C with equation

$$y = \frac{1}{2}x^2 - 10x + 22$$

- (a) Write $\frac{1}{2}x^2 - 10x + 22$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

The point M is the minimum turning point of C , as shown in Figure 3.

- (b) Deduce the coordinates of M

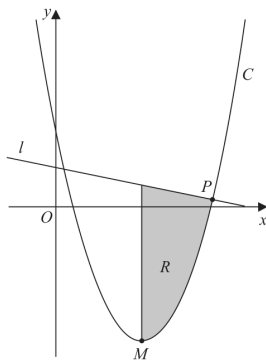
(2)

The line l is the normal to C at the point P , as shown in Figure 3.Given that l has equation $y = k - \frac{1}{8}x$, where k is a constant,

- (c) (i) find the coordinates of P

- (ii) find the value of k

(6)

Question 9 continues on the next page**Figure 4**Figure 4 is a copy of Figure 3. The finite region R , shown shaded in Figure 4, is bounded by l , C and the line through M parallel to the y -axis.

- (d) Identify the inequalities that define R .

(3)

(Total for Question 9 is 14 marks)

10.

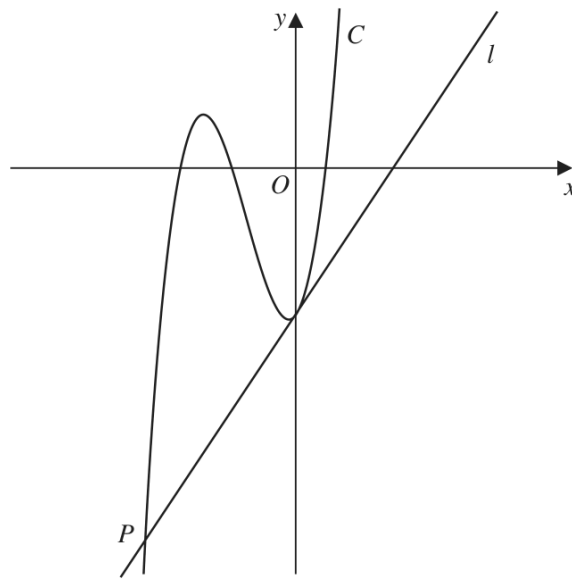
**Figure 4**

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x + 20)(x + 6)(2x - 3)$$

- (a) Use the given information to state the values of x for which

$$f(x) > 0$$

(2)

- (b) Expand $(3x + 20)(x + 6)(2x - 3)$, writing your answer as a polynomial in simplest form.

(3)

The straight line l is the tangent to C at the point where C cuts the y -axis.

Given that l cuts C at the point P , as shown in Figure 4,

- (c) find, using algebra, the x coordinate of P

(Solutions based on calculator technology are not acceptable.)

(5)

(Total for Question 10 is 10 marks)

Q35 **PAST PAPER** Pearson IAL WMA11 · January 2024 · Q3**P1 · TOPIC 12** **7 marks**

3. The curve C has equation

$$y = \frac{5x^3 - 8}{2x^2} \quad x > 0$$

- (a) Find $\frac{dy}{dx}$ writing your answer in simplest form.

(4)

The point $P(2, 4)$ lies on C .

- (b) Find an equation for the tangent to C at P writing your answer in the form $ax + by + c = 0$, where a , b and c are integers.

(3)**(Total for Question 3 is 7 marks)**

Q36 PAST PAPER Pearson IAL WMA11 · January 2025 · Q5

P1 · TOPIC 12 8 marks

5. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The curve C has equation

$$y = 4x^3 + \frac{2}{x} + 9 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(2)

Given that

- the point P lies on C
- the line with equation $y = k - 5x$, where k is a constant, is the tangent to C at P

- (b) show that the x coordinate of P satisfies the equation

$$12x^4 + 5x^2 - 2 = 0$$

(2)

- (c) Hence find the value of k .

(4)

(Total for Question 5 is 8 marks)

Q37 PAST PAPER Pearson IAL WMA11 · October 2025 · Q1

P1 · TOPIC 12 6 marks

1. The curve C has equation

$$y = 6x^2 + 3\sqrt{x} + \frac{5}{8} \quad x > 0$$

- (a) Find $\frac{dy}{dx}$ writing the answer in simplest form.

(3)

The point $P\left(\frac{1}{4}, \frac{5}{2}\right)$ lies on C .

- (b) Find the equation of the tangent to C at P , writing your answer in the form $y = mx + c$ where m and c are integers.

(3)

(Total for Question 1 is 6 marks)

Q38 PAST PAPER Pearson IAL WMA11 · October 2025 · Q10

P1 · TOPIC 12 10 marks

10.

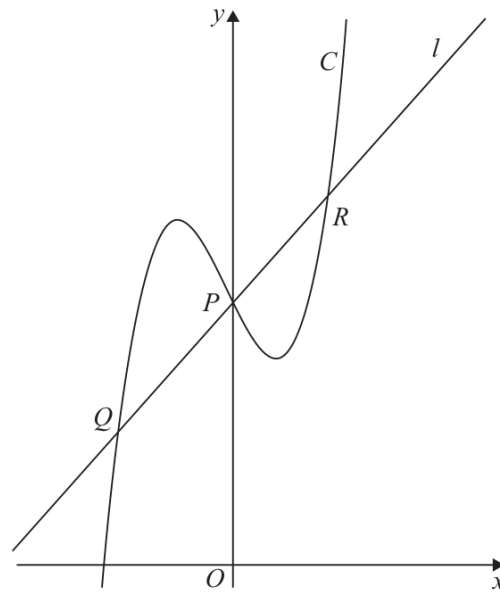


Figure 4

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 4 shows a sketch of the curve C with equation

$$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5$$

The line l is the normal to C at the point P where $x = 0$

The line l also intersects C at points Q and R as shown in Figure 4.

(a) Find, using algebra, the x coordinate of point Q .

(6)

The point T lies on C .

Given that

- the tangent to C at T is parallel to l
- the x coordinate of T is positive

(b) find, using algebra, the exact x coordinate of T .

(4)

(Total for Question 10 is 10 marks)

04 How Many Stationary Points? The Discriminant

$f'(x)$ of a cubic is a quadratic. Its discriminant tells you how many points have zero gradient.

WHEN YOU SEE → YOU DO

“exactly one point where the gradient is zero”

→ $b^2 - 4ac = 0$ for $f'(x)$

“no solutions to $f'(x) = 0$ ”

→ $b^2 - 4ac < 0$

“at least one real solution”

→ $b^2 - 4ac \geq 0$

“ $\frac{dy}{dx}$ in the form $p(x + q)^2 + r$ ”

→ complete the square: min gradient = r at $x = -q$

WORKED MODEL · NO STATIONARY POINTS

$$f(x) = x^3 - 4x^2 + 6x - 9$$

1 $f'(x) = 3x^2 - 8x + 6$

differentiate

2 $b^2 - 4ac = 64 - 72 = -8 < 0$

discriminant

ANSWER $f'(x) = 0$ has no real solutions

IDEA 04

Practise: How Many Stationary Points?

My questions first, then selected practice, then real past papers.

Q39 **DR ESLAM** Discriminant of $f'(x)$

P1 · TOPIC 12 3 marks

$f(x) = x^3 + kx^2 + 3x$. Find the values of k for which $f'(x) = 0$ has exactly one solution.

SELECTED PRACTICE**Q40** **PRACTICE** Practice · Medium

P1 · TOPIC 12 4 marks

6 (a) The function f is defined by $f(x) = x^3 - 6x^2 - cx + 12$.

Find $f'(x)$.

(2 marks)

(b) Given that the equation $f'(x) = 0$ has exactly one real solution, find the value of c .

(2 marks)

Q41**PRACTICE**

Practice · Hard

P1 · TOPIC 12**4 marks**

- 6 A curve has the equation $y = 4x^3 + bx^2 + 3x - 17$, where b is a constant. Given that there is only one point on the curve where the gradient is zero, determine the possible values of b .

(4 marks)**Q42****PRACTICE**

Practice · Very Hard

P1 · TOPIC 12**4 marks**

- 6 The function f is defined by $f(x) = x^n - x$, $n \in \mathbb{N}$, $n \geq 2$. Determine the relationship between the value of n and the number of real solutions to the equation $f'(x) = 0$.

(4 marks)

PAST PAPERS

Q43 PAST PAPER Pearson IAL WMA11 · January 2026 · Q7

P1 · TOPIC 12 10 marks

7. The curve C has equation

$$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3}$$

- (a) Show that $\frac{dy}{dx}$ can be written in the form

$$p(x + q)^2 + r$$

where p , q and r are constants to be found.

(5)

- (b) Hence state

- (i) the minimum value of $\frac{dy}{dx}$

- (ii) the value of x at which this minimum value occurs.

(2)

Given that S is the point on C at which the gradient is a minimum,

- (c) find the equation of the tangent to C at S , giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

(Total for Question 7 is 10 marks)

05 From First Principles: the Chord PQ

The gradient of a chord from P to a point h away tends to the gradient of the tangent as $h \rightarrow 0$.

WHEN YOU SEE $\rightarrow$ YOU DO

“Q with x coordinate $2 + h$ ”

$$\rightarrow \text{gradient } PQ = \frac{y_Q - y_P}{h}$$

“explain the relationship”

$\rightarrow$ as $h \rightarrow 0$, the chord gradient $\rightarrow$ the tangent gradient

WORKED MODEL · JANUARY 2020

$y = x^2 + 3x - 2$, $P(3, 16)$, Q at $x = 3 + h$.

1 $y_Q = (3 + h)^2 + 3(3 + h) - 2 = 16 + 9h + h^2$

substitute

2 gradient = $\frac{9h + h^2}{h}$

change in y over h

ANSWER $9 + h \rightarrow 9$ as $h \rightarrow 0$

IDEA 05

Practise: From First Principles

My questions first, then real past papers.

Q44 **DR ESLAM** Chord to tangent

P1 • TOPIC 12 3 marks

$y = x^2$. P is $(1, 1)$ and Q has x coordinate $1 + h$. Find the gradient of PQ and its limit as $h \rightarrow 0$.

PAST PAPERS

Q45 PAST PAPER Pearson IAL WMA11 · October 2019 · Q7

P1 · TOPIC 12 6 marks

7.

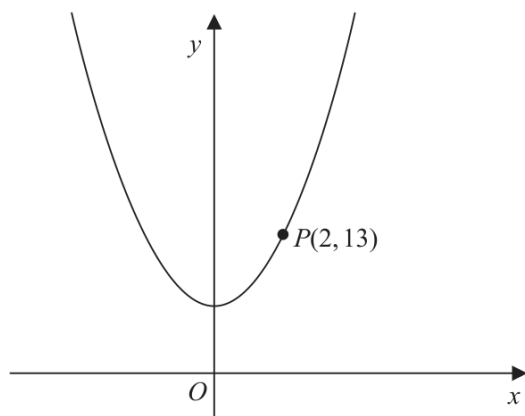


Figure 4

Figure 4 shows part of the curve with equation $y = 2x^2 + 5$

The point $P(2, 13)$ lies on the curve.

- (a) Find the gradient of the tangent to the curve at P . (2)

The point Q with x coordinate $2 + h$ also lies on the curve.

- (b) Find, in terms of h , the gradient of the line PQ . Give your answer in simplest form. (3)

- (c) Explain briefly the relationship between the answer to (b) and the answer to (a). (1)

(Total 6 marks)

Q46 PAST PAPER Pearson IAL WMA11 · January 2020 · Q3

P1 · TOPIC 12 6 marks

3.

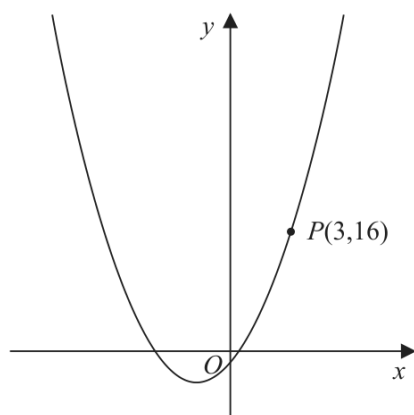
**Figure 1**

Figure 1 shows part of the curve with equation $y = x^2 + 3x - 2$

The point $P(3,16)$ lies on the curve.

(a) Find the gradient of the tangent to the curve at P .

(2)

The point Q with x coordinate $3 + h$ also lies on the curve.

(b) Find, in terms of h , the gradient of the line PQ . Write your answer in simplest form.

(3)

(c) Explain briefly the relationship between the answer to (b) and the answer to (a).

(1)**(Total 6 marks)**

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 Rewrite** Roots → fractional powers, “over” → negative powers, brackets → expand.
- 02 Use it** Gradient at x : substitute. Given gradient: $\frac{dy}{dx} = k$, solve.
- 03 Tangent / normal** $m_T = \frac{dy}{dx}$, $m_N = -\frac{1}{m_T}$, $y - y_1 = m(x - x_1)$.
- 04 How many?** Discriminant of $f'(x)$. Min gradient: complete the square.
- 05 First principles** $\frac{f(x+h) - f(x)}{h} \rightarrow f'(x)$.

BOTTOM LINE Rewrite → differentiate → substitute or solve → “tangent” means the derivative, “normal” means $-\frac{1}{\text{derivative}}$.

Answers

Differentiation • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • REWRITE FIRST, THEN DIFFERENTIATE

- Q1** Dr Eslam • Roots become powers
(a) $\frac{1}{2}x^{-\frac{1}{2}}$ (b) $-12x^{-4}$ (c) $-2x^{-\frac{4}{3}}$
- Q2** Dr Eslam • Split the fraction
 $\frac{5}{2}x^{\frac{3}{2}} - 2x^{-\frac{1}{2}}$
- Q3** Practice • Easy
(i) 5 (ii) $6x^2$ (iii) $\frac{1}{2}x^{-\frac{1}{2}}$
- Q4** Practice • Medium
(a) $6x^2 - 4x^{-\frac{1}{2}}$ (b) $12x + 2x^{-\frac{3}{2}}$
- Q5** Practice • Very Hard
 $-x^{-2} + 3x^{-\frac{5}{2}} - 2x^{-3}$
- Q6** Pearson IAL WMA11 • January 2023 • Q1
(a) $5x^{-\frac{1}{2}} - 3x^{\frac{1}{2}}$ (b) $-\frac{\sqrt{2}}{2}$
- Q7** Pearson IAL WMA11 • June 2023 • Q4
(a) $\frac{5}{2}x^{\frac{5}{3}} + \frac{1}{2}x^{\frac{7}{6}}$ (b) $\frac{25}{6}x^{\frac{2}{3}} + \frac{7}{12}x^{\frac{1}{6}}$
- Q8** Pearson IAL WMA11 • October 2023 • Q1
(a) $15x^2 - 6x^{-3} - 7$ (b) $30x + 18x^{-4}$

IDEA 02 • USING THE GRADIENT

- Q9** Dr Eslam · Gradient given $2x - 6 = 4 \Rightarrow (5, -4)$
- Q10** Dr Eslam · Stationary points $(1, 0)$ and $(-1, 4)$
- Q11** Practice · Easy (a) 0 (b) 48, 766, $\frac{1}{4}$
- Q12** Practice · Medium (a) $y = x^3 - 3x^2 + x - 3$ (b) $3x^2 - 6x + 1$ (c) $(1, -4)$
- Q13** Practice · Very Hard $a = 2, b = -1, c = 1$
- Q14** Pearson IAL WMA11 · January 2020 · Q9 $x = \frac{\sqrt{3}}{2}$
- Q15** Pearson IAL WMA11 · January 2021 · Q7 (a) $x = \frac{49}{4}$ (b) $x = \frac{1}{4}$
- Q16** Pearson IAL WMA11 · October 2021 · Q2 $x = \pm \frac{1}{\sqrt{5}}$
- Q17** Pearson IAL WMA11 · January 2022 · Q7 (a) $p = 22$ (b) $q = -4, 2, 4.5$ (c) $6x^2 - 10x - 34$
(d) $-1 < x < \frac{8}{3}$
- Q18** Pearson IAL WMA11 · June 2024 · Q7 (a) $6x^2 - 2kx + 14, 12x - 2k$ (b) $k = 13$ (c) $B(\frac{4}{3}, -10)$
- Q19** Pearson IAL WMA11 · October 2024 · Q3 (a) $\frac{2}{5}x^{\frac{5}{2}} + \frac{20}{3}x^{\frac{3}{2}} + 50x^{\frac{1}{2}} + c$ (b)(ii) $x = \frac{5}{3}$
- Q20** Pearson IAL WMA11 · October 2024 · Q9 (a) $x \geq -5$ (b) $9x^2 + 22x$ (c) $x = \frac{14}{9}$ (d) $(-1, 84), (-4, 336)$
- Q21** Pearson IAL WMA11 · June 2025 · Q2 (a) $5x^{\frac{3}{2}} - 4$ (b) $k = \frac{4}{3}$
- Q22** Pearson IAL WMA11 · June 2026 · Q4 (a) $x = \frac{25}{9}$ (b)(ii) $\sqrt{x} = 4$ ($\sqrt{x} = -\frac{11}{4}$ rejected)

IDEA 03 · TANGENTS AND NORMALS

- Q23** Dr Eslam · Tangent $y = 10x - 16$
- Q24** Dr Eslam · Normal, integer form gradient -2 , normal gradient $\frac{1}{2}$: $x - 2y + 6 = 0$
- Q25** Practice · Easy (i) 12 (ii) $y = 12x - 21$
- Q26** Practice · Hard $y = \frac{13}{2}x - 11$
- Q27** Practice · Very Hard (a) $P(2^{-\frac{1}{3}}, 2^{\frac{1}{6}})$ (b) $y = \sqrt{2}x$
- Q28** Pearson IAL WMA11 · October 2019 · Q5 (a) $\frac{1}{2}x^2 + 2x^{-\frac{1}{2}}$ (b) $x + 9y - 37 = 0$
- Q29** Pearson IAL WMA11 · January 2021 · Q1 (a) $6x^2 - 10x + \frac{3}{2}x^{-2}$ (b) $2x + 5y - 16 = 0$
- Q30** Pearson IAL WMA11 · June 2021 · Q1 (a) $\frac{2}{3}x - 2x^{-\frac{3}{2}} - \frac{8}{3}x^{-2}$ (b) $4x + 9y - 43 = 0$
- Q31** Pearson IAL WMA11 · June 2022 · Q10 (a) $\frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}$ (c) $x = \frac{3}{2}$ (d) $k = \frac{5}{4}$
- Q32** Pearson IAL WMA11 · October 2022 · Q1 (a) $\frac{3}{4}x^2 - 2x - 17x^{-2}$ (b) $21x + 4y - 68 = 0$
- Q33** Pearson IAL WMA11 · October 2022 · Q9 (a) $\frac{1}{2}(x - 10)^2 - 28$ (b) $M(10, -28)$ (c) $P(18, 4)$, $k = \frac{25}{4}$ (d) $10 \leq x \leq 18$, curve $\leq y \leq$ line
- Q34** Pearson IAL WMA11 · January 2023 · Q10 (a) $-\frac{20}{3} < x < -6$ or $x > \frac{3}{2}$ (b) $6x^3 + 67x^2 + 126x - 360$ (c) $x = -\frac{67}{6}$
- Q35** Pearson IAL WMA11 · January 2024 · Q3 (a) $\frac{5}{2} + 8x^{-3}$ (b) $7x - 2y - 6 = 0$
- Q36** Pearson IAL WMA11 · January 2025 · Q5 (a) $12x^2 - 2x^{-2}$ (c) $x = \frac{1}{2}$, $k = 16$
- Q37** Pearson IAL WMA11 · October 2025 · Q1 (a) $12x + \frac{3}{2}x^{-\frac{1}{2}}$ (b) $y = 6x + 1$
- Q38** Pearson IAL WMA11 · October 2025 · Q10 (a) $x = -\frac{5}{4}$ (b) $x = \frac{-1 + \sqrt{61}}{12}$

IDEA 04 · HOW MANY STATIONARY POINTS?

- Q39** Dr Eslam · Discriminant of $f'(x)$ $4k^2 - 36 = 0 \Rightarrow k = \pm 3$
- Q40** Practice · Medium (a) $3x^2 - 12x - c$ (b) $c = -12$
- Q41** Practice · Hard $b = \pm 6$
- Q42** Practice · Very Hard n even: one solution · n odd: two solutions
- Q43** Pearson IAL WMA11 · January 2026 · Q7 (a) $2(x-4)^2 + 11$ (b) 11 at $x = 4$ (c) $y = 11x + 36$

IDEA 05 · FROM FIRST PRINCIPLES

- Q44** Dr Eslam · Chord to tangent $2 + h \rightarrow 2$
- Q45** Pearson IAL WMA11 · October 2019 · Q7 (a) 8 (b) $8+2h$ (c) as $h \rightarrow 0$ the chord gradient $\rightarrow 8$
- Q46** Pearson IAL WMA11 · January 2020 · Q3 (a) 9 (b) $9 + h$ (c) as $h \rightarrow 0$, $9 + h \rightarrow 9$



DR ESLAM AHMED
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Integration

Integration is differentiation backwards: add one to the power, divide by the new power, and never forget +c. Then use a point to find c.

THE TOPIC ON ONE MAP

Integration

01 - Integrate powers of x

$$\int x^n dx \quad +c$$

02 - Brackets and fractions first

expand divide

03 - Find the curve

the point gives c tangent at P

04 - From $f''(x)$: integrate twice

two constants $f'(2) = 8$

THE ONE RULE

$$\int ax^n dx = \frac{ax^{n+1}}{n+1} + c \quad (n \neq -1)$$

power up by one · divide by the new power · +c

01 Integrate Powers of x

Same rewriting as differentiation, then the rule backwards. The number in front is divided by the **new** power.

WHEN YOU SEE → YOU DO

“find $\int \dots dx$ ”, “use calculus to find”

→ power +1, ÷ new power, +c

a number alone, e.g. -5

→ becomes $-5x$

“giving each term in simplest form”

→ simplify every coefficient and power

a fractional power $x^{\frac{1}{2}}$

→ $\div \frac{3}{2}$ means $\times \frac{2}{3}$

WORKED MODEL • JANUARY 2020

$$\int \left(\frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 \right) dx$$

$$1 \quad \frac{8}{3}x^3 - \frac{1}{2}x^{-\frac{1}{2}} - 5$$

rewrite

$$2 \quad \frac{8}{3} \cdot \frac{x^4}{4} - \frac{1}{2} \cdot \frac{x^{1/2}}{1/2} - 5x$$

power up, divide

ANSWER $\frac{2}{3}x^4 - x^{\frac{1}{2}} - 5x + c$

× **TRAP** $\frac{x^{\frac{3}{2}}}{\frac{3}{2}}$ is $\frac{2}{3}x^{\frac{3}{2}}$ — dividing by a fraction multiplies by its flip.

IDEA 01

Practise: Integrate Powers of x

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Power up, divide, +c

P1 • TOPIC 13 3 marks

Find (a) $\int 6x^2 dx$ (b) $\int \frac{4}{x^3} dx$ (c) $\int \sqrt{x} dx$

Q2 DR ESLAM Numbers alone

P1 • TOPIC 13 2 marks

Find $\int (5 - 3x^2) dx$.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 13 3 marks

1 Integrate

(i) $2x$,

(ii) $6x^2$,

(iii) $\frac{1}{2}x^{\frac{1}{2}}$.

(3 marks)

Q4**PRACTICE**

Practice · Medium

P1 · TOPIC 13 3 marks

2 Given $y = 2t^2 + t^{-\frac{1}{2}} - 3t^{-4}$, find $\int y \, dt$.

(3 marks)

Q5**PRACTICE**

Practice · Very Hard

P1 · TOPIC 13 3 marks

- 1 Use calculus to find

$$\int \left(2\sqrt{x} + 5x^{\frac{1}{3}} \right) dx$$

(3 marks)**PAST PAPERS****Q6****PAST PAPER**

Pearson IAL WMA11 · January 2020 · Q1

P1 · TOPIC 13 4 marks

1. Find, in simplest form,

$$\int \left(\frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 \right) dx$$

(4)**(Total 4 marks)**

Q7 **PAST PAPER** Pearson IAL WMA11 · October 2021 · Q1

P1 · TOPIC 13 5 marks

1. Find

$$\int 12x^3 + \frac{1}{6\sqrt{x}} - \frac{3}{2x^4} dx$$

giving each term in simplest form.

(5)

(Total 5 marks)

Q8 **PAST PAPER** Pearson IAL WMA11 · January 2022 · Q1

P1 · TOPIC 13 4 marks

1. Find

$$\int \left(\frac{8x^3}{5} - \frac{2}{3x^4} - 1 \right) dx$$

giving each term in simplest form.

(4)

(Total 4 marks)

Q9 PAST PAPER Pearson IAL WMA11 · June 2022 · Q1

P1 · TOPIC 13 4 marks

1. Find

$$\int \left(10x^5 + 6x^3 - \frac{3}{x^2} \right) dx$$

giving your answer in simplest form.

(4)

(Total 4 marks)

Q10 PAST PAPER Pearson IAL WMA11 · June 2024 · Q1

P1 · TOPIC 13 3 marks

1. Find

$$\int \left(10x^4 - \frac{3}{2x^3} - 7 \right) dx$$

giving each term in simplest form.

(3)

(Total for Question 1 is 3 marks)

Q11 **PAST PAPER** Pearson IAL WMA11 · January 2025 · Q1**P1 · TOPIC 13** **4 marks**

1. Find

$$\int \left(8x^3 - 6\sqrt{x} - \frac{2}{5x^3} \right) dx$$

giving your answer in simplest form.

(4)**(Total for Question 1 is 4 marks)**

Q12 **PAST PAPER** Pearson IAL WMA11 • January 2026 • Q1**P1 • TOPIC 13** **6 marks**

1. $y = 4x^3 + \frac{3}{2\sqrt{x}} - 8 \quad x > 0$

Find, in simplest form,

(a) $\frac{dy}{dx}$ (3)

(b) $\int y \, dx$ (3)

(Total for Question 1 is 6 marks)

02 Brackets and Fractions First

You cannot integrate a product or a quotient term by term. Expand the brackets, split the fraction, then integrate.

WHEN YOU SEE → YOU DO

$$(x-3)^2(2x+5), (3x+\frac{2}{x})^2$$

→ expand first

$$\frac{(x+3)^2}{3\sqrt{x}}$$

→ expand the top, then $\div 3x^{\frac{1}{2}}$ term by term

“given $(x-5)(2x+1)(x+3) \equiv ax^3 + \dots$, hence find”

→ use the expansion you just found

WORKED MODEL • JUNE 2025

$$\int \frac{(x+3)^2}{3\sqrt{x}} dx$$

$$1 \quad \frac{x^2 + 6x + 9}{3x^{\frac{1}{2}}} = \frac{1}{3}x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + 3x^{-\frac{1}{2}}$$

expand, split

$$2 \quad \frac{1}{3} \cdot \frac{2}{5}x^{\frac{5}{2}} + 2 \cdot \frac{2}{3}x^{\frac{3}{2}} + 3 \cdot 2x^{\frac{1}{2}}$$

integrate

$$\text{ANSWER} \quad \frac{2}{15}x^{\frac{5}{2}} + \frac{4}{3}x^{\frac{3}{2}} + 6x^{\frac{1}{2}} + C$$

IDEA 02

Practise: Brackets and Fractions First

My questions first, then selected practice, then real past papers.

Q13 DR ESLAM Expand first

P1 • TOPIC 13 3 marks

Find $\int (x + 2)(x - 5) dx$.

Q14 DR ESLAM Split the fraction

P1 • TOPIC 13 4 marks

Find $\int \frac{x^2 - 4}{\sqrt{x}} dx$.

$x^{\frac{3}{2}} - 4x^{-\frac{1}{2}}$ first.

SELECTED PRACTICE

Q15**PRACTICE**

Practice · Easy

P1 • TOPIC 13 5 marks

3 (a) Show that

$$\frac{3x^3 + 4x^6}{x^2}$$

can be written as $3x^a + 4x^b$, where a and b are constants to be found.

(2 marks)**(b)** Hence find

$$\int \frac{3x^2 + 4x^6}{x^2} dx.$$

(3 marks)

Q16**PRACTICE**

Practice · Hard

P1 · TOPIC 13 6 marks**4 (a)** Show that

$$(2 - x)^3 = 8 - 12x + 6x^2 - x^3$$

(3 marks)**(b)** Hence, or otherwise, work out

$$\int (2 - x)^3 dx$$

(3 marks)

Q17**PRACTICE**

Practice · Very Hard

P1 · TOPIC 13 6 marks**4 (a)** Show that

$$\left(3 - \frac{1}{2}x\right)^3 = 27 - \frac{27}{2}x + \frac{9}{4}x^2 - \frac{1}{8}x^3$$

(3 marks)**(b)** Hence, or otherwise, work out

$$\int \left(2\left(3 - \frac{1}{2}x\right)\right)^3 dx$$

(3 marks)

PAST PAPERS

Q18 PAST PAPER Pearson IAL WMA11 • January 2021 • Q9

P1 • TOPIC 13 11 marks

9. (i) Find

$$\int \frac{(3x+2)^2}{4\sqrt{x}} dx \quad x > 0$$

giving your answer in simplest form.

(5)

(ii) A curve C has equation $y = f(x)$.

Given

- $f'(x) = x^2 + ax + b$ where a and b are constants
- the y intercept of C is -8
- the point $P(3, -2)$ lies on C
- the gradient of C at P is 2

find, in simplest form, $f(x)$.

(6)

(Total 11 marks)

Q19 **PAST PAPER** Pearson IAL WMA11 · June 2021 · Q4**P1 · TOPIC 13** **6 marks**

4. Find

$$\int \frac{(3\sqrt{x} + 2)(x - 5)}{4\sqrt{x}} dx$$

writing each term in simplest form.

(6)

(Total 6 marks)

Q20 **PAST PAPER** Pearson IAL WMA11 • October 2022 • Q2**P1 • TOPIC 13 7 marks**

2. Given that

$$(x - 5)(2x + 1)(x + 3) \equiv ax^3 + bx^2 - 32x - 15$$

where a and b are constants,(a) find the value of a and the value of b .

(2)

(b) Hence find

$$\int \frac{(x - 5)(2x + 1)(x + 3)}{5\sqrt{x}} dx$$

writing each term in simplest form.

(5)

(Total for Question 2 is 7 marks)

Q21 **PAST PAPER** Pearson IAL WMA11 • January 2023 • Q3**P1 • TOPIC 13 5 marks**

3. Find

$$\int \frac{4x^5 + 3}{2x^2} dx$$

giving your answer in simplest form.

(5)

(Total for Question 3 is 5 marks)

Q22**PAST PAPER**

Pearson IAL WMA11 · January 2024 · Q1

P1 · TOPIC 13 5 marks

1. Find

$$\int (2x - 5)(3x + 2)(2x + 5)dx$$

writing your answer in simplest form.

(5)**(Total for Question 1 is 5 marks)**

Q23 **PAST PAPER** Pearson IAL WMA11 • June 2025 • Q3**P1 • TOPIC 13** **5 marks**

3. Find

$$\int \frac{(x+3)^2}{3\sqrt{x}} dx$$

writing your answer in simplest form.

(5)**(Total for Question 3 is 5 marks)**

Q24 **PAST PAPER** Pearson IAL WMA11 • October 2025 • Q2**P1 • TOPIC 13** **5 marks**

2. Find

$$\int (x-3)^2(2x+5) \, dx$$

writing the answer in simplest form.

(5)**(Total for Question 2 is 5 marks)**

Q25 **PAST PAPER** Pearson IAL WMA11 • June 2026 • Q1**P1 • TOPIC 13** **5 marks**

1. Find

$$\int \left(3x + \frac{2}{x} \right)^2 dx$$

giving the answer in simplest form.

(5)**Question 1 continued****(Total for Question 1 is 5 marks)**

03 Find the Curve: Use the Point for c

Given $f'(x)$ and one point, integrate to get $f(x)$ with $+c$, then put the point in to find c .

WHEN YOU SEE → YOU DO

“given $f'(x) = \dots$ and $P(4, 20)$ lies on C , find $f(x)$ ”

→ integrate, substitute P , find c

“the curve passes through $(4, 64)$ ”

→ same: the point fixes c

“the gradient at P ” / “tangent at P ”

→ substitute into $f'(x)$ — no integration needed

“tangent at P is $y = 10x + c$ ”

→ gradient = 10; the point is on the line

1

Integrate $f'(x)$

Term by term, with $+c$.

2

Substitute the point

$f(x_0) = y_0$ gives one equation for c .

3

Write $f(x)$ in full

With the number, simplified.

WORKED MODEL • OCTOBER 2022

$$f'(x) = \frac{12}{\sqrt{x}} + \frac{x}{3} - 4, \quad P(9, 8) \text{ on } C.$$

$$1 \quad f(x) = 24x^{\frac{1}{2}} + \frac{1}{6}x^2 - 4x + c$$

integrate

$$2 \quad 8 = 24(3) + \frac{81}{6} - 36 + c \Rightarrow c = -\frac{83}{2}$$

use P

ANSWER $f(x) = 24\sqrt{x} + \frac{1}{6}x^2 - 4x - \frac{83}{2}$

IDEA 03

Practise: Find the Curve

My questions first, then selected practice, then real past papers.

Q26 DR ESLAM One point, one c

P1 · TOPIC 13 4 marks

$f'(x) = 6x - 4$ and the curve passes through $(2, 5)$. Find $f(x)$.

$f(x) =$ _____

Q27 DR ESLAM Tangent given

P1 · TOPIC 13 4 marks

$f'(x) = 3x^2 + k$. The tangent at $P(1, 4)$ has equation $y = 5x - 1$. Find k and $f(x)$.

SELECTED PRACTICE

Q28**PRACTICE**

Practice · Easy

P1 · TOPIC 13

6 marks

4 (a) Integrate $5x^4 + 6x^2 + 2x + 3$.

(3 marks)

(b) Given that $f(x) = \int (5x^4 + 6x^2 + 2x + 3) dx$ and that the graph of $y = f(x)$ passes through the point $(1, 10)$, find an expression for $f(x)$ in terms of x only.

(3 marks)

Q29**PRACTICE**

Practice · Medium

P1 · TOPIC 13

3 marks

3 Find the equation of the curve passing through the point $(4, 64)$ and given by

$$y = \int (3 - 2x + 6x^2) dx$$

(3 marks)

Q30**PRACTICE**

Practice · Very Hard

P1 · TOPIC 13 4 marks

3 Find the equation of the curve passing through the point (4, -8) and given by

$$y = \int \left(\frac{2}{\sqrt{x}} - x - 3 \right) dx$$

(4 marks)

PAST PAPERS

Q31 PAST PAPER Pearson IAL WMA11 · June 2021 · Q6

P1 · TOPIC 13 8 marks

6. The curve C has equation $y = f(x)$, $x > 0$

Given that

- C passes through the point $P(8, 2)$
- $f'(x) = \frac{32}{3x^2} + 3 - 2(\sqrt[3]{x})$

- (a) find the equation of the tangent to C at P . Write your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

- (b) Find, in simplest form, $f(x)$.

(5)

(Total 8 marks)

Q32 **PAST PAPER** Pearson IAL WMA11 • October 2021 • Q10**P1 • TOPIC 13 7 marks**

10. A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = ax - 12x^{\frac{1}{3}}$, where a is a constant
- $f''(x) = 0$ when $x = 27$
- the curve passes through the point $(1, -8)$

(a) find the value of a .

(3)

(b) Hence find $f(x)$.

(4)

(Total 7 marks)

Q33 **PAST PAPER** Pearson IAL WMA11 • January 2022 • Q6**P1 • TOPIC 13 11 marks**

6. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \frac{(x+3)^2}{x\sqrt{x}}$
- the point $P(4, 20)$ lies on C

- (a) (i) find the value of the gradient at P

- (ii) Hence find the equation of the tangent to C at P , giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

- (b) Find $f(x)$, simplifying your answer.

(7)

(Total 11 marks)

Q34 **PAST PAPER** Pearson IAL WMA11 · June 2022 · Q7**P1 · TOPIC 13** **9 marks**

7. The curve C has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = \frac{2}{\sqrt{x}} + \frac{A}{x^2} + 3$, where A is a constant
- $f''(x) = 0$ when $x = 4$

- (a) find the value of A .

(4)

Given also that

- $f(x) = 8\sqrt{3}$, when $x = 12$

- (b) find $f(x)$, giving each term in simplest form.

(5)

(Total 9 marks)

Q35**PAST PAPER**

Pearson IAL WMA11 · October 2022 · Q5

P1 · TOPIC 13 9 marks

5. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = \frac{12}{\sqrt{x}} + \frac{x}{3} - 4$
- the point $P(9, 8)$ lies on C

(a) find, in simplest form, $f(x)$

(5)

The line l is the normal to C at P

(b) Find the coordinates of the point at which l crosses the y -axis.

(4)

(Total for Question 5 is 9 marks)

Q36 **PAST PAPER** Pearson IAL WMA11 • June 2023 • Q8**P1 • TOPIC 13 10 marks**

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (a) Find the equation of the tangent to the curve with equation

$$y = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

at the point $P(4, 12)$

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(5)

The curve with equation $y = f(x)$ also passes through the point $P(4, 12)$

Given that

$$f'(x) = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

- (b) find $f(x)$ giving the coefficients in simplest form.

(5)

(Total for Question 8 is 10 marks)

Q37 **PAST PAPER** Pearson IAL WMA11 • October 2023 • Q7**P1 • TOPIC 13 10 marks**

7. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \frac{4x^2 + 10 - 7x^{\frac{1}{2}}}{4x^{\frac{1}{2}}}$
- the point $P(4, -1)$ lies on C

(a) (i) find the value of the gradient of C at P

(ii) Hence find the equation of the normal to C at P , giving your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(4)

(b) Find $f(x)$.

(6)

(Total for Question 7 is 10 marks)

Q38 **PAST PAPER** Pearson IAL WMA11 · January 2024 · Q10**P1 · TOPIC 13** **8 marks**

10. In this question you must show all stages of your working.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- the point $P(2, 8\sqrt{2})$ lies on C
- $f'(x) = 4\sqrt{x^3} + \frac{k}{x^2}$ where k is a constant
- $f''(x) = 0$ at P

(a) find the exact value of k ,

(4)

(b) find $f(x)$, giving your answer in simplest form.

(4)

First released on AP - Edexcel Discord
<https://sites.google.com/view/ap-edexcel>

(Total for Question 10 is 8 marks)

Q39**PAST PAPER**

Pearson IAL WMA11 • June 2024 • Q10

P1 • TOPIC 13 10 marks

10. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = 6x - \frac{(2x-1)(3x+2)}{2\sqrt{x}}$

- the point $P(4, 12)$ lies on C

(a) find the equation of the normal to C at P , giving your answer in the form $y = mx + c$ where m and c are integers to be found,

(4)

(b) find $f(x)$, giving each term in simplest form.

(6)

(Total for Question 10 is 10 marks)

Q40 **PAST PAPER** Pearson IAL WMA11 • October 2024 • Q8**P1 • TOPIC 13 7 marks**

8. A curve C has equation $y = f(x)$.

The point P with x coordinate 3 lies on C

Given

- $f'(x) = 4x^2 + kx + 3$ where k is a constant
- the normal to C at P has equation $y = -\frac{1}{24}x + 5$

(a) show that $k = -5$

(3)

(b) Hence find $f(x)$.

(4)

(Total for Question 8 is 7 marks)

Q41 PAST PAPER Pearson IAL WMA11 • January 2025 • Q6

P1 • TOPIC 13 10 marks

6. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- the point $P(4, -5)$ lies on C
- $f'(x) = \frac{2x^2 + ax + b}{4\sqrt{x}}$, where a and b are constants
- the gradient of the tangent to C at P is 7

(a) show that

$$4a + b = 24 \quad (2)$$

Given also that $a + b = -9$

(b) find, in simplest form, $f(x)$ (7)

Curve C is transformed to the curve with equation $y = f(x - 3)$

Given that point P is transformed to the point Q ,

(c) state the coordinates of Q . (1)

(Total for Question 6 is 10 marks)

Q42 **PAST PAPER** Pearson IAL WMA11 • June 2025 • Q8**P1 • TOPIC 13 11 marks**

8. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

A curve has equation $y = f(x)$, $x > 0$

The point $P(4, 12)$ lies on the curve.

Given that

- $f'(x) = 3\sqrt{x} + kx^2$ where k is a constant
- the equation of the tangent to the curve at P has equation $y = 10x + c$ where c is a constant

(a) (i) show that $k = \frac{1}{4}$

(ii) find the value of c

(4)

(b) Hence find the value of $f''(x)$ at P .

(3)

(c) Find $f(x)$.

(4)

(Total for Question 8 is 11 marks)

Q43**PAST PAPER**

Pearson IAL WMA11 · October 2025 · Q8

P1 · TOPIC 13 8 marks

8. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = 2x + \frac{8}{x^2} + k$, where k is a constant
- the equation of the tangent to the curve at $x = \sqrt{2}$ is $y = 5x - 3\sqrt{2}$

(a) find the exact value of k .

(2)

(b) Find an equation of the normal to the curve at $x = \sqrt{2}$

(2)

(c) Find $f(x)$, writing your answer in simplest form.

(4)

(Total for Question 8 is 8 marks)

Q44 PAST PAPER Pearson IAL WMA11 • January 2026 • Q8

P1 • TOPIC 13 12 marks

8. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

A curve C has equation $y = f(x)$, $x > 0$

A point P lies on C .

Given that

- $f'(x) = 4x^2 + \frac{6}{x^2} - 4$
- the equation of the tangent to C at P is $y = 10x - 6\sqrt{3}$

- (a) (i) verify that $\sqrt{3}$ is a possible x coordinate of P ,
(ii) find, using algebra, the other possible x coordinate of P .

(6)

Given that the x coordinate of P is $\sqrt{3}$

- (b) find an equation of the normal to C at P .

(2)

- (c) Find $f(x)$, writing your answer in simplest form.
You must show each stage of your working.

(4)

(Total for Question 8 is 12 marks)

Q45 **PAST PAPER** Pearson IAL WMA11 • June 2026 • Q8**P1 • TOPIC 13** **11 marks**

8. A curve C has equation $y = f(x)$, $x > 0$

The point P with x coordinate $\frac{1}{4}$ lies on C .

Given that

- $f'(x) = 10 + \frac{5\sqrt{x} - 4}{2x^2}$
- the tangent to C at P has equation $y = kx + 2$, where k is a constant

(a) (i) find the value of k

(ii) find the equation of the normal to C at P .

Give the answer in the form $y = mx + c$, where m and c are constants.

(6)

(b) Find $f(x)$.

(5)

Question 8 continued

(Total for Question 8 is 11 marks)

04 From $f''(x)$: Integrate Twice

Two integrations, two constants — so you need two facts: a gradient and a point.

WHEN YOU SEE → YOU DO

$$“f''(x) = \dots”, “\frac{d^2y}{dx^2} = \dots”$$

→ integrate twice

$$“f'(x) = 5 \text{ at } P”, “f'(2) = 8”$$

→ fixes the **first** constant

$$“P \text{ lies on the curve}”, “f(3) = 20”$$

→ fixes the **second** constant

$$“f''(x) = 0 \text{ when } x = 27”, “\text{find } a / k”$$

→ differentiate the given f' , set = 0

$$“(2x - 1) \text{ and } (3x + 2) \text{ are factors of } f(x)”$$

→ $f(\frac{1}{2}) = 0$ and $f(-\frac{2}{3}) = 0$

WORKED MODEL • JANUARY 2020

$$f''(x) = \frac{6}{\sqrt{x^3}} + x, \quad f'(4) = -4, \quad P(4, -50).$$

$$1 \quad f'(x) = -12x^{-\frac{1}{2}} + \frac{1}{2}x^2 + c_1; \quad -4 = -6 + 8 + c_1 \Rightarrow c_1 = -6$$

first integration

$$2 \quad f(x) = -24x^{\frac{1}{2}} + \frac{1}{6}x^3 - 6x + c_2$$

second integration

$$3 \quad -50 = -48 + \frac{32}{3} - 24 + c_2 \Rightarrow c_2 = \frac{34}{3}$$

use P

$$\text{ANSWER} \quad f(x) = \frac{1}{6}x^3 - 24\sqrt{x} - 6x + \frac{34}{3}$$

IDEA 04

Practise: From $f''(x)$: Integrate Twice

My questions first, then selected practice, then real past papers.

Q46 DR ESLAM Integrate twice

P1 · TOPIC 13 5 marks

$f''(x) = 12x$, $f'(1) = 10$ and $f(1) = 3$. Find $f(x)$.

$f(x) =$ _____

SELECTED PRACTICE

Q47**PRACTICE**

Practice · Medium

P1 · TOPIC 13 7 marks

5 (a) Given $\frac{d^2 y}{dx^2} = 30x$ and that when $x = 1$, $\frac{dy}{dx} = 17$, show that

$$\frac{dy}{dx} = 15x^2 + 2$$

(4 marks)

(b) Find an equation for y in terms of x , given that when $x = 2$, $y = 40$.

(3 marks)

Q48**PRACTICE**

Practice · Hard

P1 · TOPIC 13**5 marks**

5 A function, $f(x)$, has second derivative given by

$$f''(x) = 6(x - 2).$$

Given that $f(3) = 20$, and $f'(2) = 8$, find $f(x)$.

(5 marks)

Q49**PRACTICE**

Practice · Very Hard

P1 · TOPIC 13 6 marks

5 A function, $f(x)$, has second derivative given by

$$f''(x) = 2(18x - 5).$$

Given that $(2x - 1)$ and $(3x + 2)$ are factors of $f(x)$, find $f(x)$.

(6 marks)

PAST PAPERS

Q50 **PAST PAPER** Pearson IAL WMA11 · October 2019 · Q11**P1 · TOPIC 13** **10 marks**

11. A curve has equation $y = f(x)$.

The point $P\left(4, \frac{32}{3}\right)$ lies on the curve.

Given that

- $f''(x) = \frac{4}{\sqrt{x}} - 3$
- $f'(x) = 5$ at P

find

(a) the equation of the tangent to the curve at P , writing your answer in the form $y = mx + c$, where m and c are constants to be found,

(2)

(b) $f(x)$.

(8)

(Total 10 marks)

Q51 **PAST PAPER** Pearson IAL WMA11 • January 2020 • Q11**P1 • TOPIC 13** **11 marks**

11. A curve has equation $y = f(x)$, where

$$f''(x) = \frac{6}{\sqrt{x^3}} + x \quad x > 0$$

The point $P(4, -50)$ lies on the curve.

Given that $f'(x) = -4$ at P ,

- (a) find the equation of the normal at P , writing your answer in the form $y = mx + c$, where m and c are constants,

(3)

- (b) find $f(x)$.

(8)

(Total 11 marks)

Q52 **PAST PAPER** Pearson IAL WMA11 · January 2023 · Q11**P1 · TOPIC 13** **8 marks**

11. A curve C has equation $y = f(x)$, $x > 0$

Given that

- $f''(x) = 4x + \frac{1}{\sqrt{x}}$
- the point P has x coordinate 4 and lies on C
- the tangent to C at P has equation $y = 3x + 4$

(a) find an equation of the normal to C at P

(2)

(b) find $f(x)$, writing your answer in simplest form.

(6)

(Total for Question 11 is 8 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 The rule** Power +1, $\div$ new power, +c. Rewrite roots and fractions first.
- 02 Brackets / fractions** Expand, split, then integrate term by term.
- 03 Find the curve** Integrate $f'(x)$, the point gives c.
- 04 From $f''(x)$** Integrate twice: gradient fact $\rightarrow c_1$, point $\rightarrow c_2$.

BOTTOM LINE Rewrite $\rightarrow$ power up, divide $\rightarrow$ +c $\rightarrow$ a point finds c; a gradient finds the c of f' .

Answers

Integration • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • INTEGRATE POWERS OF X

- Q1** Dr Eslam • Power up, divide, +c (a) $2x^3 + c$ (b) $-2x^{-2} + c$ (c) $\frac{2}{3}x^{\frac{3}{2}} + c$
- Q2** Dr Eslam • Numbers alone $5x - x^3 + c$
- Q3** Practice • Easy (i) $x^2 + c$ (ii) $2x^3 + c$ (iii) $\frac{1}{3}x^{\frac{3}{2}} + c$
- Q4** Practice • Medium $\frac{2}{3}t^3 + 2t^{\frac{1}{2}} + t^{-3} + c$
- Q5** Practice • Very Hard $\frac{4}{3}x^{\frac{3}{2}} + \frac{15}{4}x^{\frac{4}{3}} + c$
- Q6** Pearson IAL WMA11 • January 2020 • Q1 $\frac{2}{3}x^4 - x^{\frac{1}{2}} - 5x + c$
- Q7** Pearson IAL WMA11 • October 2021 • Q1 $3x^4 + \frac{1}{3}x^{\frac{1}{2}} + \frac{1}{2}x^{-3} + c$
- Q8** Pearson IAL WMA11 • January 2022 • Q1 $\frac{2}{5}x^4 + \frac{2}{9}x^{-3} - x + c$
- Q9** Pearson IAL WMA11 • June 2022 • Q1 $\frac{5}{3}x^6 + \frac{3}{2}x^4 + 3x^{-1} + c$
- Q10** Pearson IAL WMA11 • June 2024 • Q1 $2x^5 + \frac{3}{4}x^{-2} - 7x + c$
- Q11** Pearson IAL WMA11 • January 2025 • Q1 $2x^4 - 4x^{\frac{3}{2}} + \frac{1}{5}x^{-2} + c$
- Q12** Pearson IAL WMA11 • January 2026 • Q1 (a) $12x^2 - \frac{3}{4}x^{-\frac{3}{2}}$ (b) $x^4 + 3x^{\frac{1}{2}} - 8x + c$

IDEA 02 • BRACKETS AND FRACTIONS FIRST

- Q13** Dr Eslam • Expand first $\frac{1}{3}x^3 - \frac{3}{2}x^2 - 10x + c$
- Q14** Dr Eslam • Split the fraction $\frac{2}{5}x^{\frac{5}{2}} - 8x^{\frac{1}{2}} + c$
- Q15** Practice • Easy (a) $a = 1, b = 4$ (b) $3x + \frac{4}{5}x^5 + c$
- Q16** Practice • Hard (b) $8x - 6x^2 + 2x^3 - \frac{1}{4}x^4 + c$
- Q17** Practice • Very Hard (b) $216x - 54x^2 + 6x^3 - \frac{1}{4}x^4 + c$
- Q18** Pearson IAL WMA11 • January 2021 • Q9 (i) $\frac{9}{10}x^{\frac{5}{2}} + 2x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + c$ (ii) $f(x) = \frac{1}{3}x^3 - 2x^2 + 5x - 8$
- Q19** Pearson IAL WMA11 • June 2021 • Q4 $\frac{3}{8}x^2 + \frac{1}{3}x^{\frac{3}{2}} - \frac{15}{4}x - 5x^{\frac{1}{2}} + c$
- Q20** Pearson IAL WMA11 • October 2022 • Q2 (a) $a = 2, b = -3$ (b) $\frac{4}{35}x^{\frac{7}{2}} - \frac{6}{25}x^{\frac{5}{2}} - \frac{64}{15}x^{\frac{3}{2}} - 6x^{\frac{1}{2}} + c$
- Q21** Pearson IAL WMA11 • January 2023 • Q3 $\frac{1}{2}x^4 - \frac{3}{2}x^{-1} + c$
- Q22** Pearson IAL WMA11 • January 2024 • Q1 $3x^4 + \frac{8}{3}x^3 - \frac{75}{2}x^2 - 50x + c$
- Q23** Pearson IAL WMA11 • June 2025 • Q3 $\frac{2}{15}x^{\frac{5}{2}} + \frac{4}{3}x^{\frac{3}{2}} + 6x^{\frac{1}{2}} + c$
- Q24** Pearson IAL WMA11 • October 2025 • Q2 $\frac{1}{2}x^4 - \frac{7}{3}x^3 - 6x^2 + 45x + c$
- Q25** Pearson IAL WMA11 • June 2026 • Q1 $3x^3 + 12x - \frac{4}{x} + c$

IDEA 03 • FIND THE CURVE

- Q26** Dr Eslam • One point, one c $3x^2 - 4x + 1$
- Q27** Dr Eslam • Tangent given $k = 2; f(x) = x^3 + 2x + 1$
- Q28** Practice • Easy (a) $x^5 + 2x^3 + x^2 + 3x + c$ (b) $f(x) = x^5 + 2x^3 + x^2 + 3x + 3$
- Q29** Practice • Medium $y = 2x^3 - x^2 + 3x - 60$
- Q30** Practice • Very Hard $y = 4\sqrt{x} - \frac{1}{2}x^2 - 3x + 4$
- Q31** Pearson IAL WMA11 • June 2021 • Q6 (a) $y = -\frac{5}{6}x + \frac{26}{3}$ (b) $f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{\frac{4}{3}} + \frac{10}{3}$
- Q32** Pearson IAL WMA11 • October 2021 • Q10 (a) $a = \frac{4}{9}$ (b) $f(x) = \frac{2}{9}x^2 - 9x^{\frac{4}{3}} + \frac{7}{9}$
- Q33** Pearson IAL WMA11 • January 2022 • Q6 (a) $\frac{49}{8}; 49x - 8y - 36 = 0$ (b) $f(x) = \frac{2}{3}x^{\frac{3}{2}} + 12x^{\frac{1}{2}} - 18x^{-\frac{1}{2}} - \frac{1}{3}$
- Q34** Pearson IAL WMA11 • June 2022 • Q7 (a) $A = -4$ (b) $f(x) = 4\sqrt{x} + \frac{4}{x} + 3x - \frac{109}{3}$
- Q35** Pearson IAL WMA11 • October 2022 • Q5 (a) $f(x) = 24\sqrt{x} + \frac{1}{6}x^2 - 4x - \frac{83}{2}$ (b) $(0, 11)$
- Q36** Pearson IAL WMA11 • June 2023 • Q8 (a) $12x - y - 36 = 0$ (b) $f(x) = \frac{1}{16}x^4 - 16x^{\frac{1}{2}} + 28$
- Q37** Pearson IAL WMA11 • October 2023 • Q7 (a) $\frac{15}{2}; 2x + 15y + 7 = 0$ (b) $f(x) = \frac{2}{5}x^{\frac{5}{2}} - \frac{7}{4}x + 5x^{\frac{1}{2}} - \frac{84}{5}$
- Q38** Pearson IAL WMA11 • January 2024 • Q10 (a) $k = 24\sqrt{2}$ (b) $f(x) = \frac{8}{5}x^{\frac{5}{2}} - \frac{24\sqrt{2}}{x} + \frac{68\sqrt{2}}{5}$
- Q39** Pearson IAL WMA11 • June 2024 • Q10 (a) $y = 2x + 4$ (b) $f(x) = 3x^2 - \frac{6}{5}x^{\frac{5}{2}} - \frac{1}{3}x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + \frac{16}{15}$
- Q40** Pearson IAL WMA11 • October 2024 • Q8 (b) $f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x - \frac{141}{8}$
- Q41** Pearson IAL WMA11 • January 2025 • Q6 (b) $a = 11, b = -20: f(x) = \frac{1}{5}x^{\frac{5}{2}} + \frac{11}{6}x^{\frac{3}{2}} - 10x^{\frac{1}{2}} - \frac{91}{15}$ (c) $Q(7, -5)$
- Q42** Pearson IAL WMA11 • June 2025 (a) $k = \frac{11}{4}, c = -28$ (b) $\frac{11}{4}$ (c) $f(x) = 2x^{\frac{3}{2}} + \frac{1}{12}x^3 - \frac{28}{3}$

IDEA 04 • FROM $f''(x)$: INTEGRATE TWICE

- Q46** Dr Eslam • Integrate twice $2x^3 + 4x - 3$
- Q47** Practice • Medium (b) $y = 5x^3 + 2x - 4$
- Q48** Practice • Hard $f(x) = x^3 - 6x^2 + 20x - 13$
- Q49** Practice • Very Hard $f(x) = 6x^3 - 5x^2 - 3x + 2$
- Q50** Pearson IAL WMA11 • October 2019 • Q11 (a) $y = 5x - \frac{28}{3}$ (b) $f(x) = \frac{16}{3}x^{\frac{3}{2}} - \frac{3}{2}x^2 + x - 12$
- Q51** Pearson IAL WMA11 • January 2020 • Q11 (a) $y = \frac{1}{4}x - 51$ (b) $f(x) = \frac{1}{6}x^3 - 24\sqrt{x} - 6x + \frac{34}{3}$
- Q52** Pearson IAL WMA11 • January 2023 • Q11 (a) $y = -\frac{1}{3}x + \frac{52}{3}$ (b) $f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{\frac{3}{2}} - 33x + \frac{284}{3}$